THE VACUUM BELL CORRELATION OF THE MASSIVE FREE SCALAR FIELD ATTAINS THE CLASSICAL BOUND FOR BOUNDED REGIONS BEYOND A FINITE SEPARATION

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Abstract. Summers and Werner showed that the vacuum of a massive relativistic quantum field violates the Clauser–Horne–Shimony–Holt (CHSH) inequality maximally for suitable spacelike separated regions, and that the violation decays as the regions are moved apart. Problem 12 of the IQOQI Vienna Open Quantum Problems list asks whether some violation is possible for regions arbitrarily far apart. We answer it in the negative for fixed bounded regions. For the real massive free scalar field in $3+1$ dimensions, the vacuum Bell correlation of two double cones of radius R centred at equal times, the supremum of the CHSH values of their local observables, equals the classical bound once the spatial distance between their centres exceeds a fixed multiple of R, independent of the mass. Hence the vacuum satisfies the CHSH inequality for any two bounded regions moved sufficiently far apart in a fixed spacelike direction, although it remains entangled across them.

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1. Introduction

Vacuum entanglement is a structural feature of relativistic quantum field theory. The Reeh–Schlieder theorem makes the vacuum cyclic for local algebras [29]; together with noncommutativity, this cyclicity implies entanglement across spacelike separated regions [16]. The operator-algebraic and quantitative aspects of vacuum entanglement are developed in the account of Hollands and Sanders [19], and the operator-algebraic ideas are explained informally in Witten's notes [44]. Bell correlation is a stronger property: some local measurements in the vacuum violate a Bell inequality. At large spacelike separation the vacuum stays entangled [16, 41], while its Bell correlation decays towards the classical value, as the mass-gap bound below shows.

Summers and Werner had established maximal vacuum violation of Bell's inequalities for suitable localizations in free Bose and Fermi theories [36, 38]. Their further results had identified complementary wedges and tangent double cones as geometries with maximal Bell correlation; under the hypotheses of those results, maximality holds in every normal state [39, 40].

At positive separation, a mass gap instead gives the upper bound on the Bell correlation (2.4)

$$\beta\big(\omega;\mathfrak{A}(O_1),\mathfrak{A}(O_2)\big)\le 1+2e^{-md(O_1,O_2)}.$$

Here ω is the vacuum state of a theory with lowest mass $m>0$, the normalization has classical bound 1, and $d(O_1,O_2)$ is the maximal timelike distance through which one region can be translated while preserving spacelike separation [37], [35, Proposition 4.5]. That is, $d(O_1,O_2)$ is the largest

τ such that $O_1 + (t, 0)$ and O_2 are spacelike separated for all $|t| < \tau$; for the two double cones considered below, it equals $L - 2R$.

The relation between Bell correlation and statistical independence had been systematized by Summers [34]. His 2008 survey, subsequently published as [35], had noted that model-independent lower bounds on the distant vacuum Bell correlation were not yet available. Whether two distant regions show Bell correlation depends on the state: Landau's result gives states with maximal violation for commuting nonabelian algebras satisfying the Schlieder property [24]. Halvorson and Clifton had proved generic Bell correlation of states for spacelike local algebras [16]; they had noted that none of their results "decide the question of whether the vacuum state always retains some Bell correlation across arbitrary spacelike separated regions" [16, Section IV]. Verch and Werner had further proved vacuum distillability across arbitrarily large spacelike distances [41].

Reznik, Retzker and Silman [30], and later Matsumura and Nambu [26], had obtained Bell–CHSH violation after filtering locally coupled probes, in [30] at every separation of the probes. Their unfiltered detector states satisfy CHSH in the regimes analysed. Filtering changes the experiment by selecting an output state, whereas the regional Bell correlation considered here retains every outcome of the original vacuum measurement. Verch and Werner had remarked that the authors of [30] do not demonstrate Bell correlations of the vacuum over arbitrarily separated and arbitrarily small regions "in the sense of proving a violation of the CHSH inequalities by the quantum field observables themselves" [41, Section 7].

The question whether the vacuum can violate the CHSH inequality for regions arbitrarily far apart, with every outcome retained, is Problem 12 of the IQOQI Vienna Open Quantum Problems list [21] and of the collection of open problems in quantum information of Krüger and Werner [22], where the problem is dated 22 January 2002 and has R. Verch as contact; its statement is reproduced verbatim.[1]

Problem 12: Bell Inequalities for long range vacuum correlations (IQOQI Vienna Open Quantum Problems [21])

It is well known [1] that vacuum fluctuations maximally violate the CHSH-Bell inequalities for suitable spacelike separated observables, and that this violation goes to zero as the two localization regions are moved apart.
Decide whether some (necessarily small) violation of the inequalities is possible for regions arbitrarily far apart. For definiteness, consider a massive scalar free relativistic Bose field.

In this paper we prove that the answer is negative for two double cones of equal radius moved apart (Theorem 1.1 below), and hence for any two bounded regions moved apart in a fixed spacelike direction (Corollary 1.2). Problem 12 remains open for regions that grow in proportion to their separation, at ratios that Theorem 1.1 does not cover, and for wedges at a positive gap, as does the relation of these results to Bell tests after local filtering (Section 9).

For $R > 0$, the double cone

$$D_R = \{(x^0, \mathbf{x}) \in \mathbb{R}^{1+3} : |x^0| + |\mathbf{x}| < R\}$$

has base $B_R(0)$ at time zero. We consider the real free scalar field of mass $m > 0$, its vacuum state, and the two local von Neumann algebras (defined in Section 2)

$$\mathfrak{A}_m(D_R), \qquad \mathfrak{A}_m(D_R + (0, L\mathbf{e}_1)), \qquad L > 2R.$$

Thus L is the spatial distance between the centres of the two double cones, and their bases are separated by the gap $L - 2R$.

[1]The box quotes the IQOQI list, which gives the same date and contact. The reference "[1]" in the quoted problem statement is the paper of Summers and Werner [36].

The Summers–Werner Bell correlation of the two algebras in the vacuum state is denoted by $\beta_m(R, L)$, with classical bound 1 and Tsirelson bound $\sqrt{2}$; its algebraic definition is (2.4). The supremum defining $\beta_m(R, L)$ includes all binary local measurements, represented by the self-adjoint contractions in (2.4). The measurements may depend on L, and their outcome probabilities may be arbitrarily small.

Theorem 1.1 (Separation threshold)**.** *There is a finite constant $C_* > 2$, independent of m and R, such that*

$$\forall m, R > 0, \qquad L \geq C_* R \quad \Longrightarrow \quad \beta_m(R, L) = 1. \tag{1.1}$$

The constant produced by the argument is astronomically large: for the parameters in Table 1 of Section 8, $C_* \geq C_{\mathrm{prob}} = Q_0 \cdot 10^{362\,232.284166\ldots}$, where the numerical enclosure of the correlation coefficient Q_0 of (8.13) remains open. Theorem 1.1 is therefore a pure existence statement about eventual behaviour, with no information at physically relevant separations (Section 9, item (4)).

Corollary 1.2 (Fixed bounded regions)**.** *For every $m > 0$, any two fixed bounded localization regions O_1, O_2, and any fixed spacelike translation direction, the vacuum Bell correlation of the local algebras of O_1 and of the translate of O_2 equals 1 for all sufficiently large translations of O_2 in that direction.*

Corollary 1.2 follows from Theorem 1.1 by Poincaré covariance and isotony (Section 8.4).

1.1. **The mechanism of the threshold.** The central estimates compare a joint event probability with the product of its marginals. For projections $E \in \mathfrak{A}_m(D_R)$ and $P \in \mathfrak{A}_m(D_R + (0, L\mathbf{e}_1))$, with subscripts A and B for the two double cones, we write

$$e_A = \omega(E), \qquad e_B = \omega(P), \qquad w = \omega(EP).$$

With the constants k, λ of (3.7), and with the exponents a, b and $\eta = 1 - (1 - b)/(ab^2)$ of (3.2), the two event bounds have the form

$$w \leq k(e_A e_B)^b, \qquad w \geq \lambda(e_A e_B)^{1/a}, \qquad 0 < a < \tfrac{1}{2} < b < 1, \qquad 1 - \frac{1-b}{ab^2} > 0.$$

The constants k, λ approach 1 as L/R increases, uniformly in m (see (8.11)). By (8.3), the bounds on $k - 1$ and $1 - \lambda$ decay like powers of R/L, more slowly than the mass-gap bound above, because the uniformity extends to small mR.

In a violating CHSH experiment, the two event bounds force a positive lower bound on the least likely winning cell, a joint outcome probability defined in Step 2 of the proof of Theorem 3.1. The correlation estimate (3.3), supplied by (8.13), forces that same cell to have probability tending to zero as L/R grows, and beyond a finite separation the two requirements are incompatible (Theorem 3.1). The upper event bound and the correlation estimate are quantitative forms of clustering; the lower event bound, which controls the joint probability of arbitrarily rare events through the joint covariance norm (Sections 5 and 7), is the ingredient specific to this argument, and it excludes a violation carried by events of very small probability.

The two event bounds arise from opposite orders of comparison between a finite Gaussian vacuum restriction ρ_F and the product σ_F of its marginals, through the sandwiched Rényi quantities $\widetilde{Q}_{r_+}(\rho_F \| \sigma_F)$ in (8.5) and $\widetilde{Q}_{r_-}(\sigma_F \| \rho_F)$ in (8.8). The forward comparison uses the uncertainty reserves (2.9) of the two balls (Section 6). The reverse comparison uses the uncertainty reserve of their union in the joint covariance norm (Section 7). Estimates on the complete source spaces precede all finite-dimensional restrictions, and compression preserves the bounds. Hence the constants are independent of the selected modes. Buchholz's construction of normal product states for separated local algebras [7] supplies the broader split-property context for the comparison with the product of the marginals, which the argument carries out on finite source systems (Section 8).

2. Setting and notation

2.1. **The field, local algebras, the Bell correlation, and scaling.** We use units $\hbar = c = 1$ and metric signature $(+,-,-,-)$. Thus $x \cdot y = x^0 y^0 - \mathbf{x}\cdot\mathbf{y}$. Momentum components are written $p_0, \mathbf{p} = (p_1, p_2, p_3)$ without index lowering. Spatial Fourier transforms are

$$\widehat{f}(\mathbf{p}) = \int_{\mathbb{R}^3} e^{-i\mathbf{p}\cdot\mathbf{x}} f(\mathbf{x})\, d^3x, \qquad f(\mathbf{x}) = \frac{1}{(2\pi)^3}\int_{\mathbb{R}^3} e^{i\mathbf{p}\cdot\mathbf{x}}\widehat{f}(\mathbf{p})\, d^3p.$$

The unitary Fourier transform is $\mathcal{F}f = (2\pi)^{-3/2}\widehat{f}$. Euclidean norms on complex spacetime are written $|\cdot|_{\mathrm{E}}$.

For mass $\mu > 0$, the positive mass shell and one-particle space are

$$H_\mu^+ = \{p \in \mathbb{R}^{1+3} : p\cdot p = \mu^2,\ p_0 > 0\}, \qquad \mathcal{H}_{1,\mu} = L^2(H_\mu^+, d\mathfrak{m}_\mu), \qquad d\mathfrak{m}_\mu(p) = \frac{d^3\mathbf{p}}{2(2\pi)^3 p_0}.$$

The vacuum representation acts on the symmetric Fock space $\mathscr{F}_\mu = \Gamma_{\mathrm{s}}(\mathcal{H}_{1,\mu})$, with vacuum vector Ω_μ. We abbreviate its vector state by $\omega(X) = \langle \Omega_\mu, X\Omega_\mu\rangle$ when the mass is fixed.

The time-zero field and its conjugate momentum, smeared with real $f, g \in C_c^\infty(\mathbb{R}^3)$, are denoted by $\phi_\mu(f)$ and $\pi_\mu(g)$. For a source pair $\mathsf{x} = (f, g)$, we set

$$h_\mu = (-\nabla^2 + \mu^2)^{1/2}, \qquad \Phi_\mu(\mathsf{x}) = \phi_\mu(f) + \pi_\mu(g), \qquad \mathsf{W}_\mu(\mathsf{x}) = e^{i\Phi_\mu(\mathsf{x})}.$$

The symplectic form and vacuum covariance are

$$\varsigma(\mathsf{x},\mathsf{y}) = \int_{\mathbb{R}^3} (fg' - gf')\, d^3x, \qquad \mathsf{y} = (f', g'), \tag{2.1}$$

$$\mathsf{v}_{\mu,U}(\mathsf{x},\mathsf{y}) = \frac{1}{2}\left(\langle f, h_\mu^{-1} f'\rangle + \langle g, h_\mu g'\rangle\right) \quad \text{for sources supported in } U. \tag{2.2}$$

The subscript U in (2.2) labels the source Hilbert space introduced in Section 2.2; the right-hand side is defined for any pair of sources and is independent of U. The free-field Weyl algebra satisfies

$$\mathsf{W}_\mu(\mathsf{x})\mathsf{W}_\mu(\mathsf{y}) = e^{-i\varsigma(\mathsf{x},\mathsf{y})/2}\mathsf{W}_\mu(\mathsf{x}+\mathsf{y}), \qquad \mathsf{W}_\mu(\mathsf{x})^* = \mathsf{W}_\mu(-\mathsf{x}).$$

The vacuum is the centred quasi-free state characterized by

$$\omega(\mathsf{W}_\mu(\mathsf{x})) = \exp\left[-\frac{1}{2}\mathsf{v}_{\mu,U}(\mathsf{x},\mathsf{x})\right].$$

These formulas are the usual free-field and Weyl conventions [15, 32]; the covariances of [32] are twice ours.

For a real $F \in C_c^\infty(\mathbb{R}^{1+3})$, the spacetime-smeared field is $\phi_\mu(F) = \int \phi_\mu(x)F(x)\, d^4x$, with spacetime Weyl operator $\mathsf{W}_\mu(F) = e^{i\phi_\mu(F)}$. With $\partial_0 = \partial/\partial x^0$, let $E_{\mathrm{ret}}F$ and $E_{\mathrm{adv}}F$ be the solutions v of $(\partial_0^2 - \nabla^2 + \mu^2)v = F$ that vanish at all sufficiently early and at all sufficiently late times, respectively. Their difference $v_F = E_{\mathrm{ret}}F - E_{\mathrm{adv}}F$ solves the homogeneous equation, and the field equation expresses $\phi_\mu(F)$ through the Cauchy data of v_F at time zero:

$$\phi_\mu(F) = \Phi_\mu(\mathsf{x}), \qquad \mathsf{x} = \big(\partial_0 v_F(0,\cdot),\ -v_F(0,\cdot)\big).$$

By finite propagation speed, v_F vanishes outside the causal future and past of $\operatorname{supp} F$, so the time-zero pair x is supported in the base $B_R(\mathbf{x}_0)$ of any double cone $D_R + (0, \mathbf{x}_0)$ containing $\operatorname{supp} F$.

For a spacetime region O, the notation $\mathfrak{A}_\mu(O)$ denotes the von Neumann algebra generated by the Weyl operators of real spacetime test functions supported in O. Araki's time-zero identification [1], together with the free Bose-field modular description [12], gives

$$\mathfrak{A}_\mu(D_R) = \{\mathsf{W}_\mu(f,g) : f, g \in C_c^\infty(B_R(0);\mathbb{R})\}''. \tag{2.3}$$

Finite propagation gives the inclusion from spacetime smearings to time-zero sources: if $\operatorname{supp} F \subset D_R$, the formula for $\phi_\mu(F)$ above gives $\mathsf{W}_\mu(F) = \mathsf{W}_\mu(\mathsf{x})$ with x supported in $B_R(0)$, so $\mathfrak{A}_\mu(D_R)$ lies in the right-hand side of (2.3).

For the converse inclusion, compact Cauchy data strictly inside $B_R(0)$ determine a homogeneous Klein–Gordon solution. Let $f, g \in C_c^\infty(B_R(0); \mathbb{R})$, choose $0 < \delta < R$ with $\operatorname{supp} f \cup \operatorname{supp} g \subset \overline{B_{R-\delta}(0)}$, and let u be the solution of $(\partial_0^2 - \nabla^2 + \mu^2)u = 0$ with $u(0,\cdot) = -g$ and $\partial_0 u(0,\cdot) = f$. By finite propagation speed, $u(x^0,\cdot)$ is supported in $\overline{B_{R-\delta+|x^0|}(0)}$.

Multiplying u by a temporal cutoff that changes from 0 to 1 in a sufficiently short interval around zero, and then applying the Klein–Gordon operator, produces a test function supported in D_R. Let $\chi \in C^\infty(\mathbb{R})$ vanish on $(-\infty, -\epsilon]$ and equal 1 on $[\epsilon, \infty)$, where $0 < 2\epsilon < \delta$. With $\chi = \chi(x^0)$,

$$F = (\partial_0^2 - \nabla^2 + \mu^2)(\chi u) = \chi'' u + 2\chi'\, \partial_0 u$$

is real and smooth, and vanishes unless $|x^0| \le \epsilon$ and $|\mathbf{x}| \le R - \delta + \epsilon$. On that set $|x^0| + |\mathbf{x}| \le R - \delta + 2\epsilon < R$, so $\operatorname{supp} F \subset D_R$.

The causal solution v_F of F recovers the original data. The functions χu and $(\chi - 1)u$ solve $(\partial_0^2 - \nabla^2 + \mu^2)v = F$ and vanish for $x^0 \le -\epsilon$ and for $x^0 \ge \epsilon$, respectively, so $E_{\mathrm{ret}} F = \chi u$, $E_{\mathrm{adv}} F = (\chi - 1)u$ and $v_F = u$. The formula for $\phi_\mu(F)$ above gives

$$\phi_\mu(F) = \Phi_\mu\big(\partial_0 u(0,\cdot),\, -u(0,\cdot)\big) = \Phi_\mu(f, g),$$

so $\mathsf{W}_\mu(f,g) = \mathsf{W}_\mu(F) \in \mathfrak{A}_\mu(D_R)$. Since f and g were arbitrary, the converse inclusion in (2.3) follows.

An *effect* is an operator E with $0 \le E \le I$. A self-adjoint contraction A represents the binary measurement with effects $(I \pm A)/2$. For commuting von Neumann algebras $\mathfrak{A}, \mathfrak{B}$, we use the Summers–Werner normalization

$$\beta(\omega; \mathfrak{A}, \mathfrak{B}) = \sup_{\substack{A_i = A_i^* \in \mathfrak{A},\ \|A_i\| \le 1 \\ B_j = B_j^* \in \mathfrak{B},\ \|B_j\| \le 1}} \frac{1}{2} \left| \omega\big(A_0(B_0 + B_1) + A_1(B_0 - B_1)\big) \right|. \tag{2.4}$$

The classical CHSH bound is 1 [11, 37], and Tsirelson's bound is $\sqrt{2}$ [10]. The identities $A_0 = A_1 = B_0 = B_1 = I$ give the value 1, so $\beta(\omega; \mathfrak{A}, \mathfrak{B}) \ge 1$. Some authors, including Verch and Werner [41], omit the factor $1/2$, so their classical and Tsirelson bounds are 2 and $2\sqrt{2}$, respectively. In the normalization (2.4),

$$\beta_m(R, L) = \beta\big(\omega; \mathfrak{A}_m(D_R), \mathfrak{A}_m(D_R + (0, L\mathbf{e}_1))\big).$$

Lemma 2.1 (Dilation)**.** *The source transformation*

$$\mu = mR, \qquad s = L/R, \qquad \widetilde{f}(\mathbf{y}) = R^2 f(R\mathbf{y}), \qquad \widetilde{g}(\mathbf{y}) = R g(R\mathbf{y}) \tag{2.5}$$

preserves the symplectic form and vacuum covariance, with mass m replaced by μ. The transformation induces a vacuum-preserving isomorphism of the corresponding local Weyl algebras and their von Neumann closures. Consequently,

$$\beta_m(R, L) = \beta_{mR}(1, L/R).$$

Proof. Step 1: the symplectic form. Let $\mathsf{x} = (f, g)$ and $\mathsf{y} = (f', g')$ be real smooth sources supported in a bounded open set U, and let $\widetilde{\mathsf{x}} = (\widetilde{f}, \widetilde{g})$ and $\widetilde{\mathsf{y}} = (\widetilde{f'}, \widetilde{g'})$ be their transforms by (2.5), supported in $\widetilde{U} = \{\mathbf{y} : R\mathbf{y} \in U\}$. A change of variables in (2.1) preserves the symplectic form: with $\mathbf{x} = R\mathbf{y}$,

$$\int_{\mathbb{R}^3} \widetilde{f}\, \widetilde{g'}\, d^3y = R^3 \int_{\mathbb{R}^3} f(R\mathbf{y})\, g'(R\mathbf{y})\, d^3y = \int_{\mathbb{R}^3} f g'\, d^3x,$$

and likewise for $\widetilde{g}\, \widetilde{f'}$, so that $\varsigma(\widetilde{\mathsf{x}}, \widetilde{\mathsf{y}}) = \varsigma(\mathsf{x}, \mathsf{y})$.

Step 2: the covariance. The same change of variables in the Fourier integral gives

$$\widehat{\widetilde{f}}(\mathbf{q}) = R^{-1} \widehat{f}(\mathbf{q}/R), \qquad \widehat{\widetilde{g}}(\mathbf{q}) = R^{-2} \widehat{g}(\mathbf{q}/R).$$

The Fourier multiplier h_m in (2.2) satisfies $h_{mR}(\mathbf{q}) = R\, h_m(\mathbf{q}/R)$. With Plancherel's formula and the substitution $\mathbf{q} = R\mathbf{p}$, the factors R^2 and R in (2.5) therefore preserve the two covariance terms:

$$\begin{aligned}
\langle \widetilde{f}, h_{mR}^{-1}\widetilde{f}'\rangle &= \frac{1}{(2\pi)^3}\int_{\mathbb{R}^3} \frac{\overline{\widehat{\widetilde{f}}(\mathbf{q})}\,\widehat{\widetilde{f}}'(\mathbf{q})}{h_{mR}(\mathbf{q})}\, d^3q \\
&= R^{-2}R^{-1}R^3\,\frac{1}{(2\pi)^3}\int_{\mathbb{R}^3} \frac{\overline{\widehat{f}(\mathbf{p})}\,\widehat{f}'(\mathbf{p})}{h_m(\mathbf{p})}\, d^3p = \langle f, h_m^{-1} f'\rangle, \\
\langle \widetilde{g}, h_{mR}\widetilde{g}'\rangle &= \frac{1}{(2\pi)^3}\int_{\mathbb{R}^3} \overline{\widehat{\widetilde{g}}(\mathbf{q})}\,\widehat{\widetilde{g}}'(\mathbf{q})\, h_{mR}(\mathbf{q})\, d^3q \\
&= R^{-4}R\,R^3\,\frac{1}{(2\pi)^3}\int_{\mathbb{R}^3} \overline{\widehat{g}(\mathbf{p})}\,\widehat{g}'(\mathbf{p})\, h_m(\mathbf{p})\, d^3p = \langle g, h_m g'\rangle.
\end{aligned}$$

Hence $\mathsf{v}_{mR,\widetilde{U}}(\widetilde{\mathsf{x}},\widetilde{\mathsf{y}}) = \mathsf{v}_{m,U}(\mathsf{x},\mathsf{y})$.

Step 3: the unitary. The Weyl relations and the quasi-free characteristic function are therefore preserved: $\mathsf{x} \mapsto \widetilde{\mathsf{x}}$ is a real linear bijection of the smooth sources, with inverse $f(\mathbf{x}) = R^{-2}\widetilde{f}(\mathbf{x}/R)$, $g(\mathbf{x}) = R^{-1}\widetilde{g}(\mathbf{x}/R)$, and

$$\langle \Omega_{mR}, \mathsf{W}_{mR}(\widetilde{\mathsf{x}})\Omega_{mR}\rangle = \exp\Big[-\frac{1}{2}\mathsf{v}_{mR,\widetilde{U}}(\widetilde{\mathsf{x}},\widetilde{\mathsf{x}})\Big] = \exp\Big[-\frac{1}{2}\mathsf{v}_{m,U}(\mathsf{x},\mathsf{x})\Big] = \langle \Omega_m, \mathsf{W}_m(\mathsf{x})\Omega_m\rangle.$$

In both representations the vacuum vector is cyclic for the Weyl operators of the smooth sources. Uniqueness of the GNS representation therefore gives a unitary $\mathsf{V}\colon \mathscr{F}_m \to \mathscr{F}_{mR}$ with

$$\mathsf{V}\Omega_m = \Omega_{mR}, \qquad \mathsf{V}\mathsf{W}_m(\mathsf{x})\mathsf{V}^* = \mathsf{W}_{mR}(\widetilde{\mathsf{x}})$$

for every real smooth source x.

Step 4: the two regions. The transformation maps the sources supported in $B_R(0)$ onto those supported in $B_1(0)$, and those supported in $B_R(L\mathbf{e}_1)$ onto those supported in $B_1(s\mathbf{e}_1)$. By (2.3), and by the same identity for the translated double cones with bases $B_R(L\mathbf{e}_1)$ and $B_1(s\mathbf{e}_1)$, conjugation by V therefore maps $\mathfrak{A}_m(D_R)$ onto $\mathfrak{A}_{mR}(D_1)$ and $\mathfrak{A}_m(D_R + (0, L\mathbf{e}_1))$ onto $\mathfrak{A}_{mR}(D_1 + (0, s\mathbf{e}_1))$. This conjugation preserves vacuum expectations, $\langle \Omega_{mR}, \mathsf{V}X\mathsf{V}^*\Omega_{mR}\rangle = \langle \Omega_m, X\Omega_m\rangle$. Formula (2.4) then gives the stated identity: $A \mapsto \mathsf{V}A\mathsf{V}^*$ is a bijection between the self-adjoint contractions of corresponding algebras and leaves the expression under the supremum in (2.4) unchanged, so $\beta_m(R, L) = \beta_{mR}(1, L/R)$. □

Lemma 2.1 reduces Theorem 1.1 to double cones of unit radius, whose algebras are generated by the Weyl operators of sources in unit balls by (2.3). Sections 2–5 use an arbitrary mass $\mu > 0$ and arbitrary length scales; from Section 6 on, μ denotes the dimensionless mass mR and $s = L/R$ the dimensionless separation of unit balls.

2.2. **Source Hilbert spaces.** For a nonempty bounded open $U \subset \mathbb{R}^3$, let $\mathcal{X}_{\mu,U}$ be the completion of the real smooth source pairs supported in U, in the norm of (2.2). Its elements are distributions in $H^{-1/2} \oplus H^{1/2}$ supported in $\overline{U}$. Here $H^\alpha(\mathbb{R}^3)$ is the Sobolev space defined by the Fourier weight $(1 + |\mathbf{p}|^2)^\alpha$. The complexification is $\mathcal{H}_{\mu,U}$, with conjugation c_U. Inner products are conjugate-linear in the first variable.

The complexified source map is

$$(\kappa_U(f, g))(p) = \widehat{f}(\mathbf{p}) + ip_0\widehat{g}(\mathbf{p}).$$

For real sources $\mathsf{x} = (f, g)$ and $\mathsf{y} = (f', g')$, the measure $d\mathfrak{m}_\mu$ and Plancherel's formula give

$$\begin{aligned}\langle \kappa_U \mathsf{x}, \kappa_U \mathsf{y}\rangle &= \frac{1}{(2\pi)^3}\int_{\mathbb{R}^3}\Big(\frac{\overline{\widehat{f}}\,\widehat{f'}}{2p_0} + \frac{p_0\,\overline{\widehat{g}}\,\widehat{g'}}{2} + \frac{i}{2}(\overline{\widehat{f}}\,\widehat{g'} - \overline{\widehat{g}}\,\widehat{f'})\Big)\,d^3p \\ &= \frac{1}{2}(\langle f, h_\mu^{-1} f'\rangle + \langle g, h_\mu g'\rangle) + \frac{i}{2}\int_{\mathbb{R}^3}(fg' - gf')\,d^3x,\end{aligned}$$

that is,

$$\langle \kappa_U \mathsf{x}, \kappa_U \mathsf{y}\rangle = \mathsf{v}_{\mu,U}(\mathsf{x}, \mathsf{y}) + \frac{i}{2}\varsigma(\mathsf{x}, \mathsf{y}). \tag{2.6}$$

Thus the real source map is isometric: $\|\kappa_U \mathsf{x}\|^2 = \mathsf{v}_{\mu,U}(\mathsf{x}, \mathsf{x})$, since $\varsigma(\mathsf{x}, \mathsf{x}) = 0$. Its complexification is bounded: with the norm $\|\mathsf{x} + i\mathsf{y}\|^2 = \mathsf{v}_{\mu,U}(\mathsf{x}, \mathsf{x}) + \mathsf{v}_{\mu,U}(\mathsf{y}, \mathsf{y})$ of $\mathcal{H}_{\mu,U}$,

$$\|\kappa_U(\mathsf{x} + i\mathsf{y})\|^2 = \|\kappa_U \mathsf{x}\|^2 + \|\kappa_U \mathsf{y}\|^2 - 2\,\mathrm{Im}\langle \kappa_U \mathsf{x}, \kappa_U \mathsf{y}\rangle \overset{(2.6)}{=} \|\mathsf{x} + i\mathsf{y}\|^2 - \varsigma(\mathsf{x}, \mathsf{y}) \le 2\,\|\mathsf{x} + i\mathsf{y}\|^2,$$

because $|\varsigma(\mathsf{x}, \mathsf{y})| = 2|\mathrm{Im}\langle \kappa_U \mathsf{x}, \kappa_U \mathsf{y}\rangle| \le 2\,\|\kappa_U \mathsf{x}\|\,\|\kappa_U \mathsf{y}\| \le \|\mathsf{x} + i\mathsf{y}\|^2$.

Lemma 2.2 (Standardness and injectivity)**.** *For every nonempty bounded open U, the map κ_U from $\mathcal{H}_{\mu,U}$ to $\mathcal{H}_{1,\mu}$ is injective and has dense complex range. Its real range $\mathcal{K}_U = \kappa_U \mathcal{X}_{\mu,U}$ is closed and satisfies*

$$\mathcal{K}_U \cap i\mathcal{K}_U = \{0\}, \qquad \overline{\mathcal{K}_U + i\mathcal{K}_U} = \mathcal{H}_{1,\mu}. \tag{2.7}$$

Proof. Step 1: entire transforms. The components of an element of $\mathcal{H}_{\mu,U}$ are distributions supported in the compact set $\overline{U}$, and by the Paley–Wiener–Schwartz theorem their Fourier transforms extend from $\mathbb{R}^3$ to entire functions on $\mathbb{C}^3$.

Step 2: injectivity. If $\kappa_U(f, g) = 0$, then on real momenta

$$\widehat{f}(\mathbf{p}) + i\sqrt{\mu^2 + |\mathbf{p}|^2}\,\widehat{g}(\mathbf{p}) = 0.$$

For fixed real transverse momentum $\mathbf{p}_\perp$, squaring and continuing in $z = p_1$ gives

$$\widehat{f}(z, \mathbf{p}_\perp)^2 = -(z^2 + M^2)\widehat{g}(z, \mathbf{p}_\perp)^2, \qquad M^2 = \mu^2 + |\mathbf{p}_\perp|^2 > 0,$$

first for real z, and then for all $z \in \mathbb{C}$, because both sides are entire functions of z by Step 1.

At $z = iM$, the factor $z^2 + M^2 = (z - iM)(z + iM)$ has a simple zero, since $M > 0$. So a nonzero right-hand side has odd vanishing order, whereas the left-hand side, a square, has even vanishing order. Both sides therefore vanish identically. Since $\mathbf{p}_\perp$ was arbitrary, $f = g = 0$.

Step 3: a vector orthogonal to the range. For density, the equivalent spatial L^2 realization, obtained with the unitary map $\psi \mapsto \mathcal{F}^{-1}((2\pi)^{-3/2}(2p_0)^{-1/2}\psi)$ of $\mathcal{H}_{1,\mu}$ onto $L^2(\mathbb{R}^3)$, is

$$\kappa_U(f, g) = 2^{-1/2}(h_\mu^{-1/2} f + i h_\mu^{1/2} g).$$

A vector v perpendicular to the complex local range gives a distribution $u = h_\mu^{-1/2} v$ with $u = h_\mu u = 0$ on U: for all $f, g \in C_c^\infty(U)$, orthogonality to $\kappa_U(f, 0)$ and to $\kappa_U(0, g)$ reads

$$\langle u, f\rangle = \langle h_\mu^{-1/2} v, f\rangle = 0, \qquad \langle h_\mu u, g\rangle = \langle h_\mu^{1/2} v, g\rangle = 0,$$

where $u \in H^{1/2}(\mathbb{R}^3)$ and $h_\mu u \in H^{-1/2}(\mathbb{R}^3)$.

Step 4: the elliptic continuation. The function $w(\tau) = e^{-\tau h_\mu} u$, $\tau > 0$, is smooth and solves

$$(-\partial_\tau^2 - \nabla^2 + \mu^2)w = 0,$$

and, as $\tau \downarrow 0$, $w(\tau) \to u$ in $H^{1/2}(\mathbb{R}^3)$ and $\partial_\tau w(\tau) = -h_\mu w(\tau) \to -h_\mu u$ in $H^{-1/2}(\mathbb{R}^3)$. Its Dirichlet and Neumann traces, u and $-h_\mu u$, vanish on any ball B contained in U.

Extension by zero across the boundary patch $\{0\} \times B$ is a weak solution of the same elliptic equation on the connected open set $\mathcal{D} = ((0, \infty) \times \mathbb{R}^3) \cup (\mathbb{R} \times B)$. Indeed, for $\varphi \in C_c^\infty(\mathcal{D})$ and $\delta > 0$,

integration by parts on $[\delta,\infty)\times\mathbb{R}^3$ moves the operator onto w, where it gives zero, and leaves the boundary terms at $\tau=\delta$:

$$\int_\delta^\infty \langle w(\tau),(-\partial_\tau^2-\nabla^2+\mu^2)\varphi(\tau)\rangle\,d\tau = \langle w(\delta),\partial_\tau\varphi(\delta)\rangle - \langle \partial_\tau w(\delta),\varphi(\delta)\rangle.$$

As $\delta\downarrow 0$, the left-hand side tends to the pairing of the extension with $(-\partial_\tau^2-\nabla^2+\mu^2)\varphi$, and the right-hand side tends to $\langle u,\partial_\tau\varphi(0)\rangle+\langle h_\mu u,\varphi(0)\rangle$, a sum of pairings of $H^{1/2}$ with $H^{-1/2}$, which vanishes by Step 3 because $\varphi(0)$ and $\partial_\tau\varphi(0)$ are supported in $B\subset U$.

The extension vanishes on the open subset $(-\infty,0)\times B$ of $\mathcal{D}$. Analytic elliptic regularity and uniqueness [20] imply that it vanishes on the connected domain. Hence $w(\tau)=0$ for $\tau>0$, $u=\lim_{\tau\downarrow 0} w(\tau)=0$ in $H^{1/2}(\mathbb{R}^3)$, and $v=h_\mu^{1/2}u=0$.

Step 5: standardness. The real isometry has closed range, because $\mathcal{X}_{\mu,U}$ is complete. Injectivity of the complexification gives the first relation in (2.7): if $\kappa_U\mathsf{x}=i\kappa_U\mathsf{y}$ with $\mathsf{x},\mathsf{y}\in\mathcal{X}_{\mu,U}$, then $\kappa_U(\mathsf{x}-i\mathsf{y})=0$, so $\mathsf{x}=\mathsf{y}=0$ by Step 2. Density gives the second, since $\mathcal{K}_U+i\mathcal{K}_U=\kappa_U\mathcal{H}_{\mu,U}$, and this complex range is dense by Steps 3 and 4.

Standardness of local subspaces of the free field is known; for regions with C^1 boundary, Figliolini and Guido prove it in a similar way, from the anti-locality of h_μ [13, Propositions 2.4 and 2.7]. □

2.3. **Finite canonical systems and local approximation.** A finite-dimensional real subspace of smooth local sources is a *canonical source system* when the restriction of (2.1) is nondegenerate. A symplectic basis gives quadratures with commutation matrix

$$J_n=\bigoplus_{j=1}^n\begin{pmatrix}0&1\\-1&0\end{pmatrix}.$$

By the Stone–von Neumann theorem, the regular Weyl representation of a canonical source system is the oscillator representation with a possible multiplicity [15]. The vacuum restriction is a Gaussian density operator. Its covariance V satisfies

$$V+\frac{i}{2}J_n>0. \tag{2.8}$$

For the symplectic basis $\mathsf{x}_1,\dots,\mathsf{x}_{2n}$, supported in a bounded open set U, the covariance is $V_{kl}=\mathsf{v}_{\mu,U}(\mathsf{x}_k,\mathsf{x}_l)$, since $\omega(\mathsf{W}_\mu(\sum_k\xi_k\mathsf{x}_k))=\exp(-\frac12\sum_{k,l}\xi_k\xi_l\mathsf{v}_{\mu,U}(\mathsf{x}_k,\mathsf{x}_l))$ for $\xi\in\mathbb{R}^{2n}$, and $\varsigma(\mathsf{x}_k,\mathsf{x}_l)=(J_n)_{kl}$; so

$$\Big(V+\frac{i}{2}J_n\Big)_{kl}=\mathsf{v}_{\mu,U}(\mathsf{x}_k,\mathsf{x}_l)+\frac{i}{2}\varsigma(\mathsf{x}_k,\mathsf{x}_l)\overset{(2.6)}{=}\langle\kappa_U\mathsf{x}_k,\kappa_U\mathsf{x}_l\rangle.$$

For $c\in\mathbb{C}^{2n}\setminus\{0\}$, the vector $\sum_l c_l\mathsf{x}_l$ of $\mathcal{H}_{\mu,U}$ is nonzero; since κ_U is injective by Lemma 2.2, $c^*(V+\frac{i}{2}J_n)c=\|\kappa_U\sum_l c_l\mathsf{x}_l\|^2>0$.

Williamson's normal form [43] writes

$$V=S\left(\bigoplus_{j=1}^n \upsilon_j I_2\right)S^T,\qquad SJ_nS^T=J_n.$$

The numbers υ_j are the symplectic eigenvalues. Relation (2.8) therefore gives $\upsilon_j>1/2$, so the Gaussian density operator is faithful.

Every finite collection of smooth sources supported in an open set U lies in a finite canonical system of sources supported in U. The form ς is nondegenerate on the smooth sources supported in U: a nonzero $\mathsf{x}=(f,g)$ has the partner $(0,f)$ if $f\neq 0$, and $(-g,0)$ if $f=0$, with

$$\varsigma((f,g),(0,f))=\int_{\mathbb{R}^3}f^2\,d^3x,\qquad \varsigma((0,g),(-g,0))=\int_{\mathbb{R}^3}g^2\,d^3x.$$

Let $\mathcal{C}$ be the span of the collection, $\mathsf{r}_1,\dots,\mathsf{r}_k$ a basis of its radical $\{\mathsf{x}\in\mathcal{C}:\varsigma(\mathsf{x},\mathsf{y})=0 \text{ for all } \mathsf{y}\in\mathcal{C}\}$, and $\mathcal{C}_0$ a complement of the radical in $\mathcal{C}$, on which ς is nondegenerate. Since $\mathcal{C}_0$ is finite-dimensional,

ς is also nondegenerate on the symplectic complement $\mathcal{C}_0^\varsigma$ of $\mathcal{C}_0$ among the smooth sources supported in U, which contains the radical; so there are local partners $\mathsf{y}_1, \dots, \mathsf{y}_k \in \mathcal{C}_0^\varsigma$ with $\varsigma(\mathsf{r}_i, \mathsf{y}_j) = \delta_{ij}$. The matrix of ς on $\mathsf{r}_1, \dots, \mathsf{r}_k, \mathsf{y}_1, \dots, \mathsf{y}_k$ is

$$\begin{pmatrix} 0 & I_k \\ -I_k & * \end{pmatrix},$$

which is invertible. So these vectors form a basis of their span, ς is nondegenerate on the span, and the span is ς-orthogonal to $\mathcal{C}_0$. Hence ς is nondegenerate on the sum of $\mathcal{C}_0$ and the span, a finite canonical system that contains $\mathcal{C}$. The finite canonical systems in U form a directed family: two of them lie in the system obtained in this way from their sum.

Lemma 2.3 (Passage to all local effects)**.** *Suppose an inequality involving*

$$\omega(E), \qquad \omega(P), \qquad \omega(EP)$$

holds, with constants independent of the canonical systems, for every effect E of the von Neumann algebra generated by the Weyl operators of a finite canonical source system in a first region and every effect P of such an algebra in a second region. If the inequality has the form $\Phi(\omega(E), \omega(P), \omega(EP)) \leq 0$ with a continuous function Φ, it holds for every pair of effects in the full regional algebras, generated by the Weyl operators of all smooth sources in the two regions (for two balls, the double-cone algebras of (2.3)*). The same conclusion holds for a bound on $|\omega(AB) - \omega(A)\omega(B)|$, with pairs of self-adjoint contractions A, B in place of the effects E, P.*

Proof. Step 1: a dense union. Let $\mathfrak{M}$ be the regional algebra of the first region and $\mathfrak{M}_0$ the union of the von Neumann algebras of the finite canonical source systems in that region. The finite canonical Weyl algebras generate the local algebras: every smooth source of the region lies in a finite canonical system of the region, so $\mathfrak{M}_0 \subset \mathfrak{M}$ contains the Weyl operators that generate $\mathfrak{M}$. The directed union $\mathfrak{M}_0$ is a strongly dense unital $*$-algebra: two finite canonical systems lie in a third, so sums and products of elements of $\mathfrak{M}_0$ lie in $\mathfrak{M}_0$, and by the bicommutant theorem the strong closure of $\mathfrak{M}_0$ is $\mathfrak{M}_0'' = \mathfrak{M}$.

Step 2: effects of the union. For an effect $E \in \mathfrak{M}$, the operator $2E - I$ is a self-adjoint contraction in $\mathfrak{M}$. Kaplansky density [5, Theorem 2.4.16] gives a net of self-adjoint contractions $X_\alpha \in \mathfrak{M}_0$ with $X_\alpha \to 2E - I$ strongly, and the affine map $t \mapsto (1+t)/2$ turns them into effects $E_\alpha = (I + X_\alpha)/2$ of $\mathfrak{M}_0$ with $E_\alpha \to E$ strongly. Each E_α is an effect of the von Neumann algebra of one finite canonical system.

Step 3: joint convergence. The approximations in the two regions can be chosen independently: let $P_\beta \to P$ be obtained in the same way in the second region. Products of bounded strongly convergent nets converge strongly. So the three vacuum expectations converge along the pairs (α, β): $\omega(E_\alpha) \to \omega(E)$, $\omega(P_\beta) \to \omega(P)$ and $\omega(E_\alpha P_\beta) \to \omega(EP)$.

Step 4: the limit. By hypothesis, $\Phi(\omega(E_\alpha), \omega(P_\beta), \omega(E_\alpha P_\beta)) \leq 0$ for all α, β, and the continuity of Φ gives $\Phi(\omega(E), \omega(P), \omega(EP)) \leq 0$. The argument for self-adjoint contractions is the same, with Kaplansky density applied to A and B themselves. □

2.4. **Modular operators and uncertainty reserves.** A closed real subspace satisfying (2.7) is *standard*. The Tomita operator and modular operator of a standard subspace, written here for $\mathcal{K}_U$, are

$$\mathsf{s}_U(u + iv) = u - iv \quad (u, v \in \mathcal{K}_U), \qquad \Delta_U = \mathsf{s}_U^* \mathsf{s}_U.$$

We use the standard-subspace formulation of modular theory [12, 31].

For an operator T on the source space, $\overline{T} = \mathsf{c}_U T \mathsf{c}_U$. The source Gram operators and their *uncertainty reserve* are

$$G_U = \kappa_U^* \kappa_U, \qquad \overline{G}_U = 2I - G_U, \qquad \mathsf{R}_U = (G_U \overline{G}_U)^{1/2}. \tag{2.9}$$

The middle formula, an identity for $\overline{G}_U = \mathsf{c}_U G_U \mathsf{c}_U$, follows from (2.6): for real x, y, the conjugation is antiunitary and fixes real vectors, so

$$\langle \mathsf{x}, \overline{G}_U \mathsf{y}\rangle = \overline{\langle \mathsf{x}, G_U \mathsf{y}\rangle} = \mathsf{v}_{\mu,U}(\mathsf{x},\mathsf{y}) - \frac{i}{2}\varsigma(\mathsf{x},\mathsf{y}) = 2\mathsf{v}_{\mu,U}(\mathsf{x},\mathsf{y}) - \langle \mathsf{x}, G_U \mathsf{y}\rangle = \langle \mathsf{x}, (2I - G_U)\mathsf{y}\rangle,$$

and complex linearity extends the identity to $\mathcal{H}_{\mu,U}$.

Both Gram operators are positive and injective, and they commute: $\langle \mathsf{x}, G_U \mathsf{x}\rangle = \|\kappa_U \mathsf{x}\|^2$ and $\langle \mathsf{x}, \overline{G}_U \mathsf{x}\rangle = \|\kappa_U \mathsf{c}_U \mathsf{x}\|^2$, where κ_U is injective by Lemma 2.2, and $\overline{G}_U = 2I - G_U$ commutes with G_U. So the spectrum of G_U lies in $[0, 2]$, and

$$G_U \overline{G}_U = G_U(2I - G_U) = I - (I - G_U)^2 \leq I.$$

Since G_U and $2I - G_U$ are injective, $G_U \overline{G}_U$ is injective. Hence $0 < \mathsf{R}_U \leq I$; the spectrum of R_U can accumulate at zero. The reserve measures the room left by the uncertainty relation: R_U equals I on source directions along which the symplectic form vanishes relative to the covariance, where $G_U = I$, and R_U is small on directions along which the covariance nearly saturates the uncertainty bound, where G_U is close to 0 or to 2.

In the polar decomposition $\kappa_U = \mathcal{U}_U G_U^{1/2}$, the operator $\mathcal{U}_U$ is unitary by Lemma 2.2, since κ_U is injective and has dense range. The relation $\mathsf{s}_U \kappa_U \mathsf{x} = \kappa_U \mathsf{c}_U \mathsf{x}$, which is the definition of s_U applied to $u = \kappa_U \mathsf{x}_1$ and $v = \kappa_U \mathsf{x}_2$ for $\mathsf{x} = \mathsf{x}_1 + i\mathsf{x}_2$ with real $\mathsf{x}_1, \mathsf{x}_2$, gives, for every $\mathsf{x} \in \mathcal{H}_{\mu,U}$,

$$\left\|\Delta_U^{1/2} \kappa_U \mathsf{x}\right\|^2 = \|\mathsf{s}_U \kappa_U \mathsf{x}\|^2 = \|\kappa_U \mathsf{c}_U \mathsf{x}\|^2 \overset{(2.9)}{=} \langle \mathsf{c}_U \mathsf{x}, G_U \mathsf{c}_U \mathsf{x}\rangle = \langle \mathsf{x}, \overline{G}_U \mathsf{x}\rangle.$$

The first equality is the polar decomposition of s_U, whose antiunitary factor preserves norms, and the last holds because c_U is antiunitary and $\overline{G}_U = \mathsf{c}_U G_U \mathsf{c}_U$.

Let $\overline{G}_U G_U^{-1}$ and its powers be defined by the joint spectral calculus of the commuting operators G_U and $\overline{G}_U$. With $\xi = G_U^{1/2}\mathsf{x}$, so that $\kappa_U \mathsf{x} = \mathcal{U}_U \xi$ and $\mathsf{x} = G_U^{-1/2}\xi$, the identity for $\|\Delta_U^{1/2}\kappa_U \mathsf{x}\|^2$ reads

$$\left\|\left(\mathcal{U}_U^* \Delta_U \mathcal{U}_U\right)^{1/2} \xi\right\|^2 = \left\|\Delta_U^{1/2} \mathcal{U}_U \xi\right\|^2 = \langle G_U^{-1/2}\xi, \overline{G}_U G_U^{-1/2}\xi\rangle = \left\|\left(\overline{G}_U G_U^{-1}\right)^{1/2}\xi\right\|^2$$

for $\xi \in \operatorname{Ran} G_U^{1/2} = \operatorname{Dom} G_U^{-1/2}$. This space is the form domain of both operators: that of $\mathcal{U}_U^* \Delta_U \mathcal{U}_U$ is $\mathcal{U}_U^* \operatorname{Dom} \Delta_U^{1/2} = \mathcal{U}_U^* \operatorname{Dom} \mathsf{s}_U = \mathcal{U}_U^* \kappa_U \mathcal{H}_{\mu,U} = \operatorname{Ran} G_U^{1/2}$, and that of $\overline{G}_U G_U^{-1}$ is $\operatorname{Dom} G_U^{-1/2}$, because $(2-\lambda)/\lambda \leq 2/\lambda = (2-\lambda)/\lambda + 1$ for $0 < \lambda \leq 2$. Positive self-adjoint operators with the same form domain and the same quadratic form coincide, so $\mathcal{U}_U^* \Delta_U \mathcal{U}_U = \overline{G}_U G_U^{-1}$, and

$$\Delta_U = \mathcal{U}_U \overline{G}_U G_U^{-1} \mathcal{U}_U^*, \qquad \Delta_U^c \kappa_U = \mathcal{U}_U h_{c,U}, \qquad h_{c,U} = G_U^{1/2-c} \overline{G}_U^c. \tag{2.10}$$

For $0 \leq c \leq 1/2$, $h_{c,U}$ is bounded, because the function $\lambda^{1/2-c}(2-\lambda)^c$ is bounded on $[0,2]$. The second formula holds on the whole source space: $\kappa_U \mathcal{H}_{\mu,U} = \operatorname{Dom} \Delta_U^{1/2} \subset \operatorname{Dom} \Delta_U^c$, and the first formula gives

$$\Delta_U^c \kappa_U = \mathcal{U}_U \left(\overline{G}_U G_U^{-1}\right)^c \mathcal{U}_U^* \mathcal{U}_U G_U^{1/2} = \mathcal{U}_U G_U^{1/2-c} \overline{G}_U^c.$$

In the notation of Rieffel and van Daele [31, Sections 2 and 3], with P and Q the real orthogonal projections onto $\mathcal{K}_U$ and $i\mathcal{K}_U$, their operators $R = P + Q$, $2 - R$ and $T = R^{1/2}(2-R)^{1/2}$ are $\mathcal{U}_U G_U \mathcal{U}_U^*$, $\mathcal{U}_U \overline{G}_U \mathcal{U}_U^*$ and $\mathcal{U}_U \mathsf{R}_U \mathcal{U}_U^*$, because $P + Q = \kappa_U \kappa_U^*$, and the first formula in (2.10) is their relation $\Delta = (2-R)R^{-1}$. For quasi-free states, Lechner and Sanders obtain the same two formulas from the same relation between the Tomita operator and the conjugation, in their notation with $G_U = 1 + \Sigma$ and $\overline{G}_U = 1 - \Sigma$ [25, Proposition 5.2 and the proof of Proposition 5.3].

Lemma 2.4 (Containing-region inequality)**.** *If $\mathcal{K}_U \subset \mathcal{K}_W$ are standard real subspaces, Δ_W is the modular operator of $\mathcal{K}_W$, and $0 < c < 1/2$, then every $\mathsf{x} \in \mathcal{H}_{\mu,U}$ satisfies*

$$\left\|\Delta_W^c \kappa_U \mathsf{x}\right\| \leq \left\|h_{c,U}\mathsf{x}\right\|. \tag{2.11}$$

Proof. Step 1: the form order. The inclusion gives $\mathsf{s}_U \subset \mathsf{s}_W$: the domain $\mathcal{K}_U + i\mathcal{K}_U$ of s_U lies in the domain $\mathcal{K}_W + i\mathcal{K}_W$ of s_W, and both operators map $u + iv$ to $u - iv$. The quadratic form of Δ_U is the restriction of the form of Δ_W: by the polar decompositions of s_U and s_W, every $\xi \in \operatorname{Dom} \Delta_U^{1/2} = \operatorname{Dom} \mathsf{s}_U$ satisfies

$$\left\|\Delta_U^{1/2}\xi\right\| = \|\mathsf{s}_U\xi\| = \|\mathsf{s}_W\xi\| = \left\|\Delta_W^{1/2}\xi\right\|.$$

Therefore $\Delta_U \geq \Delta_W$ in form order:

$$\operatorname{Dom} \Delta_U^{1/2} \subset \operatorname{Dom} \Delta_W^{1/2}, \quad \text{and} \quad \left\|\Delta_W^{1/2}\xi\right\| \leq \left\|\Delta_U^{1/2}\xi\right\| \quad \text{for } \xi \in \operatorname{Dom} \Delta_U^{1/2}.$$

Step 2: the Löwner–Heinz inequality. The Löwner–Heinz inequality for $2c \in (0,1)$, in Heinz's form for unbounded positive operators ordered as quadratic forms [18, Satz 2], gives $\Delta_U^{2c} \geq \Delta_W^{2c}$. In form order, the inequality $\Delta_U^{2c} \geq \Delta_W^{2c}$ says that $\operatorname{Dom} \Delta_U^{c} \subset \operatorname{Dom} \Delta_W^{c}$ and

$$\|\Delta_W^c \xi\| \leq \|\Delta_U^c \xi\|, \qquad \xi \in \operatorname{Dom} \Delta_U^c.$$

Step 3: the source bound. For $\mathsf{x} \in \mathcal{H}_{\mu,U}$, the vector $\kappa_U \mathsf{x}$ lies in $\operatorname{Dom} \mathsf{s}_U = \operatorname{Dom} \Delta_U^{1/2}$, which is contained in $\operatorname{Dom} \Delta_U^c$ since $c < 1/2$. Step 2 at $\xi = \kappa_U \mathsf{x}$ gives (2.11):

$$\|\Delta_W^c \kappa_U \mathsf{x}\| \overset{\text{Step 2}}{\leq} \|\Delta_U^c \kappa_U \mathsf{x}\| \overset{(2.10)}{=} \|\mathcal{U}_U h_{c,U} \mathsf{x}\| = \|h_{c,U}\mathsf{x}\|,$$

the last step because $\mathcal{U}_U$ is unitary. Equivalently, the closure of $\Delta_W^c \Delta_U^{-c}$ is a contraction, which is the bound of [8, Section 2] for the inclusion $\mathcal{K}_U \subset \mathcal{K}_W$ and the exponent c. □

We write $\mathcal{S}_j$ for the Schatten class, with $\|T\|_j = (\operatorname{Tr} |T|^j)^{1/j}$ for real $1 \leq j < \infty$. The unadorned operator norm is $\|T\|$. Our uses of Schatten Hölder inequalities and complex interpolation follow [33].

Convention 2.5 (Negative reserve powers)**.** An expression such as

$$\mathsf{R}_U^b X \mathsf{R}_{U'}^b, \qquad X \in \mathcal{S}_1,$$

with $b > 0$ and the uncertainty reserves (2.9) of nonempty bounded open sets $U, U' \subset \mathbb{R}^3$, denotes the bounded operator obtained by composition, and an equation between a bounded operator and such an expression is an equality of bounded operators. Such an equation takes the place of multiplying the bounded operator by R_U^{-b} on the left and by $\mathsf{R}_{U'}^{-b}$ on the right. For a linear functional ℓ, the notation $\|\ell \mathsf{R}_U^{-b}\| \leq M$ means

$$|\ell(\mathsf{x})| \leq M \left\|\mathsf{R}_U^b \mathsf{x}\right\|,$$

for every $\mathsf{x} \in \mathcal{H}_{\mu,U}$, followed by continuous extension of $\ell \mathsf{R}_U^{-b}$ from $\operatorname{Ran} \mathsf{R}_U^b$. All negative reserve powers below have this factorization interpretation.

Throughout, C denotes a positive universal constant whose value may change from line to line. Dependence on a fixed parameter is indicated by a subscript. Named constants such as C_*, C_c, C_1, C_2, C_{sub} and C_{prob} keep their values throughout, and $C_{\mu,s}$ in Section 6 is an operator.

Table 1 in Section 8 lists the constants that enter C_*. Some letters occur in two fonts with different meanings: E and $\mathcal{E}$ (an effect and the energy scale), P and $\mathfrak{P}_\mu$ (an effect and the momentum covariance kernel), $\mathcal{X}_{\mu,U}$ and $\mathfrak{X}_\mu$ (the real source space and the field covariance kernel), R and R_U (the radius and the uncertainty reserve), and J_n and J (the canonical commutation matrix and the normalized commutation form).

3. A probability criterion for the classical CHSH bound

Let ω be any state on the von Neumann algebra generated by two commuting von Neumann algebras $\mathfrak{A}$ and $\mathfrak{B}$. For $r > 1$, the event comparison constants are

$$K_{r,+} = \sup_{E,P} \frac{\omega(EP)^r}{(\omega(E)\omega(P))^{r-1}}, \qquad K_{r,-} = \sup_{E,P} \frac{(\omega(E)\omega(P))^r}{\omega(EP)^{r-1}}. \tag{3.1}$$

The suprema range over projections $E \in \mathfrak{A}$ and $P \in \mathfrak{B}$ with $\omega(E)\omega(P) > 0$. A positive numerator with zero denominator gives $+\infty$. At $E = P = I$ each quotient equals 1, so both constants are at least 1.

Theorem 3.1 (Two event bounds and clustering)**.** *Let $r_+ > 2$ and $1 < r_- < 2$, and set*

$$b = 1 - \frac{1}{r_+}, \qquad a = 1 - \frac{1}{r_-}, \qquad \eta = 1 - \frac{1-b}{ab^2}. \tag{3.2}$$

Suppose $\eta > 0$, $K_{r_+,+}, K_{r_-,-} < \infty$, and, for some $q_{\mathrm{cl}} \geq 0$,

$$|\omega(AB) - \omega(A)\omega(B)| \leq q_{\mathrm{cl}} \tag{3.3}$$

for all self-adjoint contractions $A \in \mathfrak{A}$ and $B \in \mathfrak{B}$. Define

$$\nu_+ = \frac{1+b}{ar_+b^2}, \qquad \nu_- = \frac{1-a}{a}. \tag{3.4}$$

If

$$q_{\mathrm{cl}}^{\eta} K_{r_+,+}^{\nu_+} K_{r_-,-}^{\nu_-} \leq 2^{-2/(ab)}, \tag{3.5}$$

then the Summers–Werner Bell correlation $\beta(\omega; \mathfrak{A}, \mathfrak{B})$ of (2.4) equals 1.

Proof. Step 1: reduction to quartets of reflections. Identities attain the value 1 in (2.4), so $\beta(\omega; \mathfrak{A}, \mathfrak{B}) = 1$ unless some self-adjoint contractions $A_0, A_1 \in \mathfrak{A}$ and $B_0, B_1 \in \mathfrak{B}$ give the CHSH expression an absolute value larger than 1. Since A_i and B_j commute, the expression is real, and negating both observables at one site changes its sign; a value smaller than -1 thus gives a value larger than 1.

Every self-adjoint contraction is a spectral average of reflections:

$$A = \frac{1}{2} \int_{-1}^{1} \left(2\,\mathbf{1}_{\{A \geq u\}} - I\right) du.$$

The reflections $2\,\mathbf{1}_{\{A \geq u\}} - I$ lie in every von Neumann algebra that contains A, and the integral converges in norm, because its Riemann sums are step functions of A whose scalar counterparts converge to $t \mapsto t$ uniformly on $[-1, 1]$.

The CHSH expression is separately affine in its four observables and continuous in norm. With A_1, B_0, B_1 fixed, its value at A_0 is therefore a limit of averages, with weights of total mass 1, of its values at the reflections $2\,\mathbf{1}_{\{A_0 \geq u\}} - I$, and a value larger than 1 at A_0 gives a value larger than 1 at one of these reflections. Replacing A_0, A_1, B_0 and B_1 in turn in this way turns a strict violation into a strict violation by reflections.

For a quartet of reflections $A_0, A_1 \in \mathfrak{A}$ and $B_0, B_1 \in \mathfrak{B}$, the *score* is the value

$$S = \tfrac{1}{2}\omega(A_0(B_0 + B_1) + A_1(B_0 - B_1))$$

of the CHSH expression in (2.4).

Step 2: the winning cells and their symmetries. The *cells* of the experiment are the sixteen joint probabilities $\omega(E_i^{\pm} P_j^{\pm})$ of the outcome effects $E_i^{\pm} = (I \pm A_i)/2$ and $P_j^{\pm} = (I \pm B_j)/2$, which are projections because A_i and B_j are reflections. A cell is *winning* when the product of its outcomes agrees with the sign of the corresponding correlator in the expanded score

$$S = \tfrac{1}{2}(\omega(A_0B_0) + \omega(A_0B_1) + \omega(A_1B_0) - \omega(A_1B_1)).$$

The winning cells of the setting pair (i,j) are thus $\omega(E_i^+P_j^+)$ and $\omega(E_i^-P_j^-)$ if $(i,j)\neq(1,1)$, and $\omega(E_1^+P_1^-)$ and $\omega(E_1^-P_1^+)$ if $(i,j)=(1,1)$.

Let p_{ij} be the sum of the two winning cells of the setting pair (i,j). Since $A_iB_j=(E_i^+-E_i^-)(P_j^+-P_j^-)$ and the four cells of a setting pair add up to 1, the correlator $\omega(A_iB_j)$, with its sign in S, equals $p_{ij}-(1-p_{ij})$, and the sum over the four setting pairs is

$$S=\frac{1}{2}\sum_{i,j=0}^{1}(2p_{ij}-1)=\sum_{i,j=0}^{1}p_{ij}-2,$$

so $S+2$ is the sum of the eight winning cells.

The maps that replace (A_0,A_1,B_0,B_1) by

$$(A_0,-A_1,B_1,B_0),\qquad (A_1,A_0,B_0,-B_1),\qquad (-A_0,-A_1,-B_0,-B_1)$$

carry reflection quartets to reflection quartets and permute the eight winning cells. The first exchanges the two settings in $\mathfrak{B}$, which moves the cells of the setting pair (i,j) to $(i,1-j)$, and flips the outcome of A_1, which compensates the opposite signs of $\omega(A_1B_0)$ and $\omega(A_1B_1)$ in S. The second exchanges the two settings in $\mathfrak{A}$, which moves the cells of (i,j) to $(1-i,j)$, and flips the outcome of B_1, which compensates the opposite signs of $\omega(A_0B_1)$ and $\omega(A_1B_1)$. The third flips the outcomes at both sites, which exchanges the two winning cells of each setting pair.

The three maps therefore leave S unchanged, and the eight winning cells form one orbit under them: compositions of the first two carry any setting pair to $(0,0)$, and the third exchanges the two winning cells of $(0,0)$. Replacing the quartet by its image under a composition of the three maps, we may assume that a least likely winning cell is $\omega(E_0^-P_0^-)$, the cell of the outcomes -1 of A_0 and B_0.

Step 3: the probability form of the score. By Step 2, the least probability of a winning cell is $w=\omega(E_0^-P_0^-)$, and we show that $S>1$ takes the probability form

$$w>\omega(E_0^-P_1^+)+\omega(E_1^+P_0^-)+\omega(E_1^-P_1^-). \tag{3.6}$$

We write

$$e_A=\omega(E_0^-),\qquad e_B=\omega(P_0^-),\qquad x_A=\omega(E_1^-),\qquad x_B=\omega(P_1^-).$$

Since $A_i=I-2E_i^-$ and $B_j=I-2P_j^-$, and $\omega((I-2E)(I-2P))=1-2\omega(E)-2\omega(P)+4\omega(EP)$ for all effects E,P, the four correlators are

$$\begin{aligned}
\omega(A_0B_0)&=1-2e_A-2e_B+4w, & \omega(A_0B_1)&=1-2e_A-2x_B+4\omega(E_0^-P_1^-),\\
\omega(A_1B_0)&=1-2x_A-2e_B+4\omega(E_1^-P_0^-), & \omega(A_1B_1)&=1-2x_A-2x_B+4\omega(E_1^-P_1^-).
\end{aligned}$$

We insert them into the expanded score of Step 2, in which the terms in x_A and x_B cancel, and use $\omega(E_0^-P_1^+)=e_A-\omega(E_0^-P_1^-)$ and $\omega(E_1^+P_0^-)=e_B-\omega(E_1^-P_0^-)$, which hold because $P_1^+=I-P_1^-$ and $E_1^+=I-E_1^-$:

$$\begin{aligned}
S&=1-2e_A-2e_B+2w+2\omega(E_0^-P_1^-)+2\omega(E_1^-P_0^-)-2\omega(E_1^-P_1^-)\\
&=1+2\left\{w-\omega(E_0^-P_1^+)-\omega(E_1^+P_0^-)-\omega(E_1^-P_1^-)\right\}.
\end{aligned}$$

Since $S>1$, the expression in braces is positive, and (3.6) follows. The three terms on the right of (3.6) are values of ω at products of commuting projections, hence nonnegative; so $w>0$, and each of the three terms is smaller than w.

Step 4: the two neighbouring winning cells. The cells $E_0^-P_1^-$ and $E_1^-P_0^-$ are winning, because $\omega(A_0B_1)$ and $\omega(A_1B_0)$ have the sign $+$ in S; so their probabilities are at least w. By Step 3, the first two terms on the right of (3.6) are smaller than w, so

$$\omega(E_0^-P_1^-)=e_A-\omega(E_0^-P_1^+)>e_A-w,\qquad \omega(E_1^-P_0^-)=e_B-\omega(E_1^+P_0^-)>e_B-w.$$

Hence

$$\omega(E_0^- P_1^-) \geq \max\{w, e_A - w\} \geq e_A/2, \qquad \omega(E_1^- P_0^-) \geq \max\{w, e_B - w\} \geq e_B/2.$$

Step 5: the two event bounds and the marginals x_A, x_B. From (3.1), for all projections $E \in \mathfrak{A}$ and $P \in \mathfrak{B}$,

$$\omega(EP) \leq k\big(\omega(E)\omega(P)\big)^b, \qquad \omega(EP) \geq \lambda\big(\omega(E)\omega(P)\big)^{1/a}, \tag{3.7}$$

where

$$k = K_{r_+,+}^{1/r_+}, \qquad \lambda = K_{r_-,-}^{-1/(r_- - 1)}.$$

If $\omega(E)\omega(P) > 0$, the two quotients in (3.1) at E, P are at most $K_{r_+,+}$ and $K_{r_-,-}$. The second quotient has a positive numerator; since a zero denominator would make it $+\infty$ and $K_{r_-,-} < \infty$, its denominator is positive, that is, $\omega(EP) > 0$. Multiplying out the two bounds on the quotients gives

$$\omega(EP)^{r_+} \leq K_{r_+,+}\big(\omega(E)\omega(P)\big)^{r_+ - 1}, \qquad \omega(EP)^{r_- - 1} \geq K_{r_-,-}^{-1}\big(\omega(E)\omega(P)\big)^{r_-},$$

and the roots of orders r_+ and $r_- - 1$ give (3.7), since $(r_+ - 1)/r_+ = b$ and $r_-/(r_- - 1) = 1/a$ by (3.2). If $\omega(E)\omega(P) = 0$, then $\omega(EP) = 0$, because the projection EP satisfies $EP \leq E$ and $EP \leq P$, and both bounds in (3.7) hold trivially.

The upper bound in (3.7), applied to the two neighbouring winning cells, gives

$$\frac{e_A}{2} \overset{\text{Step 4}}{\leq} \omega(E_0^- P_1^-) \overset{(3.7)}{\leq} k(e_A x_B)^b, \qquad \frac{e_B}{2} \overset{\text{Step 4}}{\leq} \omega(E_1^- P_0^-) \overset{(3.7)}{\leq} k(x_A e_B)^b.$$

Since $e_A \geq \omega(E_0^- P_0^-) = w > 0$ and likewise $e_B > 0$, division by $k e_A^b$ and by $k e_B^b$, followed by the power $1/b$, gives

$$x_B \geq (2k)^{-1/b} e_A^{(1-b)/b}, \qquad x_A \geq (2k)^{-1/b} e_B^{(1-b)/b}.$$

Step 6: a lower bound on w. At the cell $E_0^- P_0^-$, the upper bound in (3.7) reads $w \leq k(e_A e_B)^b$, that is, $e_A e_B \geq (w/k)^{1/b}$. By the lower bound in (3.7) at $E_1^- P_1^-$, by the bounds of Step 5 on x_A and x_B raised to the power $1/a$, and by the bound $e_A e_B \geq (w/k)^{1/b}$ at $E_0^- P_0^-$ raised to the power $(1-b)/(ab)$, both powers being positive,

$$\begin{aligned} w &\overset{(3.6)}{>} \omega(E_1^- P_1^-) \overset{(3.7)}{\geq} \lambda (x_A x_B)^{1/a} \overset{\text{Step 5}}{\geq} \lambda (2k)^{-2/(ab)} (e_A e_B)^{(1-b)/(ab)} \\ &\overset{(3.7)}{\geq} \lambda (2k)^{-2/(ab)} \Big(\frac{w}{k}\Big)^{(1-b)/(ab^2)} = \lambda\, 2^{-2/(ab)}\, k^{-(1+b)/(ab^2)}\, w^{(1-b)/(ab^2)}, \end{aligned}$$

where the exponents of k add up to $-2/(ab) - (1-b)/(ab^2) = -(1+b)/(ab^2)$. Since $w > 0$, division by $w^{(1-b)/(ab^2)}$, with $\eta = 1 - (1-b)/(ab^2)$ from (3.2), gives

$$w^\eta > \lambda\, 2^{-2/(ab)}\, k^{-(1+b)/(ab^2)} = 2^{-2/(ab)} K_{r_+,+}^{-\nu_+} K_{r_-,-}^{-\nu_-}.$$

Here

$$k^{-(1+b)/(ab^2)} = K_{r_+,+}^{-(1+b)/(a r_+ b^2)} = K_{r_+,+}^{-\nu_+} \quad \text{by (3.4), and} \quad \lambda = K_{r_-,-}^{-1/(r_- - 1)} = K_{r_-,-}^{-\nu_-},$$

because $r_- - 1 = a/(1-a)$ by (3.2). Multiplication by $K_{r_+,+}^{\nu_+} K_{r_-,-}^{\nu_-}$ gives

$$w^\eta K_{r_+,+}^{\nu_+} K_{r_-,-}^{\nu_-} > 2^{-2/(ab)}. \tag{3.8}$$

Step 7: an upper bound on w from clustering. Let $\mathsf{m}_i = \omega(A_i)$ and $\mathsf{n}_j = \omega(B_j)$ be the marginal means, and let S_{prod} be the score obtained by replacing each correlator $\omega(A_i B_j)$ by $\mathsf{m}_i \mathsf{n}_j$. Since $|\mathsf{m}_i| \leq 1$ and $\frac{1}{2}(|x+y| + |x-y|) = \max\{|x|, |y|\}$ for real x, y,

$$S_{\text{prod}} = \tfrac{1}{2}\{\mathsf{m}_0(\mathsf{n}_0 + \mathsf{n}_1) + \mathsf{m}_1(\mathsf{n}_0 - \mathsf{n}_1)\} \leq \tfrac{1}{2}\{|\mathsf{n}_0 + \mathsf{n}_1| + |\mathsf{n}_0 - \mathsf{n}_1|\} = \max\{|\mathsf{n}_0|, |\mathsf{n}_1|\}.$$

The correlation bound (3.3), applied to the four pairs A_i, B_j, bounds each difference $\omega(A_iB_j) - \mathsf{m}_i\mathsf{n}_j$ by q_{cl} in absolute value, so $|S - S_{\mathrm{prod}}| \le \frac{1}{2} \cdot 4q_{\mathrm{cl}} = 2q_{\mathrm{cl}}$, and

$$1 < S \le S_{\mathrm{prod}} + 2q_{\mathrm{cl}} \le \max\{|\mathsf{n}_0|, |\mathsf{n}_1|\} + 2q_{\mathrm{cl}}.$$

The *minority outcome* of B_j is the less probable of its two effects $P_j^{\pm}$; since $\omega(P_j^{\pm}) = (1 \pm \mathsf{n}_j)/2$, its probability is $(1 - |\mathsf{n}_j|)/2$. Thus $S > 1$ forces one of the minority outcome probabilities of B_0, B_1 to be smaller than q_{cl}: for j with $|\mathsf{n}_j| = \max\{|\mathsf{n}_0|, |\mathsf{n}_1|\}$, the last display gives $(1 - |\mathsf{n}_j|)/2 < q_{\mathrm{cl}}$. A winning cell lies in the minority outcome of B_j: if the outcome is $P_j^{\pm}$, the cell $\omega(E_0^{\pm}P_j^{\pm})$ with the same sign is winning, because $\omega(A_0B_j)$ has the sign + in S. Since $E_0^{\pm}P_j^{\pm} \le P_j^{\pm}$, the minimality of w gives

$$w \le \omega(E_0^{\pm}P_j^{\pm}) \le \omega(P_j^{\pm}) < q_{\mathrm{cl}}.$$

Conclusion. Since $\eta > 0$, $0 < w < q_{\mathrm{cl}}$ and $K_{r_+,+}, K_{r_-,-} \ge 1$,

$$w^{\eta}K_{r_+,+}^{\nu_+}K_{r_-,-}^{\nu_-} < q_{\mathrm{cl}}^{\eta}K_{r_+,+}^{\nu_+}K_{r_-,-}^{\nu_-} \overset{(3.5)}{\le} 2^{-2/(ab)},$$

which contradicts (3.8). Hence no reflection quartet has $S > 1$, and $\beta(\omega; \mathfrak{A}, \mathfrak{B}) = 1$ by Step 1. □

4. Gaussian comparison and source compression

4.1. **Thermal powers and a covariance perturbation bound.** For quadratures with commutation matrix J_n, a centred Gaussian state with covariance V has characteristic function $\exp(-z^TVz/2)$. We use the Weyl and Gaussian formalism of [32], normalized so that the oscillator ground state has covariance $I/2$.

For a faithful Gaussian state ρ_V, let $V^{[\tau]}$ be the covariance of $\rho_V^{\tau}/\operatorname{Tr}\rho_V^{\tau}$. In Williamson coordinates, Proposition 1 of [32] gives, in our normalization,

$$\upsilon_j^{[\tau]} = \frac{1}{2}\coth\left(\tau \operatorname{arctanh}\frac{1}{2\upsilon_j}\right). \tag{4.1}$$

For one mode with number operator N, the thermal state $\rho_V \propto e^{-\beta N}$ has $V = \upsilon I_2$ with $\upsilon = \operatorname{Tr}(\rho_V N) + \frac{1}{2} = \frac{1}{2}\coth(\beta/2)$, and $\rho_V^{\tau} \propto e^{-\tau\beta N}$ has $V^{[\tau]} = \frac{1}{2}\coth(\tau\beta/2)I_2$, which is (4.1) because $\beta/2 = \operatorname{arctanh}(1/(2\upsilon))$.

For $0 < \tau < 1$, the thermal increment is

$$\mathsf{H}_{\tau}(V) = V^{[\tau]} - V > 0.$$

In Williamson coordinates, $\mathsf{H}_{\tau}(V)$ is diagonal with the entries $\upsilon_j^{[\tau]} - \upsilon_j$, which are positive because $\upsilon_j = \frac{1}{2}\coth(\operatorname{arctanh}\frac{1}{2\upsilon_j})$ and coth decreases on $(0, \infty)$.

For faithful density operators and $r > 1$, the sandwiched Rényi quantity is

$$\widetilde{Q}_r(\rho\|\sigma) = \operatorname{Tr}\left(\sigma^{-(r-1)/(2r)}\rho\sigma^{-(r-1)/(2r)}\right)^r. \tag{4.2}$$

Its logarithm divided by $r - 1$ is the sandwiched Rényi divergence, introduced independently by Wilde, Winter and Yang [42] and by Müller-Lennert, Dupuis, Szehr, Fehr and Tomamichel [27]. The value in (4.2) is $+\infty$ when the positive sandwich $\sigma^{-(r-1)/(2r)}\rho\sigma^{-(r-1)/(2r)}$ does not lie in $\mathcal{S}_r$. Theorem 4.1 bounds $\widetilde{Q}_r$ with constants independent of the number of oscillators and of their thermal parameters.

Theorem 4.1 (Gaussian Rényi comparison)**.** *Let ρ, σ be centred faithful finite-mode Gaussian states on the same canonical system, with covariance matrices V_ρ, V_σ. For $r > 1$, set*

$$\tau = \frac{r-1}{r}, \qquad k_r = \max\{2, \tau^{-1}\}, \qquad \mathsf{Z} = \mathsf{H}_{\tau}(V_\sigma)^{-1/2}(V_\rho - V_\sigma)\mathsf{H}_{\tau}(V_\sigma)^{-1/2}.$$

If $\|\mathsf{Z}\| \le (4k_r)^{-1}$, *then*

$$\log \widetilde{Q}_r(\rho\|\sigma) \le 2rk_r \|\mathsf{Z}\|_1 . \tag{4.3}$$

Proof. Step 1: independence of the quadrature coordinates. For a real invertible matrix S, a positive matrix A and a real symmetric matrix X of the same size, the matrix $W = (SAS^T)^{-1/2}SA^{1/2}$ satisfies $WW^T = I$, so it is orthogonal, and

$$(SAS^T)^{-1/2}\, SXS^T\, (SAS^T)^{-1/2} = W(A^{-1/2}XA^{-1/2})W^T.$$

A real change of quadrature coordinates by S, with the commutation form transformed at the same time, replaces V_ρ, V_σ and $\mathsf{H}_\tau(V_\sigma)$ by $SV_\rho S^T$, $SV_\sigma S^T$ and $S\mathsf{H}_\tau(V_\sigma)S^T$. With $A = \mathsf{H}_\tau(V_\sigma)$ and $X = V_\rho - V_\sigma$, the identity above shows that Z changes to $W\mathsf{Z}W^T$. Hence $\|\mathsf{Z}\|$, $\|\mathsf{Z}\|_1$ and the norm hypothesis do not depend on the quadrature coordinates; neither does $\widetilde{Q}_r(\rho\|\sigma)$, which is defined by the two states.

Step 2: the sandwich formulas. In Williamson coordinates for σ, which Step 1 allows, we write

$$V_0 = V_\sigma, \qquad \mathsf{T}_\tau = V_0^{[\tau]}, \qquad \mathsf{H} = \mathsf{T}_\tau - V_0,$$
$$\mathsf{T}_{1-\tau} = V_0^{[1-\tau]}, \qquad \mathsf{L} = (\mathsf{T}_\tau^2 - I/4)^{1/2}.$$

For the sandwich $\Theta = \sigma^{-\tau/2}\rho\sigma^{-\tau/2}$ and the covariance V_τ of the normalized sandwich $\Theta/\operatorname{Tr}\Theta$, Appendix A gives the Gaussian sandwich formulas

$$\operatorname{Tr}\Theta = \frac{\operatorname{Tr}\sigma^{1-\tau}}{\sqrt{\det(I-\mathsf{Z})}}, \qquad V_\tau = -\mathsf{T}_\tau + \mathsf{L}(\mathsf{T}_\tau - V_\rho)^{-1}\mathsf{L}, \tag{4.4}$$

whenever $V_\rho < \mathsf{T}_\tau$; under this condition Seshadreesan, Lami and Wilde give $\widetilde{Q}_r(\rho\|\sigma)$ in closed form [32, Theorem 22].

The norm hypothesis implies the condition $V_\rho < \mathsf{T}_\tau$: since $\|\mathsf{Z}\| \le (4k_r)^{-1} < 1$,

$$\mathsf{T}_\tau - V_\rho = \mathsf{H} - (V_\rho - V_0) = \mathsf{H}^{1/2}(I-\mathsf{Z})\mathsf{H}^{1/2} > 0.$$

Then, by Appendix A, Θ is a positive trace-class operator with the trace (A.4).

Step 3: the perturbation of the covariance. At $V_\rho = V_0$, the sandwich is $\sigma^{1-\tau}$, and the normalized state $\sigma^{1-\tau}/\operatorname{Tr}\sigma^{1-\tau}$ has covariance $\mathsf{T}_{1-\tau}$; so (4.4) at $V_\rho = V_0$ reads $\mathsf{T}_{1-\tau} = -\mathsf{T}_\tau + \mathsf{L}\mathsf{H}^{-1}\mathsf{L}$. For general V_ρ, Step 2 gives $(\mathsf{T}_\tau - V_\rho)^{-1} = \mathsf{H}^{-1/2}(I-\mathsf{Z})^{-1}\mathsf{H}^{-1/2}$. Subtracting the formula at $V_\rho = V_0$ from (4.4) gives

$$V_\tau - \mathsf{T}_{1-\tau} \overset{(4.4)}{=} \mathsf{L}\big((\mathsf{T}_\tau - V_\rho)^{-1} - \mathsf{H}^{-1}\big)\mathsf{L} \overset{\text{Step 2}}{=} \mathsf{L}\mathsf{H}^{-1/2}\big((I-\mathsf{Z})^{-1} - I\big)\mathsf{H}^{-1/2}\mathsf{L}.$$

In Williamson coordinates V_0 is diagonal, and so are T_τ (by (4.1)), H and L. Hence $\mathsf{P} = \mathsf{L}\mathsf{H}^{-1/2}$ is a diagonal matrix equal to $\mathsf{H}^{-1/2}\mathsf{L}$, so $\mathsf{P}^2 = \mathsf{L}\mathsf{H}^{-1}\mathsf{L}$, and the formula at $V_\rho = V_0$ and the last display give

$$\mathsf{P}^2 = \mathsf{T}_{1-\tau} + \mathsf{T}_\tau, \qquad V_\tau - \mathsf{T}_{1-\tau} = \mathsf{P}\big((I-\mathsf{Z})^{-1} - I\big)\mathsf{P}.$$

For each thermal mode, with $u = \operatorname{arctanh}(1/(2\upsilon_j))$, formula (4.1) gives the entries $\frac{1}{2}\coth(\tau u)$ of T_τ and $\frac{1}{2}\coth((1-\tau)u)$ of $\mathsf{T}_{1-\tau}$, the two terms of P^2. If $\tau \le 1/2$, then $\tau u \le (1-\tau)u$, and since $x\coth x$ increases on $(0,\infty)$, $\tau u\coth(\tau u) \le (1-\tau)u\coth((1-\tau)u)$. If $\tau \ge 1/2$, then $\tau u \ge (1-\tau)u$, and since coth decreases, $\coth(\tau u) \le \coth((1-\tau)u)$. The two cases give

$$\frac{\mathsf{T}_\tau}{\mathsf{T}_{1-\tau}} = \frac{\coth(\tau u)}{\coth((1-\tau)u)} \le \max\{1, (1-\tau)/\tau\},$$

and hence, since $1 + \max\{1,(1-\tau)/\tau\} = \max\{2,\tau^{-1}\} = k_r$, $\mathsf{T}_{1-\tau} + \mathsf{T}_\tau \le k_r\mathsf{T}_{1-\tau}$.

For

$$\delta = \|\mathsf{Z}\|, \qquad \mathsf{Y} = \mathsf{T}_{1-\tau}^{-1/2}(V_\tau - \mathsf{T}_{1-\tau})\mathsf{T}_{1-\tau}^{-1/2},$$

the identity $(I-\mathsf{Z})^{-1}-I=\mathsf{Z}(I-\mathsf{Z})^{-1}$ gives the first formula below, and the bound $\mathsf{T}_{1-\tau}+\mathsf{T}_\tau\le k_r\mathsf{T}_{1-\tau}$ gives the second, since P and $\mathsf{T}_{1-\tau}$ are diagonal:

$$\mathsf{Y}=\mathsf{T}_{1-\tau}^{-1/2}\mathsf{P}\,\mathsf{Z}(I-\mathsf{Z})^{-1}\,\mathsf{P}\mathsf{T}_{1-\tau}^{-1/2},$$
$$\left\|\mathsf{T}_{1-\tau}^{-1/2}\mathsf{P}\right\|^2=\left\|\mathsf{T}_{1-\tau}^{-1}(\mathsf{T}_{1-\tau}+\mathsf{T}_\tau)\right\|\le k_r.$$

The eigenvalues of Z lie in $[-\delta,\delta]$, so $\|\mathsf{Z}(I-\mathsf{Z})^{-1}\|\le\delta/(1-\delta)$ and $\|\mathsf{Z}(I-\mathsf{Z})^{-1}\|_1\le\|\mathsf{Z}\|_1/(1-\delta)$. We obtain

$$\|\mathsf{Y}\|\le\frac{k_r\delta}{1-\delta},\qquad\|\mathsf{Y}\|_1\le\frac{k_r}{1-\delta}\|\mathsf{Z}\|_1. \tag{4.5}$$

The norm hypothesis gives $k_r\delta\le1/4$, hence $\delta\le1/8$ since $k_r\ge2$, and (4.5) gives $\|\mathsf{Y}\|\le\frac14\cdot\frac87=\frac27$.

Step 4: the purity function and its derivative. For a physical covariance, define

$$\mathscr{P}_r(V)=\log\operatorname{Tr}\rho_V^r=-\sum_j\log f_r(\upsilon_j),\qquad f_r(x)=(x+\tfrac12)^r-(x-\tfrac12)^r.$$

The second expression holds mode by mode: a thermal mode with symplectic eigenvalue υ is the state $(1-\zeta)\zeta^N$ with $\zeta=(\upsilon-1/2)/(\upsilon+1/2)$, and $\operatorname{Tr}((1-\zeta)\zeta^N)^r=(1-\zeta)^r/(1-\zeta^r)=1/f_r(\upsilon)$. For $x\ge1/2$, the substitution $\zeta=(x-1/2)/(x+1/2)$ gives $f_r(x)=(x+\frac12)^r(1-\zeta^r)$, $f_r'(x)=r(x+\frac12)^{r-1}(1-\zeta^{r-1})$ and $x=(x+\frac12)(1+\zeta)/2$, hence

$$0\le\frac{xf_r'(x)}{f_r(x)}=\frac{r(1+\zeta)}{2}\frac{1-\zeta^{r-1}}{1-\zeta^r}\le r,$$

where the upper bound is equivalent to $(1-\zeta)(1+\zeta^{r-1})\ge0$, since $2(1-\zeta^r)-(1+\zeta)(1-\zeta^{r-1})=(1-\zeta)(1+\zeta^{r-1})$.

The derivative of $\mathscr{P}_r$ involves the derivatives of the symplectic eigenvalues of V, which are the positive eigenvalues of iJ_nV. At a Williamson-diagonal covariance $\Upsilon=\bigoplus_j\upsilon_jI_2$ with distinct υ_j, the vector $\psi_j=(e_{2j-1}-ie_{2j})/\sqrt2$, where the e_k are the standard basis vectors, satisfies $iJ_n\psi_j=\psi_j$. Since iJ_n is self-adjoint, ψ_j is both a right and a left eigenvector of $iJ_n\Upsilon$ for its simple eigenvalue υ_j, and first-order perturbation of this eigenvalue shows that the differential of a symplectic eigenvalue is half the trace of the corresponding two-dimensional diagonal block $\dot V_{jj}$:

$$d\upsilon_j[\dot V]=\psi_j^*\,iJ_n\dot V\,\psi_j=\psi_j^*\dot V\psi_j=\tfrac12\operatorname{Tr}\dot V_{jj}.$$

With $0\le f_r'(\upsilon_j)/f_r(\upsilon_j)\le r/\upsilon_j$, the bound on $xf_r'(x)/f_r(x)$ at $x=\upsilon_j$, and since the diagonal blocks of a matrix have total trace norm at most its trace norm, it follows that

$$\begin{aligned}|d\mathscr{P}_r(\Upsilon)[\dot V]|&\le\sum_j\frac{f_r'(\upsilon_j)}{f_r(\upsilon_j)}\frac{|\operatorname{Tr}\dot V_{jj}|}{2}\le\frac r2\sum_j|\operatorname{Tr}(\upsilon_j^{-1}\dot V_{jj})|\\&\le\frac r2\left\|\Upsilon^{-1/2}\dot V\Upsilon^{-1/2}\right\|_1.\end{aligned}$$

By Williamson's normal form, a physical covariance is $V=S\Upsilon S^T$ with S symplectic. Under this symplectic congruence $\mathscr{P}_r$ is unchanged, so $d\mathscr{P}_r(V)[\dot V]=d\mathscr{P}_r(\Upsilon)[S^{-1}\dot VS^{-T}]$, and by Step 1, with $A=\Upsilon$ and $X=S^{-1}\dot VS^{-T}$, the relative perturbation $V^{-1/2}\dot VV^{-1/2}$ is orthogonally similar to $\Upsilon^{-1/2}S^{-1}\dot VS^{-T}\Upsilon^{-1/2}$, so the two matrices have the same trace norm. Therefore, when the symplectic eigenvalues of V are distinct,

$$|d\mathscr{P}_r(V)[\dot V]|\le\frac r2\left\|V^{-1/2}\dot VV^{-1/2}\right\|_1. \tag{4.6}$$

A covariance V is faithful when $V+\frac i2J_n>0$, which by Williamson's normal form, as after (2.8), holds exactly when $\upsilon_j>1/2$ for all j. The Hermitian matrix $V^{1/2}iJ_nV^{1/2}=V^{1/2}(iJ_nV)V^{-1/2}$ has

the eigenvalues of iJ_nV, which by Williamson's normal form are $\pm\upsilon_1,\dots,\pm\upsilon_n$; so

$$\mathscr{P}_r(V) = -\frac{1}{2}\operatorname{Tr} F(V^{1/2} i J_n V^{1/2}), \qquad F(x) = \log f_r(|x|).$$

For faithful V, the eigenvalues $\pm\upsilon_j$ lie in the open set $\{|x| > 1/2\}$, on which F is real-analytic, since f_r is real-analytic and positive on $(1/2,\infty)$. By the holomorphic functional calculus, $M \mapsto \operatorname{Tr} F(M)$ is smooth on the Hermitian matrices with spectrum in $\{|x| > 1/2\}$, also where eigenvalues coincide, and $V \mapsto V^{1/2}$ is smooth on positive definite matrices. Hence $\mathscr{P}_r$ is smooth on the open set of faithful covariances, and $d\mathscr{P}_r(V)$ is continuous in V there. Covariances with distinct symplectic eigenvalues are dense in the faithful covariances, and the right side of (4.6) is continuous in V; so (4.6) holds at every faithful covariance.

Step 5: integration along a segment. The segment

$$V(v) = \mathsf{T}_{1-\tau} + v(V_\tau - \mathsf{T}_{1-\tau}) = \mathsf{T}_{1-\tau}^{1/2}(I + v\mathsf{Y})\mathsf{T}_{1-\tau}^{1/2}, \qquad 0 \le v \le 1,$$

consists of physical covariances, as convex combinations of the covariances $\mathsf{T}_{1-\tau}$ and V_τ of two states. For $v < 1$ the covariances $V(v)$ are faithful: $\mathsf{T}_{1-\tau}$ is faithful, since its symplectic eigenvalues $\frac{1}{2}\coth((1-\tau)u)$ from Step 3 exceed $\frac{1}{2}$, and $V_\tau + \frac{i}{2}J_n \ge 0$, so

$$V(v) + \tfrac{i}{2}J_n = (1-v)\big(\mathsf{T}_{1-\tau} + \tfrac{i}{2}J_n\big) + v\big(V_\tau + \tfrac{i}{2}J_n\big) > 0, \qquad 0 \le v < 1.$$

The segment has the derivative $\dot V = \mathsf{T}_{1-\tau}^{1/2}\mathsf{Y}\mathsf{T}_{1-\tau}^{1/2}$. By Step 1, with $S = \mathsf{T}_{1-\tau}^{1/2}$, $A = I + v\mathsf{Y}$ and $X = \mathsf{Y}$, the matrix $V(v)^{-1/2}\dot V V(v)^{-1/2}$ is orthogonally similar to $(I+v\mathsf{Y})^{-1/2}\mathsf{Y}(I+v\mathsf{Y})^{-1/2} = \mathsf{Y}(I+v\mathsf{Y})^{-1}$, whose trace norm is at most $\|\mathsf{Y}\|_1/(1-\|\mathsf{Y}\|)$. On physical covariances, $\mathscr{P}_r = -\sum_j \log f_r(\upsilon_j)$ is continuous, because f_r is continuous and positive on $[1/2,\infty)$ and the symplectic eigenvalues depend continuously on V; so integrating (4.6) from 0 to $v < 1$, letting $v \to 1$, and using (4.5) gives

$$|\mathscr{P}_r(V_\tau) - \mathscr{P}_r(\mathsf{T}_{1-\tau})| \le \frac{r}{2}\,\frac{\|\mathsf{Y}\|_1}{1-\|\mathsf{Y}\|} \le \frac{r}{2}\,\frac{k_r}{(1-\delta)(1-\|\mathsf{Y}\|)}\,\|\mathsf{Z}\|_1\,.$$

Step 6: assembly and the constant. Since $(r-1)/(2r) = \tau/2$, and since the normalized sandwich $\Theta/\operatorname{Tr}\Theta$, a Gaussian state, has covariance V_τ,

$$\widetilde{Q}_r(\rho\|\sigma) \overset{(4.2)}{=} \operatorname{Tr}\Theta^r = (\operatorname{Tr}\Theta)^r \operatorname{Tr}\Big(\frac{\Theta}{\operatorname{Tr}\Theta}\Big)^r$$
$$\overset{(4.4)}{=} \frac{(\operatorname{Tr}\sigma^{1-\tau})^r}{\det(I-\mathsf{Z})^{r/2}}\, e^{\mathscr{P}_r(V_\tau)}.$$

Because $r(1-\tau) = 1$, the factor $(\operatorname{Tr}\sigma^{1-\tau})^r$ equals $e^{-\mathscr{P}_r(\mathsf{T}_{1-\tau})}$: the state $\sigma^{1/r}/\operatorname{Tr}\sigma^{1/r}$ has covariance $\mathsf{T}_{1-\tau}$, so

$$(\operatorname{Tr}\sigma^{1/r})^r e^{\mathscr{P}_r(\mathsf{T}_{1-\tau})} = (\operatorname{Tr}\sigma^{1/r})^r \operatorname{Tr}\Big(\frac{\sigma^{1/r}}{\operatorname{Tr}\sigma^{1/r}}\Big)^r = \operatorname{Tr}\sigma = 1.$$

Taking logarithms, with $\log\det = \operatorname{Tr}\log$, gives

$$\log\widetilde{Q}_r(\rho\|\sigma) = -\frac{r}{2}\operatorname{Tr}\log(I-\mathsf{Z}) + \mathscr{P}_r(V_\tau) - \mathscr{P}_r(\mathsf{T}_{1-\tau}).$$

The bound $|\log(1-t)| \le |t|/(1-|t|)$, $|t| < 1$, applied to the eigenvalues of Z, gives

$$|\operatorname{Tr}\log(I-\mathsf{Z})| \le \frac{\|\mathsf{Z}\|_1}{1-\delta}.$$

With Step 5, and with $1/(1-\delta) \le 8/7$ and $1/(1-\|\mathsf{Y}\|) \le 7/5$ from $\delta \le 1/8$ and $\|\mathsf{Y}\| \le 2/7$,

$$\begin{aligned}\log \widetilde{Q}_r(\rho\|\sigma) &\le \frac{r}{2}\,\frac{\|\mathsf{Z}\|_1}{1-\delta} + \frac{r}{2}\,\frac{k_r\,\|\mathsf{Z}\|_1}{(1-\delta)(1-\|\mathsf{Y}\|)} \\ &\le r\Big(\frac{4}{7}+\frac{4}{5}k_r\Big)\,\|\mathsf{Z}\|_1 \overset{k_r\ge 2}{\le} 2rk_r\,\|\mathsf{Z}\|_1\,,\end{aligned}$$

proving (4.3). □

The effect estimate below is the two-outcome consequence of data processing for the sandwiched Rényi divergence, proved for all $r > 1$ independently by Frank and Lieb [14] and by Beigi [3].

Lemma 4.2 (Effects and trace distance)**.** *Let ρ, σ be faithful density operators and $r > 1$, and write $\rho(M) = \mathrm{Tr}(\rho M)$ and $\sigma(M) = \mathrm{Tr}(\sigma M)$. If $\widetilde{Q}_r(\rho\|\sigma) < \infty$, every effect M satisfies*

$$\rho(M)^r \le \widetilde{Q}_r(\rho\|\sigma)\,\sigma(M)^{r-1}. \tag{4.7}$$

For $r \ge 2$, one also has

$$\|\rho-\sigma\|_1 \le \sqrt{\widetilde{Q}_r(\rho\|\sigma)^{1/(r-1)} - 1}. \tag{4.8}$$

Proof. Step 1: the effect bound. With $\tau = (r-1)/r$, finiteness of $\widetilde{Q}_r(\rho\|\sigma)$ means that the sandwich $\Theta = \sigma^{-\tau/2}\rho\sigma^{-\tau/2}$ of (4.4) lies in $\mathcal{S}_r$, with $\rho = \sigma^{\tau/2}\Theta\sigma^{\tau/2}$, as $\sigma^{\tau/2}$ maps every vector into the domain of $\sigma^{-\tau/2}$. Since $\sigma^{\tau/2} \in \mathcal{S}_{2/\tau}$ and $\tau + 1/r = 1$, the trace is cyclic on the products below, and the Schatten Hölder inequality with the exponents r and $1/\tau$ bounds $\rho(M)$ by the product of $\|\Theta\|_r$, which is $\widetilde{Q}_r(\rho\|\sigma)^{1/r}$ by (4.2), and the factor $\left\|\sigma^{\tau/2}M\sigma^{\tau/2}\right\|_{1/\tau}$:

$$\rho(M) = \mathrm{Tr}(\sigma^{\tau/2}\Theta\sigma^{\tau/2}M) = \mathrm{Tr}(\Theta\,\sigma^{\tau/2}M\sigma^{\tau/2}) \overset{\text{Hölder}}{\le} \|\Theta\|_r\,\left\|\sigma^{\tau/2}M\sigma^{\tau/2}\right\|_{1/\tau}.$$

The operator $\sigma^{\tau/2}M\sigma^{\tau/2}$ has the same nonzero spectrum as $M^{1/2}\sigma^{\tau}M^{1/2}$. Hansen's inequality for operator monotone functions and contractions [17, Theorem], applied to $t \mapsto t^{\tau}$ and the contraction $M^{1/2}$, gives

$$M^{1/2}\sigma^{\tau}M^{1/2} \le (M^{1/2}\sigma M^{1/2})^{\tau}.$$

Monotonicity of the trace of a positive power bounds the remaining factor by $\sigma(M)^{\tau}$:

$$\begin{aligned}\left\|\sigma^{\tau/2}M\sigma^{\tau/2}\right\|_{1/\tau} &= \left\|M^{1/2}\sigma^{\tau}M^{1/2}\right\|_{1/\tau} \overset{\text{Hansen}}{\le} \left\|(M^{1/2}\sigma M^{1/2})^{\tau}\right\|_{1/\tau} \\ &= \left(\mathrm{Tr}\,M^{1/2}\sigma M^{1/2}\right)^{\tau} = \sigma(M)^{\tau}.\end{aligned}$$

Hence $\rho(M) \le \widetilde{Q}_r(\rho\|\sigma)^{1/r}\sigma(M)^{\tau}$, and raising to the power r, with $r\tau = r-1$, proves (4.7).

Step 2: comparison of the orders 2 and r. For $r \ge 2$, Schatten interpolation [33] of

$$z \longmapsto \sigma^{-\tau z/2}\rho\sigma^{-\tau z/2}$$

between the trace norm and the r-norm, at $z = 1/(2\tau)$, gives

$$\widetilde{Q}_2(\rho\|\sigma) \le \widetilde{Q}_r(\rho\|\sigma)^{1/(r-1)}.$$

Finite spectral compressions of σ justify the interpolation for unbounded inverse powers. Let Π_n be the spectral projection of σ for $[1/n, 1]$, which has finite rank, and let $\Theta_{1/2} = \sigma^{-1/4}\rho\sigma^{-1/4}$, the sandwich of (4.4) for $r = 2$. The compressed family $z \mapsto \Pi_n\sigma^{-\tau z/2}\rho\sigma^{-\tau z/2}\Pi_n$ is entire and bounded

on the strip $0 \le \operatorname{Re} z \le 1$. Since $\sigma^{-i\tau t/2}\Pi_n$, $t \in \mathbb{R}$, is a partial isometry commuting with Π_n, the trace-norm endpoint is at most 1, and the other endpoint is a compression of the full r-sandwich:

$$\left\|\Pi_n \sigma^{-i\tau t/2}\rho\sigma^{-i\tau t/2}\Pi_n\right\|_1 \le \|\rho\|_1 = 1,$$
$$\left\|\Pi_n \sigma^{-\tau(1+it)/2}\rho\sigma^{-\tau(1+it)/2}\Pi_n\right\|_r = \|\Pi_n\Theta\Pi_n\|_r \le \|\Theta\|_r .$$

The point $\theta = 1/(2\tau)$ lies in $(0,1]$ because $r \ge 2$, and the interpolated exponent p_θ satisfies $1/p_\theta = (1-\theta) + \theta/r = 1 - \theta\tau = 1/2$. The three-lines bound at $z = \theta$ therefore gives

$$\left\|\Pi_n\Theta_{1/2}\Pi_n\right\|_2 \le \|\Theta\|_r^\theta = \widetilde{Q}_r(\rho\|\sigma)^{1/(2(r-1))}.$$

As $n \to \infty$, the left-hand side increases to $\|\Theta_{1/2}\|_2 = \widetilde{Q}_2(\rho\|\sigma)^{1/2}$, since $\|\Pi_n\Theta_{1/2}\Pi_n\|_2^2$ is the sum of the squared moduli of the matrix entries of $\Theta_{1/2}$ between eigenvectors of σ with eigenvalues at least $1/n$. The Hilbert–Schmidt limit gives $\widetilde{Q}_2(\rho\|\sigma) \le \widetilde{Q}_r(\rho\|\sigma)^{1/(r-1)}$. For density matrices, this inequality is the monotonicity in the order of the sandwiched Rényi divergence, proved independently by Beigi [3, Theorem 7] and by Müller-Lennert et al. [27, Theorem 7], and the argument above is Beigi's proof, applied to the orders 2 and r.

Step 3: the trace distance. Since $\operatorname{Tr}(\Theta_{1/2}\sigma^{1/2}) = \operatorname{Tr}(\sigma^{1/4}\Theta_{1/2}\sigma^{1/4}) = \operatorname{Tr}\rho = 1$,

$$\left\|\Theta_{1/2} - \sigma^{1/2}\right\|_2^2 = \operatorname{Tr}\Theta_{1/2}^2 - 2\operatorname{Tr}(\Theta_{1/2}\sigma^{1/2}) + \operatorname{Tr}\sigma = \widetilde{Q}_2(\rho\|\sigma) - 1.$$

The compressions Π_n of Step 2 justify this identity. Finally,

$$\rho - \sigma = \sigma^{1/4}(\Theta_{1/2} - \sigma^{1/2})\sigma^{1/4},$$

and Hölder with $\left\|\sigma^{1/4}\right\|_4 = 1$ gives

$$\begin{aligned}\|\rho - \sigma\|_1 &\le \left\|\sigma^{1/4}\right\|_4 \left\|\Theta_{1/2} - \sigma^{1/2}\right\|_2 \left\|\sigma^{1/4}\right\|_4 = \sqrt{\widetilde{Q}_2(\rho\|\sigma) - 1} \\ &\overset{\text{Step 2}}{\le} \sqrt{\widetilde{Q}_r(\rho\|\sigma)^{1/(r-1)} - 1},\end{aligned}$$

which is (4.8). □

4.2. **Thermal weights and compression.** A full source space is a space $\mathcal{X}_{\mu,U}$ of Section 2.2, and a finite source space is a finite canonical source system; both carry the covariance inner product. On a source space of either kind, the symplectic form is represented by the bounded skew-adjoint operator J with

$$\mathsf{v}_{\mu,U}(\mathsf{x}, \mathsf{J}\mathsf{y}) = \varsigma(\mathsf{x}, \mathsf{y}), \qquad \|\mathsf{J}\| \le 2,$$

the norm bound being the imaginary part of (2.6). In coordinates of a finite source space that are orthonormal for the covariance, J is the commutation matrix. On the complexification, with $G, \overline{G}, \mathsf{R}$ the operators $G_U, \overline{G}_U, \mathsf{R}_U$ of (2.9) written without the subscript, (2.6) gives $\langle \mathsf{x}, G\mathsf{y}\rangle = \mathsf{v}_{\mu,U}(\mathsf{x},\mathsf{y}) + \frac{i}{2}\mathsf{v}_{\mu,U}(\mathsf{x},\mathsf{J}\mathsf{y})$ for real x, y, hence

$$G = I + i\mathsf{J}/2, \qquad \overline{G} = I - i\mathsf{J}/2, \qquad d = |i\mathsf{J}/2|, \qquad \mathsf{R} = (G\overline{G})^{1/2} = (I - d^2)^{1/2},$$

where the last identity holds because $i\mathsf{J}/2$ is self-adjoint, and where d and R act on the complexification and preserve the real source space.

In finite canonical dimension, $0 < d < I$: d is invertible because ς is nondegenerate, and $d < I$ because G and $\overline{G}$ are positive definite by (2.8). An orthogonal change of basis then brings J to blocks $2d_jJ_1$, on which $d = d_jI_2$ with $0 < d_j < 1$, and keeps the covariance I. Rescaling the two basis vectors of a block by $(2d_j)^{-1/2}$ gives the commutation matrix J_1 and the covariance $I_2/(2d_j)$,

so the block has symplectic eigenvalue $\upsilon_j = 1/(2d_j)$, and (4.1) gives $\upsilon_j^{[\tau]} = \frac{1}{2}\coth(\tau \operatorname{arctanh} d_j)$. Undoing the rescaling multiplies covariances by $2d_j$, so formula (4.1) becomes

$$\mathsf{H}_\tau(I) = d\coth(\tau \operatorname{arctanh} d) - I.$$

For $0 < d < 1$ the scalar function $d\coth(\tau \operatorname{arctanh} d) - 1$ is positive. The function extends continuously at $d = 0$, with the value $(1-\tau)/\tau$, and scalar functional calculus defines the corresponding bounded positive operator on the full source space, with limiting value zero at $d = 1$.

Lemma 4.3 (Thermal comparison)**.** *For $0 < \tau < 1$,*

$$\mathsf{H}_\tau(I) \geq c_\tau \mathsf{R}^{2\tau}, \qquad c_\tau = \begin{cases} 2^{1-2\tau}, & 0 < \tau \leq 1/2, \\ (1-\tau)/\tau, & 1/2 \leq \tau < 1. \end{cases} \tag{4.9}$$

These constants are the infima over $0 < d < 1$ of the scalar ratios $(d\coth(\tau \operatorname{arctanh} d) - 1)(1-d^2)^{-\tau}$.

Proof. Step 1: the ratio as a function of u. For $d = \tanh u$, $u > 0$, one has $(1-d^2)^{-\tau} = \cosh(u)^{2\tau}$ and

$$\tanh u \coth(\tau u) - 1 = \frac{\sinh u \cosh(\tau u) - \cosh u \sinh(\tau u)}{\cosh u \sinh(\tau u)} = \frac{\sinh((1-\tau)u)}{\cosh u \sinh(\tau u)},$$

so the ratio in (4.9) is

$$\mathfrak{q}_\tau(u) = \frac{\sinh((1-\tau)u)}{\sinh(\tau u)} \cosh(u)^{2\tau-1}.$$

Its limits at zero and infinity are $(1-\tau)/\tau$ and $2^{1-2\tau}$; at infinity, $\sinh((1-\tau)u)/\sinh(\tau u)$ behaves as $e^{(1-2\tau)u}$ and $\cosh(u)^{2\tau-1}$ as $2^{1-2\tau}e^{(2\tau-1)u}$.

Step 2: an auxiliary convex function. For fixed $u > 0$, define $\mathfrak{f}(v) = v\coth(vu)$. Its derivative

$$\mathfrak{f}'(v) = \coth(vu) - vu\operatorname{csch}^2(vu)$$

is increasing, because

$$\mathfrak{f}''(v) = 2u\operatorname{csch}^2(vu)(vu\coth(vu) - 1) > 0.$$

For $0 < v \leq 1$, one also has $\mathfrak{f}'(v) \leq \mathfrak{f}'(1) \leq \tanh u$; the last inequality is equivalent to $\tanh u \leq u$, because

$$\tanh u - \mathfrak{f}'(1) = \frac{u}{\sinh^2 u} - (\coth u - \tanh u) = \frac{u}{\sinh^2 u} - \frac{1}{\sinh u \cosh u} = \frac{u - \tanh u}{\sinh^2 u}.$$

Step 3: monotonicity of the ratio. By the mean value theorem, $\mathfrak{f}(1-\tau) - \mathfrak{f}(\tau) = (1-2\tau)\mathfrak{f}'(\vartheta)$ for some ϑ between τ and $1-\tau$, so $0 < \vartheta < 1$. Consequently,

$$\begin{aligned} \frac{d}{du}\log \mathfrak{q}_\tau(u) &= (1-\tau)\coth((1-\tau)u) - \tau\coth(\tau u) + (2\tau-1)\tanh u \\ &= \mathfrak{f}(1-\tau) - \mathfrak{f}(\tau) + (2\tau-1)\tanh u = (1-2\tau)\big(\mathfrak{f}'(\vartheta) - \tanh u\big) \end{aligned}$$

is nonpositive for $\tau \leq 1/2$ and nonnegative for $\tau \geq 1/2$, since $\mathfrak{f}'(\vartheta) \leq \tanh u$ by Step 2. So $\mathfrak{q}_\tau$ is nonincreasing for $\tau \leq 1/2$, with infimum its limit $2^{1-2\tau}$ at infinity, and nondecreasing for $\tau \geq 1/2$, with infimum its limit $(1-\tau)/\tau$ at zero. In both cases the infimum is c_τ, so $d\coth(\tau \operatorname{arctanh} d) - 1 \geq c_\tau(1-d^2)^\tau$ for $0 < d < 1$, by continuity also for $0 \leq d \leq 1$, and the scalar functional calculus of d gives (4.9). □

For positive operators, the weighted geometric mean is

$$A\#_\tau B = A^{1/2}(A^{-1/2}BA^{-1/2})^\tau A^{1/2}, \qquad 0 < \tau < 1.$$

We write $A\#B = A\#_{1/2}B$. Noninvertible cases are defined by decreasing positive regularization. Parallel sums $(\sum_j A_j^{-1})^{-1}$ have the analogous quadratic-form interpretation. Weighted geometric means and parallel sums are Kubo–Ando operations, and we use their compression properties [23, Theorems 3.4 and 3.5].

Lemma 4.4 (Compression of thermal weights)**.** *Let ι be a real isometry from a finite source space into a full source space, with $\varsigma(\iota\mathsf{x},\iota\mathsf{y})=\varsigma(\mathsf{x},\mathsf{y})$ for all x,y, as for an inclusion, and write*

$$G_F=\iota^*G\iota,\qquad \overline{G}_F=\iota^*\overline{G}\iota,\qquad \mathsf{R}_F=(G_F\overline{G}_F)^{1/2}.$$

For $0<a\le 1/2$,

$$\mathsf{R}_F^{2a}\ge\iota^*\mathsf{R}^{2a}\iota. \tag{4.10}$$

For an integer $r\ge 2$ and $b=(r-1)/r$,

$$\mathsf{H}_b(I_F)\ge\iota^*\mathsf{H}_b(I)\iota\ge c_b\,\iota^*\mathsf{R}^{2b}\iota. \tag{4.11}$$

Proof. Step 1: compression of parallel sums and geometric means. For positive invertible A,B, the variational formula

$$\langle z,(A^{-1}+B^{-1})^{-1}z\rangle=\inf_{x+y=z}\{\langle x,Ax\rangle+\langle y,By\rangle\}$$

gives monotonicity and the compression inequality for parallel sums: restricting the vectors in the infimum to $\operatorname{Ran}\iota$ increases the value, so that

$$\iota^*(A^{-1}+B^{-1})^{-1}\iota\le\big((\iota^*A\iota)^{-1}+(\iota^*B\iota)^{-1}\big)^{-1},$$

and the analogous formula for $(\sum_j A_j^{-1})^{-1}$ gives monotonicity and the compression inequality for more than two terms. The integral representation

$$A\#_\tau B=\frac{\sin\pi\tau}{\pi}\int_0^\infty v^{\tau-1}(A^{-1}+vB^{-1})^{-1}\,dv,$$

which for commuting A and B reduces to $A^{1-\tau}B^\tau$, writes $A\#_\tau B$ through the parallel sums of A and $v^{-1}B$, and therefore gives

$$\iota^*(A\#_\tau B)\iota\le(\iota^*A\iota)\#_\tau(\iota^*B\iota).$$

Step 2: the reserve inequality (4.10)*.* For $A=G$, $B=\overline{G}$, the two operators commute both before and after compression, because their sum is $2I$ and $\iota^*(2I)\iota=2I_F$. Commuting positive operators satisfy $A\#B=(AB)^{1/2}$, so $G\#\overline{G}=\mathsf{R}$ and $G_F\#\overline{G}_F=\mathsf{R}_F$, and Step 1 at $\tau=1/2$ (for invertible operators; Step 4 removes this restriction) gives $\mathsf{R}_F\ge\iota^*\mathsf{R}\iota$. Since $0<2a\le 1$, the function x^{2a} is operator monotone and operator concave on $[0,\infty)$, and these two properties give

$$\mathsf{R}_F^{2a}\overset{\text{monotone}}{\ge}(\iota^*\mathsf{R}\iota)^{2a}\overset{\text{Jensen}}{\ge}\iota^*\mathsf{R}^{2a}\iota,$$

which is (4.10). The map $X\mapsto\iota^*X\iota$ is completely positive and unital, as required for this use of Jensen's inequality in the form of Hansen [17, Corollary].

Since $i\mathsf{J}/2$ is self-adjoint, the squares of the reserves also compare directly:

$$\begin{aligned}\mathsf{R}_F^2-\iota^*\mathsf{R}^2\iota&=\big(I_F-(\iota^*(i\mathsf{J}/2)\iota)^2\big)-\big(I_F-\iota^*(i\mathsf{J}/2)^2\iota\big)\\&=\iota^*(i\mathsf{J}/2)(I-\iota\iota^*)(i\mathsf{J}/2)\iota\ge 0.\end{aligned}$$

Applying operator monotonicity and concavity to x^a gives a second route to (4.10):

$$\mathsf{R}_F^{2a}\ge(\iota^*\mathsf{R}^2\iota)^a\ge\iota^*\mathsf{R}^{2a}\iota.$$

Step 3: the thermal weights and (4.11)*.* The thermal increment $\mathsf{H}_{(r-1)/r}(I)$ is a parallel sum of geometric means of G and $\overline{G}$:

$$\mathsf{H}_{(r-1)/r}(I)=\left[\sum_{k=1}^{r-1}(G\#_{k/r}\overline{G})^{-1}\right]^{-1}. \tag{4.12}$$

At a scalar spectral value $d \in (-1,1)$ of $i\mathsf{J}/2$, here signed, let $g_\pm = 1 \pm d$ and $\mathfrak{z} = (g_-/g_+)^{1/r}$, so that $G\#_{k/r}\overline{G}$ takes the value $g_+^{1-k/r} g_-^{k/r} = g_+\mathfrak{z}^k$. The right-hand side of (4.12) takes the value

$$\frac{g_+}{\sum_{k=1}^{r-1}\mathfrak{z}^{-k}} = \frac{g_+\mathfrak{z}^{r-1}(1-\mathfrak{z})}{1-\mathfrak{z}^{r-1}} = d\coth\!\Big(\frac{r-1}{r}\operatorname{arctanh} d\Big) - 1.$$

For the last equality, write $d = \tanh u$; then $g_+ = e^u/\cosh u$, $\mathfrak{z} = e^{-2u/r}$, $1-\mathfrak{z} = 2e^{-u/r}\sinh(u/r)$ and $1-\mathfrak{z}^{r-1} = 2e^{-(r-1)u/r}\sinh((r-1)u/r)$, so the middle expression equals

$$\sinh(u/r)/(\cosh u \sinh((r-1)u/r)),$$

which is the right-hand side by Step 1 of the proof of Lemma 4.3 at $\tau = (r-1)/r$. The expression $d\coth(\frac{r-1}{r}\operatorname{arctanh} d) - 1$ is even in d and extends continuously at zero and the endpoints. Compression of each geometric mean in (4.12), followed by monotonicity and compression of the parallel sum, gives the first inequality in (4.11), and Lemma 4.3 gives the second; with (4.12) on the finite source space, where G_F and $\overline{G}_F$ take the places of G and $\overline{G}$ (because ι preserves ς, $\iota^*\mathsf{J}\iota$ is the operator J of the finite source space, so G_F, $\overline{G}_F$ and R_F are its own G, $\overline{G}$ and R),

$$\begin{aligned}\mathsf{H}_b(I_F) &\overset{(4.12)}{=} \Big[\sum_{k=1}^{r-1}(G_F\#_{k/r}\overline{G}_F)^{-1}\Big]^{-1} \overset{\text{Step 1}}{\geq} \Big[\sum_{k=1}^{r-1}\big(\iota^*(G\#_{k/r}\overline{G})\iota\big)^{-1}\Big]^{-1}\\ &\overset{\text{Step 1}}{\geq} \iota^*\Big[\sum_{k=1}^{r-1}(G\#_{k/r}\overline{G})^{-1}\Big]^{-1}\iota \overset{(4.12)}{=} \iota^*\mathsf{H}_b(I)\iota \overset{\text{Lem. 4.3}}{\geq} c_b\,\iota^*\mathsf{R}^{2b}\iota.\end{aligned}$$

Step 4: the full source space. Positive regularization and form limits yield (4.10) and (4.11) in infinite dimension. On the full source space, G and $\overline{G}$ are injective but need not have bounded inverses. The inequalities of Steps 1 to 3 hold for the commuting invertible operators $G+\epsilon I$ and $\overline{G}+\epsilon I$, $\epsilon > 0$, in place of G and $\overline{G}$. As $\epsilon \downarrow 0$, the geometric means and parallel sums formed from $G+\epsilon I$ and $\overline{G}+\epsilon I$ decrease strongly to those of G and $\overline{G}$, which is how the noninvertible cases are defined, while their compressions $G_F+\epsilon I_F$ and $\overline{G}_F+\epsilon I_F$ converge to the invertible G_F and $\overline{G}_F$. The inequalities therefore pass to the limit, and (4.12) holds on the full source space as the scalar identity above in the spectral calculus of $i\mathsf{J}/2$. □

The comparisons of Lemmas 4.3 and 4.4 also control trace norms after normalization by a thermal increment, as in the matrix Z of Theorem 4.1. Let H_F be a positive operator on the finite source space with $H_F \geq c_a\iota^*\mathsf{R}^{2a}\iota$; in Section 8, H_F is the thermal increment $\mathsf{H}_a(I_F)$, which satisfies this hypothesis by (4.9) on the finite source space and (4.10), as displayed in (8.6). For an operator $\mathsf{R}^aY\mathsf{R}^a$ with $Y \in \mathcal{S}_1$,

$$H_F^{-1/2}\iota^*\mathsf{R}^aY\mathsf{R}^a\iota H_F^{-1/2} = \mathsf{N}^*Y\mathsf{N}, \qquad \mathsf{N} = \mathsf{R}^a\iota H_F^{-1/2}, \qquad \|\mathsf{N}\| \leq c_a^{-1/2},$$

where the norm bound follows from the hypothesis, conjugated by $H_F^{-1/2}$:

$$\mathsf{N}^*\mathsf{N} = H_F^{-1/2}\iota^*\mathsf{R}^{2a}\iota H_F^{-1/2} \leq c_a^{-1}I_F.$$

The trace norm of $\mathsf{N}^*Y\mathsf{N}$ is therefore at most $\|\mathsf{N}\|^2\,\|Y\|_1 \leq c_a^{-1}\,\|Y\|_1$ (the *one-space normalization bound*).

For two full source spaces with reserves $\mathsf{R}_A, \mathsf{R}_B$ and real isometries ι_A, ι_B into them that preserve ς, as in Lemma 4.4, let $H_{A,F}$ and $H_{B,F}$ be positive operators with $H_{A,F} \geq c_b\iota_A^*\mathsf{R}_A^{2b}\iota_A$ and $H_{B,F} \geq c_b\iota_B^*\mathsf{R}_B^{2b}\iota_B$; in Section 8, $H_{A,F}$ and $H_{B,F}$ are the thermal increments $\mathsf{H}_b(I_F)$ of the two finite source spaces, which satisfy these hypotheses by (4.11). Conjugation by $H_{A,F}^{-1/2}$ and $H_{B,F}^{-1/2}$, as for N, bounds the norms of $\mathsf{N}_A = \mathsf{R}_A^b\iota_A H_{A,F}^{-1/2}$ and $\mathsf{N}_B = \mathsf{R}_B^b\iota_B H_{B,F}^{-1/2}$ by $c_b^{-1/2}$. The operator in (4.13) below is

$\mathsf{N}_A^* X \mathsf{N}_B$, so its trace norm is at most $\|\mathsf{N}_A\| \, \|X\|_1 \, \|\mathsf{N}_B\|$, and the bounds on $\|\mathsf{N}_A\|$ and $\|\mathsf{N}_B\|$ give

$$\left\| H_{A,F}^{-1/2} \iota_A^* \mathsf{R}_A^b X \mathsf{R}_B^b \iota_B H_{B,F}^{-1/2} \right\|_1 \le c_b^{-1} \|X\|_1 , \qquad X \in \mathcal{S}_1. \tag{4.13}$$

5. Bounded localization and Fourier evaluation

The complex mass shell is

$$\mathcal{Q}_\mu = \{ z \in \mathbb{C}^4 : z_0^2 - z_1^2 - z_2^2 - z_3^2 = \mu^2 \}.$$

For a compactly supported source $\mathsf{x} = (f, g)$, we use the entire function

$$\Psi_{\mathsf{x}}(z) = \widehat{f}(\mathbf{z}) + i z_0 \widehat{g}(\mathbf{z}).$$

The spacetime product $z \cdot z$ is Lorentzian, while $\mathbf{z} \cdot \mathbf{z}$ is the complex bilinear Euclidean spatial product.

For a nonempty bounded open $U \subset \mathbb{R}^3$, $\mathsf{x} = (f, g) \in \mathcal{H}_{\mu,U}$ and $z \in \mathbb{C}^4$, the number $\Psi_{\mathsf{x}}(z)$ depends continuously on x. Indeed, $\widehat{f}(\mathbf{z})$ and $\widehat{g}(\mathbf{z})$ are the values of the distributions $f \in H^{-1/2}$ and $g \in H^{1/2}$ on a function in $C_c^\infty(\mathbb{R}^3)$ equal to $e^{-i\mathbf{z}\cdot\mathbf{x}}$ near $\overline{U}$, so the two values are bounded by $\|f\|_{H^{-1/2}}$ and $\|g\|_{H^{1/2}}$ times the $H^{1/2}$ and $H^{-1/2}$ norms of that function. Since $(\mu^2 + |\mathbf{p}|^2)/(1 + |\mathbf{p}|^2)$ lies between $\min\{1, \mu^2\}$ and $\max\{1, \mu^2\}$, the norms $\|f\|_{H^{-1/2}}$ and $\|g\|_{H^{1/2}}$ are at most a factor depending only on μ times the norm of x in $\mathcal{H}_{\mu,U}$.

5.1. Containing wedges and weighted complex-shell evaluation.

For $\ell \ge 0$, let

$$\mathcal{W}_\ell = \{ (x^0, \mathbf{x}) : x^1 < \ell - |x^0| \}.$$

The real subspace $\mathcal{K}_{\mathcal{W}_\ell}$ of the wedge is the closed real linear span of the subspaces $\mathcal{K}_U$ of Lemma 2.2 over the bounded open sets U of its time-zero base, the half-space $\{x^1 < \ell\}$; by finite propagation speed and the time-slice construction that proves (2.3), the same subspace is also the closed real span of the one-particle vectors of the real test functions supported in $\mathcal{W}_\ell$.

The subspace $\mathcal{K}_{\mathcal{W}_\ell}$ is standard. It contains $\mathcal{K}_U$ for a ball U, so the complex span of $\mathcal{K}_{\mathcal{W}_\ell}$ is dense by Lemma 2.2. Let $\xi \in \mathcal{K}_{\mathcal{W}_\ell} \cap i\mathcal{K}_{\mathcal{W}_\ell}$, so that $-i\xi$ lies in $\mathcal{K}_{\mathcal{W}_\ell}$ as well, and let y be a smooth real source supported in a bounded open $U' \subset \{x^1 > \ell\}$. Sources supported in the two half-spaces have vanishing symplectic form, so by (2.6) the numbers $\langle \xi, \kappa_{U'} \mathsf{y} \rangle$ and $\langle -i\xi, \kappa_{U'} \mathsf{y} \rangle = i \langle \xi, \kappa_{U'} \mathsf{y} \rangle$ are real; hence $\langle \xi, \kappa_{U'} \mathsf{y} \rangle = 0$. The vectors $\kappa_{U'} \mathsf{y}$ span a dense complex subspace by Lemma 2.2, so $\xi = 0$. We write $\Delta_{\mathcal{W}_\ell}$ for the modular operator of $\mathcal{K}_{\mathcal{W}_\ell}$.

The Bisognano–Wichmann theorem identifies the wedge modular group with Lorentz boosts [4], and for the free field the same identification holds in the one-particle space [6, Remark 3.6(2)]: for the left wedge $\mathcal{W}_0$,

$$\Delta_{\mathcal{W}_0}^{iy} = \mathsf{U}(\Lambda_1(2\pi y)), \qquad (\Lambda_1(\chi)p)_0 = p_0 \cosh\chi + p_1 \sinh\chi, \quad (\Lambda_1(\chi)p)_1 = p_1 \cosh\chi + p_0 \sinh\chi,$$

where U is the one-particle representation, $(\mathsf{U}(\Lambda)\psi)(p) = \psi(\Lambda^{-1}p)$, and $\Lambda_1(\chi)$ is the boost of rapidity χ along x^1, which preserves $\mathcal{W}_0$. For the right wedge the sign of the rapidity is reversed; the choice of the left wedge fixes the sign of the imaginary shift below. Analytic continuation to $y = -ic$, $0 < c < 1/2$, replaces the real boost by the boost of imaginary rapidity $i\phi$, $\phi = 2\pi c \in (0, \pi)$: on one-particle wave functions ψ in the domain of $\Delta_{\mathcal{W}_0}^c$, $\Delta_{\mathcal{W}_0}^c \psi = \psi \circ b_\phi$ with

$$b_\phi(p) = \Lambda_1(i\phi)p = (p_0 \cos\phi + i p_1 \sin\phi,\ p_1 \cos\phi + i p_0 \sin\phi,\ p_2, p_3).$$

With the Fourier convention of Section 2,

$$\left\| \Delta_{\mathcal{W}_\ell}^c \kappa_U \mathsf{x} \right\|^2 = \int_{H_\mu^+} e^{-2\ell p_0 \sin\phi} |\Psi_{\mathsf{x}}(b_\phi(p))|^2 \, d\mathfrak{m}_\mu(p) \tag{5.1}$$

for every $\mathsf{x} \in \mathcal{H}_{\mu,U}$, U a bounded open subset of the base $\{x^1 < \ell\}$. A proper orthochronous Lorentz transformation $\mathcal{L}$ replaces the argument $b_\phi(p)$ of Ψ_{x} by $\mathcal{L} b_\phi(p)$ for the wedge $\mathcal{L}\mathcal{W}_\ell$.

Steps 1 to 5 below prove (5.1) and its form for $\mathcal{L}\mathcal{W}_\ell$, reading the contour and its damping directly from the one-particle representation, as in the similar computation of Buchholz and Lechner for the free massive field in two dimensions at $c = 1/4$ [9, Section 4]. In rapidity coordinates

$$p_0 = m_\perp \cosh\chi, \qquad p_1 = m_\perp \sinh\chi, \qquad m_\perp^2 = \mu^2 + p_2^2 + p_3^2,$$

the invariant measure is $d\mathfrak{m}_\mu = (2(2\pi)^3)^{-1} d\chi\, d^2p_\perp$, since $dp_1 = p_0\, d\chi$.

Step 1: the modular operator as a shift of rapidity. Boosts translate χ: $(\Delta_{\mathcal{W}_0}^{iy}\psi)(\chi, \mathbf{p}_\perp) = \psi(\chi - 2\pi y, \mathbf{p}_\perp)$. The Fourier transform $\int e^{-ik\chi}\psi(\chi, \mathbf{p}_\perp)\, d\chi$ turns $\Delta_{\mathcal{W}_0}^{iy}$ into multiplication by $e^{-2\pi i y k}$, so $\Delta_{\mathcal{W}_0}^{c}$ is multiplication by $e^{-\phi k}$. By the Paley–Wiener theorem for a strip, applied at almost every $\mathbf{p}_\perp$, a wave function ψ lies in the domain of $\Delta_{\mathcal{W}_0}^{c}$, with

$$(\Delta_{\mathcal{W}_0}^{c}\psi)(\chi, \mathbf{p}_\perp) = \psi(\chi + i\phi, \mathbf{p}_\perp),$$

whenever each function $\chi \mapsto \psi(\chi, \mathbf{p}_\perp)$ extends holomorphically to the strip $0 < \operatorname{Im}\chi < \phi$ and continuously to its closure, with $|\psi(\chi + i\phi', \mathbf{p}_\perp)| \le G(\chi, \mathbf{p}_\perp)$ for $0 \le \phi' \le \phi$ and a function G square integrable in $d\chi\, d^2p_\perp$. Since $\Lambda_1(\chi + i\phi) = \Lambda_1(i\phi)\Lambda_1(\chi)$, the shift $\chi \mapsto \chi + i\phi$ replaces p by $b_\phi(p)$.

Step 2: smooth sources in the base of $\mathcal{W}_0$. Let $\mathsf{x} = (f, g)$ be smooth with compact support in $\{x^1 < 0\}$, and let $z = \Lambda_1(\chi + i\phi')(m_\perp, 0, p_2, p_3)$ with $0 \le \phi' \le \phi$. Then $\operatorname{Im}\mathbf{z} = (m_\perp \cosh\chi \sin\phi', 0, 0)$ has a nonnegative first entry, so $|e^{-i\mathbf{z}\cdot\mathbf{x}}| = e^{x^1 \operatorname{Im} z_1} \le 1$ on the support of x. Integration by parts in $\widehat{f}(\mathbf{z}) = \int e^{-i\mathbf{z}\cdot\mathbf{x}} f(\mathbf{x})\, d^3x$ and in $\widehat{g}(\mathbf{z})$, together with $|z_0|^2 = |\mu^2 + \mathbf{z}\cdot\mathbf{z}| \le (\mu + |\mathbf{z}|)^2$, gives for every N a constant $C_{N,\mathsf{x}}$ with

$$|\Psi_{\mathsf{x}}(z)| \le C_{N,\mathsf{x}}(1 + |\mathbf{z}|)^{-N}.$$

Since $|\mathbf{z}| \ge \max\{|z_1|, |\mathbf{p}_\perp|\}$ with $|z_1| = m_\perp|\sinh(\chi + i\phi')| \ge m_\perp|\sinh\chi|$, and $m_\perp \ge \mu > 0$, the right-hand side is at most a function of $(\chi, \mathbf{p}_\perp)$ alone that is square integrable in $d\chi\, d^2p_\perp$ for $N > 2$. The function $\chi \mapsto \Psi_{\mathsf{x}}(\Lambda_1(\chi)(m_\perp, 0, p_2, p_3))$ is entire and equals $\kappa_U\mathsf{x}$ at real χ, so Step 1 gives $\Delta_{\mathcal{W}_0}^{c}\kappa_U\mathsf{x} = \Psi_{\mathsf{x}} \circ b_\phi$.

Step 3: translation of the wedge edge. Translation of the wedge edge by ℓ contributes the modulus $e^{-\ell p_0 \sin\phi}$. Since $\mathcal{W}_\ell = \mathcal{W}_0 + \ell\mathbf{e}_1$, one has $\Delta_{\mathcal{W}_\ell} = \mathsf{U}(\ell\mathbf{e}_1)\Delta_{\mathcal{W}_0}\mathsf{U}(\ell\mathbf{e}_1)^*$, where $(\mathsf{U}(\ell\mathbf{e}_1)\psi)(p) = e^{-i\ell p_1}\psi(p)$ represents the translation by $\ell\mathbf{e}_1$. The inner factor $\mathsf{U}(\ell\mathbf{e}_1)^*$ maps $\kappa_U\mathsf{x}$ to the one-particle vector of the source translated by $-\ell\mathbf{e}_1$, whose Fourier transform is $e^{i\ell p_1}\Psi_{\mathsf{x}}(p)$; the continuation $p_1 \mapsto p_1\cos\phi + ip_0\sin\phi$ turns the factor $e^{i\ell p_1}$ into $e^{i\ell p_1\cos\phi}e^{-\ell p_0\sin\phi}$, and the outer translation is unitary. For smooth x with compact support in $\{x^1 < \ell\}$, Step 2 applies to the translated source, and

$$|(\Delta_{\mathcal{W}_\ell}^{c}\kappa_U\mathsf{x})(p)| = e^{-\ell p_0\sin\phi}|\Psi_{\mathsf{x}}(b_\phi(p))|,$$

which gives (5.1) for such x.

Step 4: Lorentz-moved wedges. For a proper orthochronous $\mathcal{L}$, the subspace $\mathcal{K}_{\mathcal{L}\mathcal{W}_\ell} = \mathsf{U}(\mathcal{L})\mathcal{K}_{\mathcal{W}_\ell}$ is standard, with modular operator $\Delta_{\mathcal{L}\mathcal{W}_\ell} = \mathsf{U}(\mathcal{L})\Delta_{\mathcal{W}_\ell}\mathsf{U}(\mathcal{L})^*$. Let $U \subset B_T(0)$ with $D_T \subset \mathcal{L}\mathcal{W}_\ell$, let x be smooth with compact support in U, let u be the Klein–Gordon solution determined by x, and let $\mathsf{x}_\mathcal{L}$ be the time-zero pair of the solution $x \mapsto u(\mathcal{L}x)$. If $\mathbf{x}$ lies in the support of $\mathsf{x}_\mathcal{L}$, then $\mathcal{L}(0, \mathbf{x})$ lies in the support of u, whose points are, by finite propagation speed, causally related to points of $\{0\} \times \operatorname{supp}\mathsf{x} \subset D_T$; so $(0, \mathbf{x})$ is causally related to a point $q \in \mathcal{L}^{-1}D_T \subset \mathcal{W}_\ell$, and $x^1 \le q^1 + |q^0| < \ell$. Hence $\mathsf{x}_\mathcal{L}$ is smooth with compact support in a bounded open $U' \subset \{x^1 < \ell\}$, and real if x is real. By the Lorentz covariance of the free field, $\kappa_{U'}\mathsf{x}_\mathcal{L} = \mathsf{U}(\mathcal{L})^*\kappa_U\mathsf{x}$, that is, $\Psi_{\mathsf{x}_\mathcal{L}}(p) = \Psi_{\mathsf{x}}(\mathcal{L}p)$ on H_μ^+; so, for real x, $\kappa_U\mathsf{x} \in \mathsf{U}(\mathcal{L})\mathcal{K}_{U'} \subset \mathcal{K}_{\mathcal{L}\mathcal{W}_\ell}$, and by closure $\mathcal{K}_U \subset \mathcal{K}_{\mathcal{L}\mathcal{W}_\ell}$.

The functions $\Psi_{\mathsf{x}_\mathcal{L}}$ and $\Psi_{\mathsf{x}} \circ \mathcal{L}$ are holomorphic on the connected complex manifold $\mathcal{Q}_\mu$ and agree on H_μ^+, a totally real submanifold of maximal dimension, so they agree on $\mathcal{Q}_\mu$. Step 3 for $\mathsf{x}_\mathcal{L}$ therefore gives

$$|(\Delta_{\mathcal{W}_\ell}^{c}\mathsf{U}(\mathcal{L})^*\kappa_U\mathsf{x})(p)| = e^{-\ell p_0\sin\phi}|\Psi_{\mathsf{x}}(\mathcal{L}b_\phi(p))|,$$

and $\left\|\Delta_{\mathcal{L}\mathcal{W}_\ell}^{c}\kappa_U\mathsf{x}\right\| = \left\|\Delta_{\mathcal{W}_\ell}^{c}\mathsf{U}(\mathcal{L})^*\kappa_U\mathsf{x}\right\|$.

Step 5: all sources, by closure. Let $\mathsf{x} \in \mathcal{H}_{\mu,U}$, with U and $\mathcal{L}$ as in Step 4, or with U a bounded open subset of $\{x^1 < \ell\}$ and $\mathcal{L}$ the identity. The Tomita operator s of $\mathcal{K}_{\mathcal{W}_\ell}$ fixes every $\xi \in \mathcal{K}_{\mathcal{W}_\ell}$, so $\|\Delta^{1/2}_{\mathcal{W}_\ell}\xi\| = \|\mathsf{s}\xi\| = \|\xi\|$, and $\lambda^c \le 1 + \lambda^{1/2}$ for $\lambda \ge 0$ gives $\|\Delta^c_{\mathcal{W}_\ell}\xi\| \le 2\|\xi\|$. Choose smooth sources x_n with compact support in U and $\mathsf{x}_n \to \mathsf{x}$, and write $\mathsf{x}_n - \mathsf{x} = \mathsf{y}_n + i\mathsf{y}'_n$ with real $\mathsf{y}_n, \mathsf{y}'_n \to 0$. The vectors $\mathsf{U}(\mathcal{L})^*\kappa_U \mathsf{y}_n$ and $\mathsf{U}(\mathcal{L})^*\kappa_U \mathsf{y}'_n$ lie in $\mathcal{K}_{\mathcal{W}_\ell}$, since $\mathsf{U}(\mathcal{L})^*\mathcal{K}_U \subset \mathcal{K}_{\mathcal{W}_\ell}$, and have the norms of y_n and y'_n by (2.6); so, by the bound $\|\Delta^c_{\mathcal{W}_\ell}\xi\| \le 2\|\xi\|$, $\Delta^c_{\mathcal{W}_\ell}\mathsf{U}(\mathcal{L})^*\kappa_U\mathsf{x}_n$ tends to $\Delta^c_{\mathcal{W}_\ell}\mathsf{U}(\mathcal{L})^*\kappa_U\mathsf{x}$, and a subsequence converges almost everywhere. For every p, $\Psi_{\mathsf{x}_n}(\mathcal{L}b_\phi(p)) \to \Psi_{\mathsf{x}}(\mathcal{L}b_\phi(p))$ by the continuity shown at the beginning of Section 5, so the moduli of Steps 3 and 4 hold for x, and with them (5.1) for every $\mathsf{x} \in \mathcal{H}_{\mu,U}$, together with its form for $\mathcal{L}\mathcal{W}_\ell$, in which $\Psi_{\mathsf{x}}(\mathcal{L}b_\phi(p))$ replaces $\Psi_{\mathsf{x}}(b_\phi(p))$.

For $U \subset B_T(0)$ and a compact Lorentz family $\mathscr{L}$, a choice

$$\ell_{\mathscr{L}} > \sqrt{2}\sup_{\mathcal{L}\in\mathscr{L}} \left\|\mathcal{L}^{-1}\right\|_{\mathrm{E}}$$

ensures $D_T \subset \mathcal{L}\mathcal{W}_{\ell_{\mathscr{L}}T}$ for every $\mathcal{L} \in \mathscr{L}$: the double cone D_T lies in the Euclidean ball of radius T, since $(x^0)^2 + |\mathbf{x}|^2 \le (|x^0| + |\mathbf{x}|)^2 < T^2$ on D_T, so $\mathcal{L}^{-1}D_T$ lies in the ball of radius $\|\mathcal{L}^{-1}\|_{\mathrm{E}}\,T$, and every x with $|x|_{\mathrm{E}} \le r$ satisfies $x^1 + |x^0| \le \sqrt{2}\,((x^0)^2 + (x^1)^2)^{1/2} \le \sqrt{2}\,r$. For each $\mathcal{L} \in \mathscr{L}$, Step 4 therefore gives the inclusion $\mathcal{K}_U \subset \mathcal{K}_{\mathcal{L}\mathcal{W}_{\ell_{\mathscr{L}}T}}$ of standard subspaces that Lemma 2.4 requires, and Step 5 gives, for $\mathsf{x} \in \mathcal{H}_{\mu,U}$,

$$\int_{H^+_\mu} e^{-2\ell_{\mathscr{L}}Tp_0\sin\phi}|\Psi_{\mathsf{x}}(\mathcal{L}b_\phi(p))|^2\,d\mathfrak{m}_\mu(p) \overset{(5.1)}{=} \left\|\Delta^c_{\mathcal{L}\mathcal{W}_{\ell_{\mathscr{L}}T}}\kappa_U\mathsf{x}\right\|^2 \overset{\text{Lem. 2.4}}{\le} \|h_{c,U}\mathsf{x}\|^2 .$$

The integrand on the left is nonnegative and continuous in $(\mathcal{L}, p)$, so by Tonelli's theorem the left-hand side is a measurable function of $\mathcal{L}$. Integrating the bound of Lemma 2.4 against a probability measure ϑ on $\mathscr{L}$ gives

$$\int_{\mathscr{L}}\int_{H^+_\mu} e^{-2\ell_{\mathscr{L}}Tp_0\sin\phi}|\Psi_{\mathsf{x}}(\mathcal{L}b_\phi(p))|^2\,d\mathfrak{m}_\mu(p)\,d\vartheta(\mathcal{L}) \le \|h_{c,U}\mathsf{x}\|^2 . \tag{5.2}$$

Theorem 5.1 (Weighted complex-shell evaluation)**.** *For every $0 < c < 1/2$, there are finite constants $C_c, \lambda_c \ge 1$, depending only on c, such that for every $\mu, T > 0$, every nonempty bounded open $U \subset B_T(0)$, every $\mathsf{x} \in \mathcal{H}_{\mu,U}$, and every $\mathcal{E} \ge \mu$,*

$$|\Psi_{\mathsf{x}}(z)|^2 \le C_c\mathcal{E}^{-2}e^{2\lambda_c T\mathcal{E}}\,\|h_{c,U}\mathsf{x}\|^2 , \qquad z \in \mathcal{Q}_\mu, \quad |z|_{\mathrm{E}} \le 2\mathcal{E}. \tag{5.3}$$

Here $\Psi_{\mathsf{x}}(z) = \widehat{f}(\mathbf{z}) + iz_0\widehat{g}(\mathbf{z})$ for $\mathsf{x} = (f, g)$, and $h_{c,U}$ is the bounded operator of (2.10)*. The constants are independent of the shape and connectedness of U and of the dimension of any finite canonical source system in U. The choice*

$$\lambda_c = \frac{2048}{\sin(2\pi c)}$$

is admissible.

The averaged contour norm of (5.2) controls Ψ_{x} on an open shell neighbourhood (Lemma 5.2), and analytic discs reach the remaining evaluation points (Lemma 5.3).

Lemma 5.2 (Regular shell patches)**.** *For $0 < c < 1/2$ and $\phi = 2\pi c$, consider the compact normalized set*

$$\mathscr{Z} = \left\{(t, z) : 0 \le t \le 1,\ z \in \mathcal{Q}_t,\ |z|_{\mathrm{E}} \le 16,\ (\operatorname{Re} z)^2 \le -16\right\}. \tag{5.4}$$

There are a compact set $\mathscr{L}_c$ of proper orthochronous Lorentz transformations, a smooth probability density ϑ_c on $\mathscr{L}_c$ with respect to Haar measure, a neighbourhood $\mathcal{V}$ of $\mathscr{Z}$ in the shell family

$\{(t,z) : 0 \le t \le 1,\ z \in \mathcal{Q}_t\}$, and $\delta_c > 0$, such that the map $\Xi_t(\mathcal{L}, p) = \mathcal{L} b_\phi(p)$ on $\mathscr{L}_c \times H_t^+$ satisfies

$$(\Xi_t)_* \left(d\vartheta_c \otimes \mathbf{1}_{\{p_0 \le M_c\}}\, d\mathfrak{m}_t \right) \ge \delta_c\, d\,\mathrm{vol}_{\mathcal{Q}_t} \quad \text{on } \mathcal{V}_t, \qquad M_c = \frac{32}{\sin\phi}. \tag{5.5}$$

Here $d\vartheta_c$ is the probability measure with density ϑ_c, $d\,\mathrm{vol}_{\mathcal{Q}_t}$ is induced Euclidean six-dimensional volume, $\mathcal{V}_t = \{z : (t,z) \in \mathcal{V}\}$ is the slice of $\mathcal{V}$ at normalized mass t, and at $t = 0$ the shell H_0^+ and its measure $d\mathfrak{m}_0$ are given by the formulas of Section 2, so that H_0^+ is the forward light cone without its vertex. The family $\mathscr{L}_c$ can be chosen with

$$\sup_{\mathcal{L} \in \mathscr{L}_c} \left\| \mathcal{L}^{-1} \right\|_{\mathrm{E}} < \frac{32}{\sin\phi},$$

so that $\ell_c = 64/\sin\phi$ exceeds $\sqrt{2} \sup_{\mathcal{L} \in \mathscr{L}_c} \|\mathcal{L}^{-1}\|_{\mathrm{E}}$, as required of the containing-wedge parameter $\ell_{\mathscr{L}}$ for (5.2) with $\mathscr{L} = \mathscr{L}_c$. Moreover, there are $r > 0$ and $C_{\mathrm{sub}} \ge 1$ such that every $(t,z) \in \mathscr{Z}$ has a holomorphic shell chart of radius r: an injective holomorphic map $\psi_{t,z}$ from a ball of radius r in $\mathbb{C}^3$ into $\mathcal{Q}_t$ that maps the centre of the ball to z, whose image $\mathcal{B}_z$ lies in $\mathcal{V}_t$, and whose volume Jacobian with respect to Lebesgue measure lies between 1 and C_{sub}. In particular, $\mathrm{vol}(\mathcal{B}_z) \ge \mathrm{vol}(B_r)$, where B_r is a ball of radius r in $\mathbb{C}^3 = \mathbb{R}^6$, and every nonnegative plurisubharmonic function F on $\mathbb{C}^4$ satisfies

$$F(z) \le \frac{C_{\mathrm{sub}}}{\mathrm{vol}(\mathcal{B}_z)} \int_{\mathcal{B}_z} F\, d\,\mathrm{vol}_{\mathcal{Q}_t}.$$

Proof. Step 1: a preimage of every point of $\mathscr{Z}$. A point $z = u + iv \in \mathcal{Q}_t$ satisfies

$$u \cdot v = 0, \qquad u^2 - v^2 = t^2.$$

On $\mathscr{Z}$, writing $u^2 = -\rho_0^2$ with $\rho_0 > 0$, the Lorentzian length of u, gives $\rho_0 \in [4, 16]$ and $v^2 = -(t^2 + \rho_0^2)$: $\rho_0 \ge 4$ because $u^2 \le -16$, and $\rho_0^2 = |\mathbf{u}|^2 - u_0^2 \le |z|_{\mathrm{E}}^2 \le 256$, with $\mathbf{u}$ the spatial part of u.

The ordered pair u, v spans a negative-definite real two-dimensional subspace. This pair is a Lorentz image of the real and imaginary parts of $b_\phi(p)$ at

$$p_1 = p_3 = 0, \qquad p_0 = \frac{\sqrt{t^2 + \rho_0^2}}{\sin\phi}, \qquad p_2 = \sqrt{p_0^2 - t^2}.$$

Indeed, at this momentum the *reference pair*

$$\mathrm{Re}\, b_\phi(p) = (p_0 \cos\phi, 0, p_2, 0), \qquad \mathrm{Im}\, b_\phi(p) = (0, p_0 \sin\phi, 0, 0)$$

is orthogonal. Since $p_0^2 = t^2 + p_2^2$ and $p_0^2 \sin^2\phi = t^2 + \rho_0^2$, its Lorentzian squares are

$$(\mathrm{Re}\, b_\phi(p))^2 = p_0^2 \cos^2\phi - p_2^2 = t^2 \cos^2\phi - p_2^2 \sin^2\phi = t^2 - p_0^2 \sin^2\phi = -\rho_0^2$$

and $(\mathrm{Im}\, b_\phi(p))^2 = -p_0^2 \sin^2\phi = -(t^2 + \rho_0^2)$, the same as those of the *target pair* (u, v). Moreover, $p_2^2 = p_0^2 - t^2 \ge (t^2 + \rho_0^2) - t^2 = \rho_0^2$, so $|\mathbf{p}| = p_2 \ge 4$ for every $t \in [0,1]$: the preimages stay at distance at least 4 from the vertex of H_t^+.

The *source frame* $\mathcal{L}_{\mathrm{src}}$ is the proper orthochronous transformation whose columns are a future-directed unit timelike vector, the two unit vectors $\mathrm{Re}\, b_\phi(p)/\rho_0$ and $\mathrm{Im}\, b_\phi(p)/(t^2 + \rho_0^2)^{1/2}$, and a unit spacelike vector completing the orthonormal frame; the *target frame* $\mathcal{L}_{\mathrm{tar}}$ is built in the same way from u/ρ_0 and $v/(t^2 + \rho_0^2)^{1/2}$. The *matching transformation* $\mathcal{L} = \mathcal{L}_{\mathrm{tar}} \mathcal{L}_{\mathrm{src}}^{-1}$ is proper orthochronous and carries the reference pair to the target pair, so that $\Xi_t(\mathcal{L}, p) = z$.

Step 2: Ξ_t is a submersion at these preimages. The differential of Ξ_t at $(\mathcal{L}, p)$ sends $(X, \delta p)$, with X in the Lie algebra of the Lorentz group and δp tangent to H_t^+, to $Xz + \mathcal{L} b_\phi(\delta p)$. The Lorentz orbit of the ordered pair (u, v) has dimension five: the stabilizer of the pair acts on the orthogonal Lorentzian two-dimensional subspace and has dimension one, since an X with $Xu = Xv = 0$ maps that subspace into itself and is antisymmetric there. The vectors Xz therefore form a five-dimensional subspace of the tangent space of $\mathcal{Q}_t$ at z, on which the differential of the Lorentz invariant $(\mathrm{Re}\, z)^2$ vanishes.

The additional variation in p_2, with $p_1 = p_3 = 0$ and $p_0 = (t^2 + p_2^2)^{1/2}$, changes $(\operatorname{Re} b_\phi(p))^2 = t^2 \cos^2\phi - p_2^2 \sin^2\phi$ with nonzero derivative $-2p_2 \sin^2\phi$. The image of the differential thus contains that five-dimensional subspace and a vector on which the differential of $(\operatorname{Re} z)^2$ does not vanish, so the image has dimension six, the real dimension of $\mathcal{Q}_t$ at z; here z is a smooth point of $\mathcal{Q}_t$, because $d(z \cdot z) = 2z \cdot dz \neq 0$ for $z \neq 0$ and $|z|_{\mathrm{E}} \geq |\operatorname{Re} \mathbf{z}| \geq 4$ on $\mathscr{Z}$. The map Ξ_t is therefore a submersion at these preimages. Every preimage of a point of $\mathscr{Z}$ has $\mathbf{p}_\perp \neq 0$: for every $p \in H_t^+$, $(\operatorname{Re} b_\phi(p))^2 = t^2 \cos^2\phi - |\mathbf{p}_\perp|^2 \sin^2\phi$; the fold $u^2 = t^2 \cos^2\phi$, where $\mathbf{p}_\perp = 0$, is avoided on $\mathscr{Z}$ by $u^2 \leq -16$.

Step 3: bounds for the matching transformations and the preimage energies. For a Lorentz matrix $\mathcal{L}$, the Lorentz factor $\gamma_{\mathcal{L}}$ is the time component of the image of the time axis, and $\gamma_{\mathcal{L}}^2 - 1$ is the sum of the squared time components of the three spacelike columns, because $\mathcal{L}^T$ is a Lorentz matrix too, so that the first row of $\mathcal{L}$ has Lorentzian square 1. The Euclidean operator norm of a proper orthochronous $\mathcal{L}$, and of its inverse, is $\gamma_{\mathcal{L}} + (\gamma_{\mathcal{L}}^2 - 1)^{1/2}$, because $\mathcal{L} = R_1 B R_2$ with rotations R_1, R_2 of space and a boost B along x^1 of Lorentz factor $\gamma_{\mathcal{L}}$, whose Euclidean singular values are $\gamma_{\mathcal{L}} \pm (\gamma_{\mathcal{L}}^2 - 1)^{1/2}$, 1 and 1.

The target frame can be chosen with a spacelike complementary vector of zero time component: the Lorentzian complement of the target pair is two-dimensional and meets the hyperplane $\{x^0 = 0\}$ in a nonzero vector, which is spacelike. The Lorentz factor of the target frame obeys

$$\gamma_{\mathrm{tar}}^2 = 1 + \frac{u_0^2}{\rho_0^2} + \frac{v_0^2}{t^2 + \rho_0^2} \leq 16, \qquad \|\mathcal{L}_{\mathrm{tar}}\|_{\mathrm{E}} = \gamma_{\mathrm{tar}} + \sqrt{\gamma_{\mathrm{tar}}^2 - 1} \leq 4 + \sqrt{15} < 8,$$

since $|\mathbf{u}|^2 = u_0^2 + \rho_0^2$ and $|\mathbf{v}|^2 = v_0^2 + t^2 + \rho_0^2$ give $|z|_{\mathrm{E}}^2 = 2(u_0^2 + v_0^2) + 2\rho_0^2 + t^2 \leq 256$, whence $u_0^2 + v_0^2 \leq 128$, $\gamma_{\mathrm{tar}}^2 - 1 \leq (u_0^2 + v_0^2)/\rho_0^2 \leq 128/16 = 8$ and $\gamma_{\mathrm{tar}}^2 \leq 9 \leq 16$.

For the source frame, the Lorentzian complement of the reference pair is spanned by $\mathbf{e}_3 = (0, 0, 0, 1)$ and $(p_2, 0, p_0 \cos\phi, 0)$, whose Lorentzian square is $p_2^2 - p_0^2 \cos^2\phi = \rho_0^2$; the frame is completed by $(p_2, 0, p_0 \cos\phi, 0)/\rho_0$, future-directed since $p_2 > 0$, and by $\pm\mathbf{e}_3$. With this completion the time components of the three spacelike columns are $p_0 \cos\phi/\rho_0$, 0, and 0, so that

$$\gamma_{\mathrm{src}}^2 = 1 + \frac{p_0^2 \cos^2\phi}{\rho_0^2} = 1 + \cot^2\phi \left(1 + \frac{t^2}{\rho_0^2}\right) \leq 1 + \frac{17}{16} \cot^2\phi, \qquad \|\mathcal{L}_{\mathrm{src}}\|_{\mathrm{E}} \leq \frac{\sqrt{17}}{2 \sin\phi},$$

using $t \leq 1$, $\rho_0 \geq 4$, $\sqrt{1 + \cot^2\phi} = 1/\sin\phi$, and $\sqrt{1 + \frac{17}{16}\cot^2\phi} + \frac{\sqrt{17}}{4}|\cot\phi| \leq \frac{\sqrt{17}}{2}\sqrt{1 + \cot^2\phi}$. With $a = |\cot\phi|$, squaring reduces the last inequality to $\frac{\sqrt{17}}{2} a (1 + \frac{17}{16}a^2)^{1/2} \leq \frac{13}{4} + \frac{17}{8}a^2$, which holds because $(\frac{13}{4} + \frac{17}{8}a^2)^2 - \frac{17}{4}a^2(1 + \frac{17}{16}a^2) = \frac{169 + 153a^2}{16} > 0$.

Consequently the matching transformation satisfies

$$\|\mathcal{L}\|_{\mathrm{E}} = \|\mathcal{L}^{-1}\|_{\mathrm{E}} \leq \|\mathcal{L}_{\mathrm{tar}}\|_{\mathrm{E}} \, \|\mathcal{L}_{\mathrm{src}}\|_{\mathrm{E}} < \frac{4\sqrt{17}}{\sin\phi} < \frac{32}{\sin\phi}, \qquad p_0 = \frac{\sqrt{t^2 + \rho_0^2}}{\sin\phi} \leq \frac{\sqrt{257}}{\sin\phi} < \frac{32}{\sin\phi}.$$

Step 4: the shell family and the parameter family over the normalized mass. Let $\mathcal{I} \supset [0, 1]$ be an open interval. The set

$$\mathcal{Q} = \{(t, z) \in \mathcal{I} \times \mathbb{C}^4 : z \cdot z = t^2, \ z \neq 0\}$$

is a real-analytic manifold of dimension seven, because $d(z \cdot z) = 2z \cdot dz$ maps onto $\mathbb{C}$ at $z \neq 0$. For the same reason the restriction of t to $\mathcal{Q}$ is a submersion: for every δt, the equation $2z \cdot \delta z = 2t\,\delta t$ has a solution δz. Near each of its points, $\mathcal{Q}$ is therefore the product of an interval of values of t and a slice, and the slices are the shells $\mathcal{Q}_t$ without 0. In this sense, on the compact normalized region $\mathscr{Z}$, which lies in $\mathcal{Q}$ because $|z|_{\mathrm{E}} \geq 4$ there, $t = \mu/\mathcal{E} = 0$ is a regular parameter of the shell family.

The parameter family $\mathcal{P}$ consists of the triples $(t, \mathcal{L}, \mathbf{p})$ with $t \in \mathcal{I}$, $\mathcal{L}$ proper orthochronous and $\mathbf{p} \in \mathbb{R}^3 \setminus \{0\}$, with $p = ((t^2 + |\mathbf{p}|^2)^{1/2}, \mathbf{p}) \in H_t^+$; this parametrization is real-analytic in $(t, \mathbf{p})$, also at $t = 0$, and $d\mathfrak{m}_t = d^3\mathbf{p}/(2(2\pi)^3(t^2 + |\mathbf{p}|^2)^{1/2})$ depends continuously on t. The map

$\Xi(t,\mathcal{L},\mathbf{p}) = (t, \Xi_t(\mathcal{L},p))$ sends $\mathcal{P}$ into $\mathcal{Q}$. At the preimage $(t_*,\mathcal{L}_*,\mathbf{p}_*)$ of a point $(t_*,z_*)\in\mathscr{Z}$ found in Step 1, the differential of Ξ is onto: its t-component is the identity, and at fixed t it is the differential of Ξ_t, onto by Step 2. By the submersion theorem there are coordinates (t,y), $y\in\mathbb{R}^6$, on $\mathcal{Q}$ near (t_*,z_*) and (t,y,s), $s\in\mathbb{R}^3$, on $\mathcal{P}$ near $(t_*,\mathcal{L}_*,\mathbf{p}_*)$ in which $\Xi(t,y,s)=(t,y)$: complete t to coordinates (t,y) on $\mathcal{Q}$, and $(t, y\circ\Xi)$ by three functions s to coordinates on $\mathcal{P}$.

Step 5: local pushforward densities, continuous in t. The inequalities of Step 3 are strict, so the point $(t_*,\mathcal{L}_*,\mathbf{p}_*)$ has an open neighbourhood N whose closure is compact, lies in the domain of the coordinates (t,y,s) of Step 4, and carries $p_0 < 32/\sin\phi$ and $\|\mathcal{L}^{-1}\|_{\mathrm{E}} < 32/\sin\phi$. Let χ_N be a continuous function with values in $[0,1]$ and compact support in N, positive at $(t_*,\mathcal{L}_*,\mathbf{p}_*)$. With $d\mathcal{L}$ for Haar measure, the measure $\chi_N\, d\mathcal{L}\, d\mathfrak{m}_t$ on the slice of $\mathcal{P}$ at t is $m(t,y,s)\,dy\,ds$, with m continuous and compactly supported. Fubini's theorem, which is the coarea formula in these coordinates, shows that $(\Xi_t)_*(\chi_N\, d\mathcal{L}\, d\mathfrak{m}_t)$ is $\big(\int m(t,y,s)\,ds\big)\,dy$. Since $d\,\mathrm{vol}_{\mathcal{Q}_t}$ has a positive density with respect to dy that is continuous in (t,y), the pushforward has a density with respect to $d\,\mathrm{vol}_{\mathcal{Q}_t}$ that is continuous in (t,y), by dominated convergence, and positive at (t_*,z_*), because $m>0$ at $(t_*,\mathcal{L}_*,\mathbf{p}_*)$. Hence there are an open neighbourhood O_N of (t_*,z_*) in $\mathcal{Q}$ and $\delta_N>0$ such that this density is at least δ_N on O_N, for all values of t that O_N contains, $t=0$ included.

Step 6: a finite cover and one Lorentz density, which give (5.5). The set $\mathscr{Z}$ is compact, so finitely many of the neighbourhoods O_N, say $O_{N_1},\dots,O_{N_n}$, cover $\mathscr{Z}$. We put $\mathcal{V} = (O_{N_1}\cup\cdots\cup O_{N_n})\cap\{0\le t\le 1\}$. The projections of the closures of $N_1,\dots,N_n$ to the Lorentz group form a compact set $\mathscr{L}'$ inside the open set $\{\mathcal{L} : \|\mathcal{L}^{-1}\|_{\mathrm{E}} < 32/\sin\phi\}$. A smooth function $\widetilde{\vartheta}\ge 0$ on the Lorentz group, positive on $\mathscr{L}'$, whose support $\mathscr{L}_c$ is a compact subset of that open set, gives the density $\vartheta_c = \widetilde{\vartheta}/\int\widetilde{\vartheta}\,d\mathcal{L}$, and $\sup_{\mathcal{L}\in\mathscr{L}_c}\|\mathcal{L}^{-1}\|_{\mathrm{E}} < 32/\sin\phi$.

Since $p_0 < M_c$ on each N_i and $0\le\chi_{N_i}\le 1$,

$$\vartheta_c(\mathcal{L})\mathbf{1}_{\{p_0\le M_c\}} \ge \Big(\min_{\mathscr{L}'}\vartheta_c\Big)\chi_{N_i}(t,\mathcal{L},\mathbf{p}), \qquad i=1,\dots,n.$$

Pushforward preserves the order of positive measures, so by Step 5, on the slice $\{z : (t,z)\in O_{N_i}\}$,

$$(\Xi_t)_*\Big(d\vartheta_c\otimes\mathbf{1}_{\{p_0\le M_c\}}\,d\mathfrak{m}_t\Big) \ge \Big(\min_{\mathscr{L}'}\vartheta_c\Big)\delta_{N_i}\,d\,\mathrm{vol}_{\mathcal{Q}_t}.$$

These slices cover $\mathcal{V}_t$, which gives (5.5) with $\delta_c = (\min_{\mathscr{L}'}\vartheta_c)\min_i\delta_{N_i} > 0$. All choices depend only on $\phi = 2\pi c$.

Step 7: holomorphic shell charts of a common radius. Let $(t_*,z_*)\in\mathscr{Z}$. Since $|\mathrm{Re}\,\mathbf{z}_*|\ge 4$, some spatial coordinate satisfies $|z_{*k}|\ge 4/\sqrt{3}$, so $\partial_{z_k}(z\cdot z) = -2z_k \ne 0$ near z_*. The holomorphic implicit function theorem, applied to $z\cdot z=\tau$ with $\tau=t^2$ as an additional complex variable, writes $\mathcal{Q}_t$ near z_* as a graph $z_k = g_t(w)$ over the other three coordinates $w\in\mathbb{C}^3$, with g_t holomorphic. For all $(t,z)\in\mathcal{Q}$ near (t_*,z_*), the graph is defined, with $|\nabla g_t|$ bounded, on a neighbourhood of the closed ball of one fixed radius about the point w formed by those three coordinates of z. Finitely many such neighbourhoods of points (t_*,z_*) cover the compact set $\mathscr{Z}$, which gives one radius and one bound for $|\nabla g_t|$; a smaller radius $r>0$ also keeps the image of the chart about every $(t,z)\in\mathscr{Z}$ inside the slice $\mathcal{V}_t$, because the charts are uniformly Lipschitz and, $\mathscr{Z}$ being compact and $\mathcal{V}$ open in the closed shell family, all points of the shell family within some positive distance of $\mathscr{Z}$ lie in $\mathcal{V}$.

For $(t,z)\in\mathscr{Z}$ the chart about z is the injective holomorphic map $\psi_{t,z}(w') = (w', g_t(w'))$, with the coordinates in the order of z, on the ball $B_r(w)$, with $\psi_{t,z}(w)=z$ and image $\mathcal{B}_z\subset\mathcal{V}_t$. The coordinate projection $z\mapsto w$ is a Euclidean contraction and inverts $\psi_{t,z}$, so the volume Jacobian $J_{t,z}$ of $\psi_{t,z}$ with respect to Lebesgue measure on $B_r(w)\subset\mathbb{C}^3=\mathbb{R}^6$ satisfies $J_{t,z}\ge 1$; the bound for $|\nabla g_t|$ gives $J_{t,z}\le C_{\mathrm{sub}}$ for a constant C_{sub}. Hence $\mathrm{vol}(B_r)\le\mathrm{vol}(\mathcal{B}_z)\le C_{\mathrm{sub}}\,\mathrm{vol}(B_r)$. For a nonnegative plurisubharmonic F on $\mathbb{C}^4$, the function $F\circ\psi_{t,z}$ is plurisubharmonic, hence subharmonic, near the

closed ball $\overline{B_r(w)}$, and

$$F(z) \overset{\text{submean}}{\leq} \frac{1}{\operatorname{vol}(B_r)} \int_{B_r(w)} F \circ \psi_{t,z} \overset{J_{t,z}\geq 1}{\leq} \frac{1}{\operatorname{vol}(B_r)} \int_{\mathcal{B}_z} F \, d\operatorname{vol}_{\mathcal{Q}_t}$$
$$\overset{J_{t,z}\leq C_{\text{sub}}}{\leq} \frac{C_{\text{sub}}}{\operatorname{vol}(\mathcal{B}_z)} \int_{\mathcal{B}_z} F \, d\operatorname{vol}_{\mathcal{Q}_t}.$$

Finally, the bound on $\mathscr{L}_c$ in Step 6 and

$$\ell_c = \frac{64}{\sin\phi} > \sqrt{2}\,\frac{32}{\sin\phi}$$

give the stated wedge containment. □

Lemma 5.3 (Null analytic discs)**.** *If $z \in \mathcal{Q}_\mu$, $\mathcal{E} \geq \mu$, and $|z|_{\mathrm{E}} \leq 2\mathcal{E}$, there is an analytic disc $\zeta \mapsto z(\zeta)$ in $\mathcal{Q}_\mu$, $|\zeta| \leq 1$, with $z(0) = z$, such that every boundary point $z(\zeta)$, $|\zeta| = 1$, satisfies*

$$|\operatorname{Re}\mathbf{z}(\zeta)| \geq 6\mathcal{E}, \qquad |\operatorname{Re} z_0(\zeta)| \leq 2\mathcal{E}, \qquad (\operatorname{Re} z(\zeta))^2 \leq -32\mathcal{E}^2, \qquad |z(\zeta)|_{\mathrm{E}} < 14\mathcal{E}.$$

In particular, $(\mu/\mathcal{E}, z(\zeta)/\mathcal{E})$ belongs to the set $\mathscr{Z}$ of (5.4) for $|\zeta| = 1$.

Proof. There is a nonzero $\mathbf{n} \in \mathbb{C}^3$ such that

$$\mathbf{z} \cdot \mathbf{n} = 0, \qquad \mathbf{n} \cdot \mathbf{n} = 0.$$

The first relation defines a complex subspace of dimension at least two, and a quadratic form on a two-dimensional complex space has a nonzero isotropic vector. Since

$$\mathbf{n} \cdot \mathbf{n} = |\operatorname{Re}\mathbf{n}|^2 - |\operatorname{Im}\mathbf{n}|^2 + 2i \operatorname{Re}\mathbf{n} \cdot \operatorname{Im}\mathbf{n} = 0,$$

the real and imaginary parts of $\mathbf{n}$ are orthogonal and of equal positive length. After rescaling by a positive factor,

$$|\operatorname{Re}\mathbf{n}| = |\operatorname{Im}\mathbf{n}| = 1, \qquad \operatorname{Re}\mathbf{n} \cdot \operatorname{Im}\mathbf{n} = 0.$$

Hence $|\operatorname{Re}(e^{iu}\mathbf{n})|^2 = |\cos u \operatorname{Re}\mathbf{n} - \sin u \operatorname{Im}\mathbf{n}|^2 = \cos^2 u + \sin^2 u = 1$ for every real u.

The disc

$$z(\zeta) = z + 8\mathcal{E}\zeta(0, \mathbf{n}), \qquad |\zeta| \leq 1,$$

lies on $\mathcal{Q}_\mu$, because

$$z(\zeta) \cdot z(\zeta) = z \cdot z - 16\mathcal{E}\zeta\, \mathbf{z} \cdot \mathbf{n} - 64\mathcal{E}^2\zeta^2\, \mathbf{n} \cdot \mathbf{n} = \mu^2.$$

On its boundary $\zeta = e^{iu}$, the time component is unchanged, $z_0(\zeta) = z_0$, and $|\operatorname{Re}\mathbf{z}|$, $|\operatorname{Re} z_0|$ are at most $|z|_{\mathrm{E}} \leq 2\mathcal{E}$; hence

$$|\operatorname{Re}\mathbf{z}(\zeta)| \geq 8\mathcal{E}|\operatorname{Re}(e^{iu}\mathbf{n})| - |\operatorname{Re}\mathbf{z}| \geq 8\mathcal{E} - 2\mathcal{E} = 6\mathcal{E}, \qquad |\operatorname{Re} z_0(\zeta)| \leq 2\mathcal{E}.$$

These two inequalities imply the Lorentzian bound

$$(\operatorname{Re} z(\zeta))^2 = (\operatorname{Re} z_0(\zeta))^2 - |\operatorname{Re}\mathbf{z}(\zeta)|^2 \leq 4\mathcal{E}^2 - 36\mathcal{E}^2 = -32\mathcal{E}^2,$$

while $|(0, \mathbf{n})|_{\mathrm{E}} = \sqrt{2}$ and $8\sqrt{2} < 12$ give

$$|z(\zeta)|_{\mathrm{E}} \leq |z|_{\mathrm{E}} + 8\mathcal{E}|(0, \mathbf{n})|_{\mathrm{E}} \leq (2 + 8\sqrt{2})\mathcal{E} < 14\mathcal{E}.$$

The normalized boundary points lie in the set $\mathscr{Z}$ of (5.4): $t = \mu/\mathcal{E} \in (0, 1]$ because $\mathcal{E} \geq \mu$, and $w = z(\zeta)/\mathcal{E}$ satisfies $w \cdot w = t^2$, $|w|_{\mathrm{E}} < 14 \leq 16$ and $(\operatorname{Re} w)^2 \leq -32 \leq -16$. □

Proof of Theorem 5.1. Step 1: the bound at the normalized points of $\mathscr{Z}$, for smooth sources. Let x be smooth with compact support in U, and write $\mu = t\mathcal{E}$, so that $t \in (0, 1]$. The substitution $p = \mathcal{E}p'$ maps H_t^+ onto H_μ^+, and since $p_0 = \mathcal{E}p_0'$ and $d^3\mathbf{p} = \mathcal{E}^3 d^3\mathbf{p}'$, the invariant measure scales as

$$d\mathfrak{m}_\mu(\mathcal{E}p') = \mathcal{E}^2 d\mathfrak{m}_t(p').$$

For $(t,z)\in\mathscr{Z}$, let $\mathcal{B}_z\subset\mathcal{V}_t$ be the image of the holomorphic shell chart about z of Lemma 5.2. The function $F=|\Psi_{\mathsf{x}}(\mathcal{E}\,\cdot)|^2$ is plurisubharmonic on $\mathbb{C}^4$, as the squared modulus of an entire function. Then

$$\begin{aligned}
|\Psi_{\mathsf{x}}(\mathcal{E}z)|^2 &\overset{\text{Lem. 5.2}}{\leq} \frac{C_{\mathrm{sub}}}{\operatorname{vol}(\mathcal{B}_z)}\int_{\mathcal{B}_z}|\Psi_{\mathsf{x}}(\mathcal{E}z')|^2\,d\operatorname{vol}_{\mathcal{Q}_t}(z')\\
&\overset{(5.5)}{\leq} \frac{C_{\mathrm{sub}}}{\delta_c\operatorname{vol}(\mathcal{B}_z)}\int_{\mathscr{L}_c}\int_{p_0'\leq M_c}|\Psi_{\mathsf{x}}(\mathcal{E}\mathcal{L}b_\phi(p'))|^2\,d\mathfrak{m}_t(p')\,d\vartheta_c(\mathcal{L})\\
&\overset{p=\mathcal{E}p'}{=} \frac{C_{\mathrm{sub}}\mathcal{E}^{-2}}{\delta_c\operatorname{vol}(\mathcal{B}_z)}\int_{\mathscr{L}_c}\int_{p_0\leq\mathcal{E}M_c}|\Psi_{\mathsf{x}}(\mathcal{L}b_\phi(p))|^2\,d\mathfrak{m}_\mu(p)\,d\vartheta_c(\mathcal{L})\\
&\overset{p_0\leq\mathcal{E}M_c}{\leq} \frac{C_{\mathrm{sub}}\mathcal{E}^{-2}e^{2\ell_cM_cT\mathcal{E}\sin\phi}}{\delta_c\operatorname{vol}(\mathcal{B}_z)}\int_{\mathscr{L}_c}\int_{H_\mu^+}e^{-2\ell_cTp_0\sin\phi}|\Psi_{\mathsf{x}}(\mathcal{L}b_\phi(p))|^2\,d\mathfrak{m}_\mu(p)\,d\vartheta_c(\mathcal{L})\\
&\overset{(5.2)}{\leq} C_c\mathcal{E}^{-2}e^{2\ell_cM_cT\mathcal{E}\sin\phi}\,\|h_{c,U}\mathsf{x}\|^2\,.
\end{aligned}$$

The second inequality applies (5.5) on $\mathcal{B}_z\subset\mathcal{V}_t$ to the function equal to F on $\mathcal{B}_z$ and to 0 elsewhere, which is at most F because $F\geq 0$:

$$\int_{\mathcal{B}_z}F\,d\operatorname{vol}_{\mathcal{Q}_t}\leq\frac{1}{\delta_c}\int_{\mathscr{L}_c}\int_{H_t^+}\mathbf{1}_{\{p_0'\leq M_c\}}F(\mathcal{L}b_\phi(p'))\,d\mathfrak{m}_t(p')\,d\vartheta_c(\mathcal{L}),$$

and $F(\mathcal{L}b_\phi(p'))=|\Psi_{\mathsf{x}}(\mathcal{E}\mathcal{L}b_\phi(p'))|^2$. The equality substitutes $p=\mathcal{E}p'$, with $\mathcal{E}\mathcal{L}b_\phi(p')=\mathcal{L}b_\phi(\mathcal{E}p')$ because $\mathcal{L}$ and b_ϕ are linear. The third inequality inserts the factor $e^{2\ell_cT(\mathcal{E}M_c-p_0)\sin\phi}\geq 1$ on $\{p_0\leq\mathcal{E}M_c\}$ and then integrates over all of H_μ^+. The last inequality is (5.2) for $\mathscr{L}=\mathscr{L}_c$, $\vartheta=\vartheta_c$ and $\ell_{\mathscr{L}}=\ell_c$, which Lemma 5.2 permits, with

$$C_c=\max\Big\{1,\frac{C_{\mathrm{sub}}}{\delta_c\inf_{(t,z)\in\mathscr{Z}}\operatorname{vol}(\mathcal{B}_z)}\Big\},$$

which is finite because $\operatorname{vol}(\mathcal{B}_z)\geq\operatorname{vol}(B_r)>0$ for every $(t,z)\in\mathscr{Z}$ (Lemma 5.2).

The values of ℓ_c and M_c in Lemma 5.2 give

$$\ell_cM_c\sin\phi=\frac{64\cdot 32}{\sin\phi}=\frac{2048}{\sin(2\pi c)}=\lambda_c,$$

so that $|\Psi_{\mathsf{x}}(\mathcal{E}z)|^2\leq C_c\mathcal{E}^{-2}e^{2\lambda_cT\mathcal{E}}\,\|h_{c,U}\mathsf{x}\|^2$ for every $(t,z)\in\mathscr{Z}$.

Step 2: every point of the statement, through analytic discs. Let $z\in\mathcal{Q}_\mu$ with $|z|_{\mathrm{E}}\leq 2\mathcal{E}$, and let $\zeta\mapsto z(\zeta)$ be the analytic disc of Lemma 5.3. The function $\zeta\mapsto\Psi_{\mathsf{x}}(z(\zeta))$ is holomorphic near the closed unit disc, so Cauchy's mean formula gives

$$|\Psi_{\mathsf{x}}(z)|^2=\Big|\frac{1}{2\pi}\int_0^{2\pi}\Psi_{\mathsf{x}}(z(e^{iu}))\,du\Big|^2\leq\max_{0\leq u\leq 2\pi}|\Psi_{\mathsf{x}}(z(e^{iu}))|^2.$$

By Lemma 5.3, $z(e^{iu})=\mathcal{E}w_u$ with $(\mu/\mathcal{E},w_u)\in\mathscr{Z}$, so Step 1, at the same $\mathcal{E}$ and $t=\mu/\mathcal{E}$, bounds every $|\Psi_{\mathsf{x}}(z(e^{iu}))|^2$, and hence $|\Psi_{\mathsf{x}}(z)|^2$, by $C_c\mathcal{E}^{-2}e^{2\lambda_cT\mathcal{E}}\,\|h_{c,U}\mathsf{x}\|^2$, which is (5.3) for smooth sources, with the same constant.

Step 3: all sources, by density. For each z, the map $\mathsf{x}\mapsto\Psi_{\mathsf{x}}(z)$ is continuous on $\mathcal{H}_{\mu,U}$ (beginning of Section 5), $h_{c,U}$ is bounded, and smooth sources are dense in $\mathcal{H}_{\mu,U}$ by its definition. Both sides of (5.3) are therefore continuous in x, and density extends the estimate from smooth sources to $\mathcal{H}_{\mu,U}$. The set U enters only through $U\subset B_T(0)$ in (5.2), and C_c, λ_c come from Lemma 5.2 and ℓ_c alone. □

5.2. **Weighted polynomial moments.** For $1/2 \le b < 1$, we fix

$$\Lambda_b = \frac{2^{15}}{\sin(\pi b)} = 16\max\{1, \lambda_{(1-b)/2}, \lambda_{b/2}\}, \qquad B_b = 5\Lambda_b. \tag{5.6}$$

The two expressions for Λ_b agree because, for the admissible choice of λ_c in Theorem 5.1, $\lambda_{(1-b)/2} = \lambda_{b/2} = 2048/\sin(\pi b)$.

Theorem 5.1 applies to sources in a ball $B_T(\mathbf{x}_0)$ after the spatial translation by $-\mathbf{x}_0$, which preserves the symplectic form, the covariance, and hence $h_{c,U}$. For a multi-index $\alpha \in \mathbb{N}^3$, define

$$\ell^f_\alpha(f,g) = \int \mathbf{x}^\alpha f(\mathbf{x})\, d^3x, \qquad \ell^g_\alpha(f,g) = \int \mathbf{x}^\alpha g(\mathbf{x})\, d^3x.$$

We also write $\ell^f_{\mathbf{z}}(f,g) = \widehat{f}(\mathbf{z})$ and $\ell^g_{\mathbf{z}}(f,g) = \widehat{g}(\mathbf{z})$.

Lemma 5.4 (Weighted moments)**.** *Let $1/2 \le b < 1$, with Λ_b, B_b as in* (5.6)*. There is a finite M_b, depending only on b, such that for all $\mu, T > 0$, $\mathbf{x}_0 \in \mathbb{R}^3$, every nonempty bounded open $U \subset B_T(\mathbf{x}_0)$ and every $\alpha \in \mathbb{N}^3$, the moments $\ell^f_\alpha, \ell^g_\alpha$, taken in coordinates centred at $\mathbf{x}_0$, satisfy, with the uncertainty reserve R_U of* (2.9) *and in the sense of Convention 2.5,*

$$\left\| \ell^f_\alpha \mathsf{R}_U^{-b} \right\| \le M_b \frac{T}{1+\mu T} e^{\Lambda_b \mu T} (B_b T)^{|\alpha|}, \tag{5.7}$$

$$\left\| \ell^g_\alpha \mathsf{R}_U^{-b} \right\| \le M_b \frac{T^2}{(1+\mu T)^2} e^{\Lambda_b \mu T} (B_b T)^{|\alpha|}. \tag{5.8}$$

Proof. Step 1: evaluation on the complex shell, bounded by the reserve. Throughout the proof we use coordinates centred at $\mathbf{x}_0$. The spectral measure of G_U is split at 1: P_1 denotes the spectral projection of G_U for $[0,1]$, and $P_2 = I - P_1$. By (2.10) and (2.9), in the joint spectral calculus of the commuting operators $G_U, \overline{G}_U$, $h_{c,U} = G_U^{1/2-c}\overline{G}_U^{\,c}$ and $\mathsf{R}_U^b = G_U^{b/2}\overline{G}_U^{\,b/2}$.

On the range of P_1, where $G_U \le I$ and $\overline{G}_U = 2I - G_U \ge I$, we choose $c = (1-b)/2$; on the range of P_2, where $G_U \ge I$, we choose $c = b/2$. Then

$$h_{(1-b)/2,U} P_1 = G_U^{b/2}\, \overline{G}_U^{(1-b)/2} P_1 = \overline{G}_U^{1/2-b} P_1\, \mathsf{R}_U^b,$$
$$h_{b/2,U} P_2 = G_U^{(1-b)/2}\, \overline{G}_U^{b/2} P_2 = G_U^{1/2-b} P_2\, \mathsf{R}_U^b,$$

and the operators $\overline{G}_U^{1/2-b} P_1$ and $G_U^{1/2-b} P_2$ are contractions because $1/2 - b \le 0$. Both values of c lie in $(0, 1/2)$, as Theorem 5.1 requires, because $1/2 \le b < 1$.

Let $\mathsf{x} \in \mathcal{H}_{\mu,U}$, $\mathcal{E} \ge \mu$, and $z \in \mathcal{Q}_\mu$ with $|z|_{\mathrm{E}} \le 2\mathcal{E}$. Since $\mathsf{x} = P_1\mathsf{x} + P_2\mathsf{x}$ and $\Psi_{\mathsf{x}}(z)$ is linear in x, Theorem 5.1 at $c = (1-b)/2$ for $P_1\mathsf{x}$ and at $c = b/2$ for $P_2\mathsf{x}$ gives

$$\begin{aligned} |\Psi_{\mathsf{x}}(z)| &\le |\Psi_{P_1\mathsf{x}}(z)| + |\Psi_{P_2\mathsf{x}}(z)| \\ &\overset{(5.3)}{\le} \mathcal{E}^{-1}\Big(C^{1/2}_{(1-b)/2} e^{\lambda_{(1-b)/2} T\mathcal{E}} \left\| h_{(1-b)/2,U} P_1 \mathsf{x} \right\| + C^{1/2}_{b/2} e^{\lambda_{b/2} T\mathcal{E}} \left\| h_{b/2,U} P_2 \mathsf{x} \right\| \Big) \\ &\le (C^{1/2}_{(1-b)/2} + C^{1/2}_{b/2}) \mathcal{E}^{-1} e^{\max\{\lambda_{(1-b)/2}, \lambda_{b/2}\} T\mathcal{E}} \left\| \mathsf{R}_U^b \mathsf{x} \right\|, \end{aligned}$$

where the last step uses $\|h_{(1-b)/2,U} P_1 \mathsf{x}\| \le \|\mathsf{R}_U^b \mathsf{x}\|$ and $\|h_{b/2,U} P_2 \mathsf{x}\| \le \|\mathsf{R}_U^b \mathsf{x}\|$, which follow from the two contractions. This bound reads, for every $\mathcal{E} \ge \mu$,

$$\left\| \Psi_{(\cdot)}(z) \mathsf{R}_U^{-b} \right\| \le D_b^{\mathrm{ev}} \mathcal{E}^{-1} e^{\lambda_b^{\mathrm{ev}} T\mathcal{E}}, \qquad z \in \mathcal{Q}_\mu, \quad |z|_{\mathrm{E}} \le 2\mathcal{E}, \tag{5.9}$$

where $\lambda_b^{\mathrm{ev}} = \max\{\lambda_{(1-b)/2}, \lambda_{b/2}\}$ and $D_b^{\mathrm{ev}} = C^{1/2}_{(1-b)/2} + C^{1/2}_{b/2}$ is finite.

Step 2: the bound on $\widehat{f}$. For $\mathbf{z} \in \mathbb{C}^3$, with Euclidean norm $|\mathbf{z}|$, we take

$$\mathcal{E} = \mu + \frac{1}{\Lambda_b T} + |\mathbf{z}|. \tag{5.10}$$

Then $\mathcal{E} \geq \mu$ and $\mu + |\mathbf{z}| < \mathcal{E}$. Let $\mathsf{x} = (f, g) \in \mathcal{H}_{\mu,U}$. The two shell points $z_\pm = (\pm\sqrt{\mu^2 + \mathbf{z}\cdot\mathbf{z}}, \mathbf{z})$ lie on $\mathcal{Q}_\mu$, and $\Psi_{\mathsf{x}}(z_\pm) = \widehat{f}(\mathbf{z}) \pm i\sqrt{\mu^2 + \mathbf{z}\cdot\mathbf{z}}\,\widehat{g}(\mathbf{z})$. Hence

$$\widehat{f}(\mathbf{z}) = \frac{\Psi_{\mathsf{x}}(z_+) + \Psi_{\mathsf{x}}(z_-)}{2}, \qquad \Psi_{\mathsf{x}}(z_+) - \Psi_{\mathsf{x}}(z_-) = 2i\sqrt{\mu^2 + \mathbf{z}\cdot\mathbf{z}}\,\widehat{g}(\mathbf{z}).$$

Since $|\mathbf{z}\cdot\mathbf{z}| \leq |\mathbf{z}|^2$,

$$|z_\pm|_{\mathrm{E}}^2 = |\mu^2 + \mathbf{z}\cdot\mathbf{z}| + |\mathbf{z}|^2 \leq \mu^2 + 2|\mathbf{z}|^2 \leq 2(\mu + |\mathbf{z}|)^2 < 2\mathcal{E}^2,$$

so $|z_\pm|_{\mathrm{E}} < \sqrt{2}\,\mathcal{E} \leq 2\mathcal{E}$, and (5.9) at the energy $\mathcal{E}$ applies at $z_\pm$:

$$|\widehat{f}(\mathbf{z})| \leq \frac{|\Psi_{\mathsf{x}}(z_+)| + |\Psi_{\mathsf{x}}(z_-)|}{2} \overset{(5.9)}{\leq} D_b^{\mathrm{ev}}\mathcal{E}^{-1}e^{\lambda_b^{\mathrm{ev}}T\mathcal{E}}\left\|\mathsf{R}_U^b\mathsf{x}\right\|.$$

Step 3: the bound on $\widehat{g}$ across zero energy. The difference in Step 2 determines $\widehat{g}(\mathbf{z})$ only at nonzero complex energy $\sqrt{\mu^2 + \mathbf{z}\cdot\mathbf{z}}$. To obtain a bound across zero energy, z_1 is varied on the circle $z_1 + 8\mathcal{E}e^{iu}$, $u \in \mathbb{R}$, with $\mathcal{E}$ fixed by (5.10) at $\mathbf{z}$. The point $\mathbf{z}_u = (z_1 + 8\mathcal{E}e^{iu}, z_2, z_3)$ satisfies

$$\mu^2 + \mathbf{z}_u\cdot\mathbf{z}_u = 64\mathcal{E}^2e^{2iu} + 16\mathcal{E}e^{iu}z_1 + (\mu^2 + \mathbf{z}\cdot\mathbf{z}),$$

where $|z_1| \leq |\mathbf{z}| < \mathcal{E}$ and $|\mu^2 + \mathbf{z}\cdot\mathbf{z}| \leq \mu^2 + |\mathbf{z}|^2 < \mathcal{E}^2$. On this circle, therefore,

$$\begin{aligned}\left|\mu^2 + (z_1 + 8\mathcal{E}e^{iu})^2 + z_2^2 + z_3^2\right| &\geq 64\mathcal{E}^2 - 16\mathcal{E}|z_1| - |\mu^2 + \mathbf{z}\cdot\mathbf{z}| \\ &\geq 64\mathcal{E}^2 - 16\mathcal{E}^2 - \mathcal{E}^2 = 47\mathcal{E}^2.\end{aligned}$$

The two shell points $w_\pm = (\pm\sqrt{\mu^2 + \mathbf{z}_u\cdot\mathbf{z}_u}, \mathbf{z}_u)$ over $\mathbf{z}_u$ satisfy, by the bound on $|z_\pm|_{\mathrm{E}}$ in Step 2 and $|\mathbf{z}_u| \leq |\mathbf{z}| + 8\mathcal{E}$,

$$\begin{aligned}|w_\pm|_{\mathrm{E}}^2 &\leq \mu^2 + 2|\mathbf{z}_u|^2 \leq 2(\mu + |\mathbf{z}_u|)^2 \leq 2(\mu + |\mathbf{z}| + 8\mathcal{E})^2 \\ &< 2(9\mathcal{E})^2 = 162\mathcal{E}^2 < (14\mathcal{E})^2,\end{aligned}$$

so the two shell points have Euclidean norm less than $14\mathcal{E} = 2\cdot 7\mathcal{E}$. Since also $7\mathcal{E} \geq \mu$, (5.9) applies at the energy $7\mathcal{E}$. Dividing the difference of Step 2 at $\mathbf{z}_u$ by $2i\sqrt{\mu^2 + \mathbf{z}_u\cdot\mathbf{z}_u}$, whose modulus is at least $2\sqrt{47}\,\mathcal{E}$, gives

$$\begin{aligned}|\widehat{g}(\mathbf{z}_u)| = \frac{|\Psi_{\mathsf{x}}(w_+) - \Psi_{\mathsf{x}}(w_-)|}{2\,|\mu^2 + \mathbf{z}_u\cdot\mathbf{z}_u|^{1/2}} &\overset{(5.9)}{\leq} \frac{2D_b^{\mathrm{ev}}(7\mathcal{E})^{-1}e^{7\lambda_b^{\mathrm{ev}}T\mathcal{E}}}{2\sqrt{47}\,\mathcal{E}}\left\|\mathsf{R}_U^b\mathsf{x}\right\| \\ &= \frac{D_b^{\mathrm{ev}}}{7\sqrt{47}}\,\mathcal{E}^{-2}e^{7\lambda_b^{\mathrm{ev}}T\mathcal{E}}\left\|\mathsf{R}_U^b\mathsf{x}\right\|.\end{aligned}$$

The function $\zeta \mapsto \widehat{g}(\zeta, z_2, z_3)$ is entire, so the Cauchy mean

$$\widehat{g}(\mathbf{z}) = \frac{1}{2\pi}\int_0^{2\pi}\widehat{g}(\mathbf{z}_u)\,du$$

obeys the same bound.

By (5.6), $\Lambda_b \geq 16\lambda_b^{\mathrm{ev}}$, so the factor 16 covers both exponents: $\lambda_b^{\mathrm{ev}}T\mathcal{E} \leq 7\lambda_b^{\mathrm{ev}}T\mathcal{E} \leq \Lambda_bT\mathcal{E}$. Since also $7\sqrt{47} > 1$, Steps 2 and 3 yield, with $D_b' = D_b^{\mathrm{ev}}$,

$$\left\|\ell_{\mathbf{z}}^f\mathsf{R}_U^{-b}\right\| \leq D_b'\mathcal{E}^{-1}e^{\Lambda_bT\mathcal{E}}, \qquad \left\|\ell_{\mathbf{z}}^g\mathsf{R}_U^{-b}\right\| \leq D_b'\mathcal{E}^{-2}e^{\Lambda_bT\mathcal{E}}.$$

Step 4: the prefactors and the growth in $\mathbf{z}$. Formula (5.10) gives $\Lambda_bT\mathcal{E} = \Lambda_b\mu T + 1 + \Lambda_bT|\mathbf{z}| \geq 1 + \Lambda_b\mu T$. With $\Lambda_b \geq 1$, it follows that

$$\mathcal{E}^{-1} \leq \frac{\Lambda_bT}{1 + \Lambda_b\mu T} \leq \frac{\Lambda_bT}{1 + \mu T}, \qquad e^{\Lambda_bT\mathcal{E}} = e\,e^{\Lambda_b\mu T}e^{\Lambda_bT|\mathbf{z}|}.$$

Inserted in the two bounds of Step 3, these two relations produce the dimensional prefactors and the growth $e^{\Lambda_b T|\mathbf{z}|}$ in $\mathbf{z}$: for every $\mathbf{z}\in\mathbb{C}^3$,

$$\left\|\ell^f_{\mathbf{z}}\mathsf{R}_U^{-b}\right\| \le eD'_b\Lambda_b\,\frac{T}{1+\mu T}\,e^{\Lambda_b\mu T}e^{\Lambda_b T|\mathbf{z}|},$$
$$\left\|\ell^g_{\mathbf{z}}\mathsf{R}_U^{-b}\right\| \le eD'_b\Lambda_b^2\,\frac{T^2}{(1+\mu T)^2}\,e^{\Lambda_b\mu T}e^{\Lambda_b T|\mathbf{z}|}.$$

Step 5: the moments by Cauchy estimates. Differentiation of $\widehat{f}(\mathbf{z})=\int e^{-i\mathbf{z}\cdot\mathbf{x}}f(\mathbf{x})\,d^3x$ and of $\widehat{g}(\mathbf{z})$ at $\mathbf{z}=0$ gives, in the centred coordinates,

$$\partial^\alpha\widehat{f}(0)=(-i)^{|\alpha|}\ell^f_\alpha(f,g),\qquad \partial^\alpha\widehat{g}(0)=(-i)^{|\alpha|}\ell^g_\alpha(f,g).$$

By Convention 2.5, Step 4 bounds the entire functions $\widehat{f}$ and $\widehat{g}$ on all of $\mathbb{C}^3$ by multiples of $e^{\Lambda_b T|\mathbf{z}|}\left\|\mathsf{R}_U^b\mathsf{x}\right\|$.

For $n=|\alpha|\ge 1$, let $r=n/(\sqrt{3}\Lambda_b T)$. On the closed polydisc $\{|z_1|,|z_2|,|z_3|\le r\}$ one has $|\mathbf{z}|\le\sqrt{3}\,r$, hence $e^{\Lambda_b T|\mathbf{z}|}\le e^n$. The multivariable Cauchy estimate on this polydisc gives

$$\begin{aligned}|\ell^f_\alpha(f,g)| &= |\partial^\alpha\widehat{f}(0)|\le\frac{\alpha!}{r^n}\max_{|z_1|,|z_2|,|z_3|\le r}|\widehat{f}(\mathbf{z})|\\ &\overset{\text{Step 4}}{\le}\alpha!\left(\frac{\sqrt{3}\Lambda_b T}{n}\right)^n e^n\cdot eD'_b\Lambda_b\,\frac{T}{1+\mu T}\,e^{\Lambda_b\mu T}\left\|\mathsf{R}_U^b\mathsf{x}\right\|.\end{aligned}$$

The derivative factor is bounded by

$$\alpha!\left(\frac{\sqrt{3}\Lambda_b T}{n}\right)^n e^n\le n^n\left(\frac{\sqrt{3}\Lambda_b T}{n}\right)^n e^n=(e\sqrt{3}\,\Lambda_b T)^n\le(5\Lambda_b T)^n=(B_b T)^n,$$

using $\alpha!\le n!\le n^n$ and $e\sqrt{3}<5$.

The same estimates for $\widehat{g}$, with the second bound of Step 4, give the factor $eD'_b\Lambda_b^2\,T^2/(1+\mu T)^2$ in place of $eD'_b\Lambda_b\,T/(1+\mu T)$. The remaining constants are absorbed in $M_b=eD'_b\Lambda_b^2$, which depends only on b and, since $\Lambda_b\ge 1$, is at least $eD'_b\Lambda_b$; with this M_b, the resulting bounds on ℓ^f_α and ℓ^g_α are (5.7) and (5.8) for $n\ge 1$. The case $n=0$ follows directly from Step 4 at $\mathbf{z}=0$, with $\mathcal{E}$ from (5.10): there $\ell^f_\alpha(f,g)=\widehat{f}(0)$, $\ell^g_\alpha(f,g)=\widehat{g}(0)$, and $e^{\Lambda_b T|\mathbf{z}|}=1$. □

6. The forward estimate for separated balls

For unit-radius balls at separation $s>2$ and mass $\mu>0$, we set $\xi=\mu s$. Throughout this section, R_A and R_B denote the uncertainty reserves (2.9) of the two balls $B_1(0)$ and $B_1(s\mathbf{e}_1)$. The real cross-covariance operator in the separate marginal source norms,

$$C_{\mu,s}:\mathcal{H}_{\mu,B_1(s\mathbf{e}_1)}\longrightarrow\mathcal{H}_{\mu,B_1(0)},$$

is defined on real sources by

$$\langle\mathsf{x},C_{\mu,s}\mathsf{y}\rangle_{\mathsf{v}_{\mu,B_1(0)}}=\mathsf{v}_\mu(\mathsf{x},\mathsf{y}),\qquad \operatorname{supp}\mathsf{x}\subset B_1(0),\quad \operatorname{supp}\mathsf{y}\subset B_1(s\mathbf{e}_1),$$

where $\mathsf{v}_\mu(\mathsf{x},\mathsf{y})$ is the vacuum covariance form (2.2) of the two sources; by (2.6), this form is bounded by the product of the two marginal norms, and $C_{\mu,s}$ is extended complex-linearly.

Away from the diagonal, the field and momentum covariance kernels are

$$\mathfrak{X}_\mu(\mathbf{w})=\frac{\mu}{4\pi^2\varrho}K_1(\mu\varrho),\qquad \mathfrak{P}_\mu(\mathbf{w})=-\frac{\mu^2}{4\pi^2\varrho^2}K_2(\mu\varrho),\qquad \varrho=\sqrt{\mathbf{w}\cdot\mathbf{w}},\tag{6.1}$$

so that, for $\mathsf{x}=(f,g)$ and $\mathsf{y}=(f',g')$ with disjoint supports,

$$\mathsf{v}_\mu(\mathsf{x},\mathsf{y})=\iint\left[f(\mathbf{x})\mathfrak{X}_\mu(\mathbf{y}-\mathbf{x})f'(\mathbf{y})+g(\mathbf{x})\mathfrak{P}_\mu(\mathbf{y}-\mathbf{x})g'(\mathbf{y})\right]d^3x\,d^3y,$$

since the real covariance has zero mixed field–momentum kernel. The square root ϱ has positive real part on the complex separated neighbourhoods used below.

For real $\mathbf{w} \neq 0$, where $\varrho = |\mathbf{w}|$, the heat-kernel integral for $h_\mu^{-1}/2$ gives the first formula in (6.1). With $h_\mu^{-1} = \pi^{-1/2} \int_0^\infty t^{-1/2} e^{-t h_\mu^2}\, dt$ and the kernel $(4\pi t)^{-3/2} e^{-\varrho^2/(4t) - \mu^2 t}$ of $e^{-t h_\mu^2}$, the convolution kernel of $h_\mu^{-1}/2$ is

$$\frac{1}{2\sqrt{\pi}} \int_0^\infty \frac{e^{-\varrho^2/(4t) - \mu^2 t}}{t^{1/2}(4\pi t)^{3/2}}\, dt = \frac{1}{16\pi^2} \int_0^\infty t^{-2} e^{-\varrho^2/(4t) - \mu^2 t}\, dt = \frac{\mu}{4\pi^2 \varrho} K_1(\mu\varrho).$$

The last equality is the representation $K_1(x) = (x/4) \int_0^\infty t^{-2} e^{-t - x^2/(4t)}\, dt$ [28, Eq. 10.32.10] at $x = \mu\varrho$, after the substitution $t \mapsto t/\mu^2$.

Applying $-\nabla^2 + \mu^2$ to $\mathfrak{X}_\mu$ gives the second formula. Off the diagonal, the kernel of $h_\mu/2 = (-\nabla^2 + \mu^2)\, h_\mu^{-1}/2$ is $(-\nabla^2 + \mu^2)\mathfrak{X}_\mu$. On radial functions $\nabla^2 = \varrho^{-1} \partial_\varrho^2\, \varrho$, and $\varrho\, \mathfrak{X}_\mu = \mu K_1(x)/(4\pi^2)$ with $x = \mu\varrho$. Hence

$$\begin{aligned}(-\nabla^2 + \mu^2)\mathfrak{X}_\mu &= -\frac{\partial_\varrho^2(\varrho\, \mathfrak{X}_\mu)}{\varrho} + \mu^2 \mathfrak{X}_\mu = \frac{\mu^4}{4\pi^2 x}\big(K_1(x) - K_1''(x)\big) \\ &= -\frac{\mu^4}{4\pi^2 x^2} K_2(x) = \mathfrak{P}_\mu(\mathbf{w}),\end{aligned}$$

because $K_1 - K_1'' = -x^{-1} K_2$ by the modified Bessel equation $K_1'' = (1 + x^{-2}) K_1 - x^{-1} K_1'$ and the recurrence $K_1' = -K_2 + x^{-1} K_1$.

The integral representation [28, Eq. 10.32.9]

$$K_j(x) = \int_0^\infty e^{-x \cosh u} \cosh(ju)\, du$$

gives, for $x > 0$,

$$\begin{aligned}K_1(x) &\leq e^{-x} \left(x^{-1} + \sqrt{\frac{\pi}{2x}} \right), \\ K_2(x) &\leq e^{-x} \left(2x^{-2} + 4x^{-1} + \sqrt{\frac{\pi}{2x}} \right).\end{aligned}$$

Indeed, the substitution $v = \cosh u - 1$, with $dv = \sqrt{v(v+2)}\, du$, $\cosh u = 1 + v$ and $\cosh 2u = 2v^2 + 4v + 1$, gives

$$K_1(x) = e^{-x} \int_0^\infty e^{-xv} \frac{1 + v}{\sqrt{v(v+2)}}\, dv, \qquad K_2(x) = e^{-x} \int_0^\infty e^{-xv} \frac{2v^2 + 4v + 1}{\sqrt{v(v+2)}}\, dv.$$

The bound $\sqrt{v(v+2)} \geq v$ on the terms of the numerators that contain v, and the bound $\sqrt{v(v+2)} \geq \sqrt{2v}$ on their constant terms, give

$$\frac{1 + v}{\sqrt{v(v+2)}} \leq 1 + (2v)^{-1/2}, \qquad \frac{2v^2 + 4v + 1}{\sqrt{v(v+2)}} \leq 2v + 4 + (2v)^{-1/2}.$$

Integration against e^{-xv}, with $\int_0^\infty e^{-xv}\, dv = x^{-1}$, $\int_0^\infty v\, e^{-xv}\, dv = x^{-2}$ and $\int_0^\infty (2v)^{-1/2} e^{-xv}\, dv = \sqrt{\pi/(2x)}$, yields the bounds for K_1 and K_2 stated above.

Multiplying the bound for K_1 by x and the bound for K_2 by x^2 gives

$$\begin{aligned}x K_1(x) &\leq \left(1 + \sqrt{\tfrac{\pi}{2}}\, x^{1/2}\right) e^{-x} \leq \left(1 + \sqrt{\tfrac{\pi}{2}}\right)(1 + x)^{1/2} e^{-x}, \\ x^2 K_2(x) &\leq \left(2 + 4x + \sqrt{\tfrac{\pi}{2}}\, x^{3/2}\right) e^{-x} \leq \left(6 + \sqrt{\tfrac{\pi}{2}}\right)(1 + x)^{3/2} e^{-x},\end{aligned}$$

because $1, x^{1/2} \le (1+x)^{1/2}$ and $1, x, x^{3/2} \le (1+x)^{3/2}$. Since $1+\sqrt{\pi/2} < 3$ and $6+\sqrt{\pi/2} < 8$, in particular,

$$xK_1(x) \le 3(1+x)^{1/2}e^{-x}, \qquad x^2K_2(x) \le 8(1+x)^{3/2}e^{-x}. \tag{6.2}$$

For $\operatorname{Re} z > 0$, the integral representation of K_j gives $|K_j(z)| \le K_j(\operatorname{Re} z)$.

Theorem 6.1 (Marginally weighted trace estimate)**.** *For each fixed $1/2 \le b < 1$, there are finite S_b, F_b, independent of μ, such that for every $\mu > 0$ and $s \ge S_b$ the cross-covariance operator $C_{\mu,s}$ factors through the reserves $\mathsf{R}_A, \mathsf{R}_B$ of the two balls as*

$$C_{\mu,s} = \mathsf{R}_A^b X_{b,\mu,s} \mathsf{R}_B^b, \qquad X_{b,\mu,s} \in \mathcal{S}_1, \qquad \|X_{b,\mu,s}\|_1 \le F_b s^{-2}. \tag{6.3}$$

Proof. The coordinates $\mathbf{x}$ and $\mathbf{y}$ are measured from the centres 0 and $s\mathbf{e}_1$ of the two balls. With B_b as in (5.6), we set

$$\mathfrak{d}_b = 2B_b, \qquad d_{0,b} = 2\sqrt{3}\,\mathfrak{d}_b = 4\sqrt{3}B_b.$$

Step 1: the square root on the polydisc. On the six-variable polydisc $|x_j|, |y_j| \le \mathfrak{d}_b$,

$$\mathbf{w} = s\mathbf{e}_1 + \mathbf{y} - \mathbf{x}, \qquad |\mathbf{w} - s\mathbf{e}_1|_{\mathrm{E}} \le d_{0,b},$$

because each of the three components of $\mathbf{y} - \mathbf{x}$ has modulus at most $2\mathfrak{d}_b$. Let $s \ge 4d_{0,b}$, and put $q = d_{0,b}/s \le 1/4$. Then

$$\mathbf{w}\cdot\mathbf{w} = s^2(1+\epsilon), \qquad \epsilon = \frac{2(y_1 - x_1)}{s} + \frac{(\mathbf{y}-\mathbf{x})\cdot(\mathbf{y}-\mathbf{x})}{s^2},$$
$$|\epsilon| \overset{|\mathbf{y}-\mathbf{x}|_{\mathrm{E}} \le d_{0,b}}{\le} \frac{2d_{0,b}}{s} + \frac{d_{0,b}^2}{s^2} = 2q + q^2 \le \frac{9}{16}.$$

So $1+\epsilon$ lies in the disc of radius $9/16$ about 1, on which the principal square root is holomorphic, and the square root $\varrho = s\sqrt{1+\epsilon}$ in (6.1) is holomorphic on the polydisc.

The real part of ϱ satisfies

$$\operatorname{Re}\varrho \;=\; s\operatorname{Re}\sqrt{1+\epsilon} \ge s\sqrt{1-|\epsilon|} \overset{|\epsilon|\le 2q+q^2}{\ge} s\sqrt{1-2q-q^2}$$
$$\overset{q\le 2/5}{\ge} s(1-2q) = s - 2d_{0,b} \overset{q\le 1/4}{\ge} \frac{s}{2}.$$

The first inequality holds because $(\operatorname{Re}\sqrt{1+\epsilon})^2 = (|1+\epsilon| + 1 + \operatorname{Re}\epsilon)/2 \ge 1 - |\epsilon|$, and the inequality marked $q \le 2/5$ is $(1-2q)^2 \le 1 - 2q - q^2$, that is, $5q^2 \le 2q$. Moreover, $|\varrho| = s\,|1+\epsilon|^{1/2} \le s\,(1+9/16)^{1/2} = 5s/4$.

Step 2: the suprema of the two kernels. We write sup for the supremum over the polydisc. At each of its points, $\mu\operatorname{Re}\varrho > 0$ by Step 1, so the bound $|K_j(z)| \le K_j(\operatorname{Re} z)$ at $z = \mu\varrho$ and (6.2) at $x = \mu\operatorname{Re}\varrho$ give

$$|\mathfrak{X}_\mu(\mathbf{w})| \overset{(6.1)}{=} \frac{\mu|K_1(\mu\varrho)|}{4\pi^2|\varrho|} \le \frac{\mu K_1(\mu\operatorname{Re}\varrho)}{4\pi^2|\varrho|} \overset{(6.2)}{\le} \frac{3(1+\mu\operatorname{Re}\varrho)^{1/2}e^{-\mu\operatorname{Re}\varrho}}{4\pi^2|\varrho|\operatorname{Re}\varrho},$$
$$|\mathfrak{P}_\mu(\mathbf{w})| \overset{(6.1)}{=} \frac{\mu^2|K_2(\mu\varrho)|}{4\pi^2|\varrho|^2} \le \frac{\mu^2 K_2(\mu\operatorname{Re}\varrho)}{4\pi^2|\varrho|^2} \overset{(6.2)}{\le} \frac{8(1+\mu\operatorname{Re}\varrho)^{3/2}e^{-\mu\operatorname{Re}\varrho}}{4\pi^2|\varrho|^2(\operatorname{Re}\varrho)^2}.$$

By Step 1, $|\varrho| \ge \operatorname{Re}\varrho \ge s/2$, $1 + \mu\operatorname{Re}\varrho \le 1 + \mu|\varrho| \le 1 + 5\xi/4 \le 5(1+\xi)/4$, and $e^{-\mu\operatorname{Re}\varrho} \le e^{-\mu(s-2d_{0,b})} = e^{-\xi+2\mu d_{0,b}}$. Thus

$$\sup|\mathfrak{X}_\mu| \le Cs^{-2}(1+\xi)^{1/2}e^{-\xi+2\mu d_{0,b}},$$
$$\sup|\mathfrak{P}_\mu| \le Cs^{-4}(1+\xi)^{3/2}e^{-\xi+2\mu d_{0,b}}.$$

Step 3: the Taylor coefficients. By Step 1, ϱ is holomorphic on the polydisc with $\operatorname{Re}\varrho \ge s/2$, and K_1 and K_2 are holomorphic on the right half-plane. So $\mathfrak{X}_\mu(s\mathbf{e}_1 + \mathbf{y} - \mathbf{x})$ and $\mathfrak{P}_\mu(s\mathbf{e}_1 + \mathbf{y} - \mathbf{x})$

are holomorphic functions of $(\mathbf{x},\mathbf{y})$ on the polydisc; let $c^f_{\alpha\beta}$ and $c^g_{\alpha\beta}$, $\alpha,\beta\in\mathbb{N}^3$, be their Taylor coefficients at the origin. By the Cauchy estimates on the polydisc,

$$\mathfrak{X}_\mu(s\mathbf{e}_1+\mathbf{y}-\mathbf{x})=\sum_{\alpha,\beta\in\mathbb{N}^3} c^f_{\alpha\beta}\,\mathbf{x}^\alpha\mathbf{y}^\beta,\qquad |c^f_{\alpha\beta}|\le \sup|\mathfrak{X}_\mu|\,\mathfrak{d}_b^{-|\alpha|-|\beta|},$$
$$\mathfrak{P}_\mu(s\mathbf{e}_1+\mathbf{y}-\mathbf{x})=\sum_{\alpha,\beta\in\mathbb{N}^3} c^g_{\alpha\beta}\,\mathbf{x}^\alpha\mathbf{y}^\beta,\qquad |c^g_{\alpha\beta}|\le \sup|\mathfrak{P}_\mu|\,\mathfrak{d}_b^{-|\alpha|-|\beta|}.$$

On the two unit balls, $|x_j|,|y_j|<1<\mathfrak{d}_b$, so both series converge absolutely and uniformly there.

Step 4: the rank-one moment series and (6.4). Let $\mathsf{x}=(f,g)$ and $\mathsf{y}=(f',g')$ be smooth real sources supported in $B_1(0)$ and $B_1(s\mathbf{e}_1)$. In the coordinates of the two balls, the formula after (6.1) and term-by-term integration of the series of Step 3 expand the covariance into products of the source moments of Lemma 5.4, one rank-one product for each monomial kernel:

$$\begin{aligned}\mathsf{v}_\mu(\mathsf{x},\mathsf{y})&=\iint\big[f(\mathbf{x})\,\mathfrak{X}_\mu(s\mathbf{e}_1+\mathbf{y}-\mathbf{x})\,f'(s\mathbf{e}_1+\mathbf{y})\\&\qquad+g(\mathbf{x})\,\mathfrak{P}_\mu(s\mathbf{e}_1+\mathbf{y}-\mathbf{x})\,g'(s\mathbf{e}_1+\mathbf{y})\big]\,d^3x\,d^3y\\&=\sum_{\alpha,\beta\in\mathbb{N}^3}\big(c^f_{\alpha\beta}\,\ell^f_\alpha(\mathsf{x})\,\ell^f_\beta(\mathsf{y})+c^g_{\alpha\beta}\,\ell^g_\alpha(\mathsf{x})\,\ell^g_\beta(\mathsf{y})\big).\end{aligned}$$

Lemma 5.4 at $T=1$, for $U=B_1(0)$ with $\mathbf{x}_0=0$ and for $U=B_1(s\mathbf{e}_1)$ with $\mathbf{x}_0=s\mathbf{e}_1$, turns (5.7)–(5.8) into

$$\big\|\ell^f_\alpha\mathsf{R}_A^{-b}\big\|\le M_b\frac{e^{\Lambda_b\mu}}{1+\mu}B_b^{|\alpha|},\qquad \big\|\ell^g_\alpha\mathsf{R}_A^{-b}\big\|\le M_b\frac{e^{\Lambda_b\mu}}{(1+\mu)^2}B_b^{|\alpha|},$$

and into the same bounds with R_B for the moments centred at $s\mathbf{e}_1$. By Convention 2.5 and the Riesz representation, for $\bullet\in\{f,g\}$ there are vectors $\mathsf{u}^\bullet_{A,\alpha}\in\mathcal{H}_{\mu,B_1(0)}$ and $\mathsf{u}^\bullet_{B,\beta}\in\mathcal{H}_{\mu,B_1(s\mathbf{e}_1)}$ such that

$$\ell^\bullet_\alpha(\mathsf{x})=\langle\mathsf{u}^\bullet_{A,\alpha},\mathsf{R}_A^b\mathsf{x}\rangle,\qquad \ell^\bullet_\beta(\mathsf{y})=\langle\mathsf{u}^\bullet_{B,\beta},\mathsf{R}_B^b\mathsf{y}\rangle$$

for all $\mathsf{x}\in\mathcal{H}_{\mu,B_1(0)}$ and $\mathsf{y}\in\mathcal{H}_{\mu,B_1(s\mathbf{e}_1)}$, with $\|\mathsf{u}^\bullet_{A,\alpha}\|=\|\ell^\bullet_\alpha\mathsf{R}_A^{-b}\|$ and $\|\mathsf{u}^\bullet_{B,\beta}\|=\|\ell^\bullet_\beta\mathsf{R}_B^{-b}\|$. We put

$$X_{b,\mu,s}=\sum_{\bullet\in\{f,g\}}\sum_{\alpha,\beta\in\mathbb{N}^3} c^\bullet_{\alpha\beta}\,\mathsf{u}^\bullet_{A,\alpha}\langle\mathsf{u}^\bullet_{B,\beta},\cdot\,\rangle,$$

a series of rank-one operators.

The term of $(\bullet,\alpha,\beta)$ has trace norm $|c^\bullet_{\alpha\beta}|\,\|\mathsf{u}^\bullet_{A,\alpha}\|\,\|\mathsf{u}^\bullet_{B,\beta}\|$. By Step 3 and the moment bounds, with $B_b/\mathfrak{d}_b=1/2$ and $\sum_{\alpha,\beta\in\mathbb{N}^3}2^{-|\alpha|-|\beta|}=\big(\sum_{k\ge0}2^{-k}\big)^6=64$, these trace norms are summable; so the series converges absolutely in trace norm, $X_{b,\mu,s}\in\mathcal{S}_1$, and

$$\begin{aligned}\|X_{b,\mu,s}\|_1&\le\sum_{\bullet,\alpha,\beta}|c^\bullet_{\alpha\beta}|\,\big\|\mathsf{u}^\bullet_{A,\alpha}\big\|\,\big\|\mathsf{u}^\bullet_{B,\beta}\big\|\\&\le M_b^2e^{2\Lambda_b\mu}\left[\frac{\sup|\mathfrak{X}_\mu|}{(1+\mu)^2}+\frac{\sup|\mathfrak{P}_\mu|}{(1+\mu)^4}\right]\sum_{\alpha,\beta\in\mathbb{N}^3}2^{-|\alpha|-|\beta|}\\&=64\,M_b^2e^{2\Lambda_b\mu}\left[\frac{\sup|\mathfrak{X}_\mu|}{(1+\mu)^2}+\frac{\sup|\mathfrak{P}_\mu|}{(1+\mu)^4}\right].\end{aligned}$$

Inserting the bounds of Step 2 for the two suprema, with $2\Lambda_b\mu+2\mu d_{0,b}=(2\Lambda_b+8\sqrt{3}B_b)\mu$, gives

$$\|X_{b,\mu,s}\|_1\le CM_b^2e^{\mathfrak{k}_b\mu-\xi}\left[\frac{s^{-2}(1+\xi)^{1/2}}{(1+\mu)^2}+\frac{s^{-4}(1+\xi)^{3/2}}{(1+\mu)^4}\right],\qquad \mathfrak{k}_b=2\Lambda_b+8\sqrt{3}B_b.\tag{6.4}$$

Step 5: the threshold S_b and the constant F_b. A sufficient geometric condition is

$$S_b \geq \max\{4, 4d_{0,b}, 2\mathfrak{k}_b\}. \tag{6.5}$$

Let $s \geq S_b$. The entry $4d_{0,b}$ is the condition of Step 1; the entry 4 gives $s > 2$, the separation assumed throughout this section, and $s \geq 1$; the entry $2\mathfrak{k}_b$ gives $\mathfrak{k}_b\mu \leq \mu s/2 = \xi/2$. So the exponential in (6.4) is at most $e^{-\xi/2}$.

Since $1 + \mu \geq 1$ and $s^{-4} \leq s^{-2}$, the bracket in (6.4) is at most $s^{-2}\big[(1+\xi)^{1/2} + (1+\xi)^{3/2}\big] \leq 2s^{-2}(1+\xi)^{3/2}$. Hence, with C the constant of (6.4),

$$\|X_{b,\mu,s}\|_1 \leq 2CM_b^2 s^{-2}(1+\xi)^{3/2}e^{-\xi/2}.$$

The finite supremum $3^{3/2}e^{-1}$ of $(1+\xi)^{3/2}e^{-\xi/2}$ over $\xi \geq 0$, attained at $\xi = 2$, gives $F_b = 2 \cdot 3^{3/2}e^{-1}CM_b^2$ in (6.3). The numbers S_b and F_b depend only on b.

Step 6: the identification with $C_{\mu,s}$. Let x, y be smooth real sources as in Step 4. Their moments are real, so that

$$\ell^\bullet_\alpha(\mathsf{x}) = \overline{\langle \mathsf{u}^\bullet_{A,\alpha}, \mathsf{R}^b_A\mathsf{x}\rangle} = \langle \mathsf{R}^b_A\mathsf{x}, \mathsf{u}^\bullet_{A,\alpha}\rangle,$$

and

$$\begin{aligned}\langle \mathsf{x}, \mathsf{R}^b_A X_{b,\mu,s}\mathsf{R}^b_B\mathsf{y}\rangle &= \sum_{\bullet,\alpha,\beta} c^\bullet_{\alpha\beta}\,\langle \mathsf{R}^b_A\mathsf{x}, \mathsf{u}^\bullet_{A,\alpha}\rangle\,\langle \mathsf{u}^\bullet_{B,\beta}, \mathsf{R}^b_B\mathsf{y}\rangle \\ &= \sum_{\bullet,\alpha,\beta} c^\bullet_{\alpha\beta}\,\ell^\bullet_\alpha(\mathsf{x})\,\ell^\bullet_\beta(\mathsf{y}) \overset{\text{Step 4}}{=} \mathsf{v}_\mu(\mathsf{x},\mathsf{y}) = \langle \mathsf{x}, C_{\mu,s}\mathsf{y}\rangle.\end{aligned}$$

The bounded operators $\mathsf{R}^b_A X_{b,\mu,s}\mathsf{R}^b_B$ and $C_{\mu,s}$ thus have the same sesquilinear form on pairs of smooth real sources. By sesquilinearity the two forms agree on complex combinations of such sources, which are dense in the complexified source spaces (Section 2.2); so density identifies the full cross-covariance operator, $C_{\mu,s} = \mathsf{R}^b_A X_{b,\mu,s}\mathsf{R}^b_B$. With the bound of Step 5, this factorization is (6.3). □

7. The reverse estimate in the joint covariance norm

For

$$U_s = B_1(0) \cup B_1(s\mathbf{e}_1),$$

the source Hilbert space $\mathcal{H}_{\mu,U_s}$ now carries its joint vacuum covariance norm. Its uncertainty reserve is denoted by $\mathcal{R}_{\mu,s}$. The product covariance $\mathsf{v}_{\mathrm{prod},\mu,s}$ is obtained from the joint covariance v_{μ,U_s} by deleting the two cross terms between the balls. The product-minus-joint covariance operator is defined by

$$\langle \mathsf{x}, \mathsf{D}_{\mu,s}\mathsf{y}\rangle_{\mathsf{v}_{\mu,U_s}} = \mathsf{v}_{\mathrm{prod},\mu,s}(\mathsf{x},\mathsf{y}) - \mathsf{v}_{\mu,U_s}(\mathsf{x},\mathsf{y}). \tag{7.1}$$

The form $\mathsf{v}_{\mathrm{prod},\mu,s} - \mathsf{v}_{\mu,U_s}$ is bounded for all sufficiently large s (Lemma 7.1).

7.1. Smooth kernels in the joint source norm.

The multiplier inequality $h_\mu \leq \max\{1,\mu\}h_1$ gives

$$\left\|h_1^{-1/2}f\right\|_2 \leq \sqrt{\max\{1,\mu\}}\,\left\|h_\mu^{-1/2}f\right\|_2. \tag{7.2}$$

For g supported in U_s, a set of volume at most $8\pi/3$, twice the volume $4\pi/3$ of a unit ball, the unitary Fourier transform and the Cauchy–Schwarz inequality give, for every $\mathbf{p}$,

$$\begin{aligned}|\mathcal{F}g(\mathbf{p})|^2 &\leq \frac{1}{(2\pi)^3}\Big(\int_{U_s}|g|\,d^3x\Big)^2 \leq \frac{\mathrm{vol}(U_s)}{(2\pi)^3}\,\|g\|_2^2 \\ &\leq \frac{8\pi/3}{8\pi^3}\,\|g\|_2^2 = \frac{1}{3\pi^2}\,\|g\|_2^2,\end{aligned}$$

and integration over the unit ball $\{|\mathbf{p}|<1\}$, of volume $4\pi/3$, gives

$$\int_{|\mathbf{p}|<1} |\mathcal{F}g(\mathbf{p})|^2\, d^3p \le \frac{4\pi}{3}\cdot\frac{1}{3\pi^2}\,\|g\|_2^2 = \frac{4}{9\pi}\,\|g\|_2^2 < \frac{1}{2}\,\|g\|_2^2\,.$$

Consequently, by Plancherel, the part of $\|g\|_2^2$ from the frequencies $|\mathbf{p}|\ge 1$ exceeds $\frac{1}{2}\,\|g\|_2^2$, so that

$$\|g\|_2^2 \le 2\int |\mathbf{p}|\,|\mathcal{F}g(\mathbf{p})|^2\, d^3p \le 2\left\|h_\mu^{1/2} g\right\|_2^2. \tag{7.3}$$

We fix a smooth cutoff χ_0 on $\mathbb{R}^3$ with $0\le\chi_0\le 1$, equal to 1 on $B_1(0)$ and with compact support in $B_2(0)$, and put $\chi(\mathbf{x})=\chi_0(\mathbf{x})+\chi_0(\mathbf{x}-s\mathbf{e}_1)$. For $s\ge 5$, the supports of the two terms are disjoint, so $\chi=1$ on U_s, and the derivative bounds of χ are those of χ_0, independent of s.

Let K be a scalar kernel on $(B_2(0)\cup B_2(s\mathbf{e}_1))^2$, with derivatives through order two in each variable bounded by M:

$$|\partial_{\mathbf{x}}^\alpha \partial_{\mathbf{y}}^\beta K(\mathbf{x},\mathbf{y})| \le M, \qquad |\alpha|,|\beta|\le 2.$$

In the joint source norm $\|\mathsf{x}\| = \mathsf{v}_{\mu,U_s}(\mathsf{x},\mathsf{x})^{1/2}$, the field-source form and the momentum-source form of K satisfy, for smooth real sources $\mathsf{x}=(f,g)$ and $\mathsf{y}=(f',g')$ supported in U_s,

$$\left|\iint f(\mathbf{x})K(\mathbf{x},\mathbf{y})f'(\mathbf{y})\, d^3x\, d^3y\right| \le C\max\{1,\mu\}M\,\|\mathsf{x}\|\,\|\mathsf{y}\|\,,$$
$$\left|\iint g(\mathbf{x})K(\mathbf{x},\mathbf{y})g'(\mathbf{y})\, d^3x\, d^3y\right| \le CM\,\|\mathsf{x}\|\,\|\mathsf{y}\|$$

(the *smooth-kernel estimate*).

Let $K_\chi(\mathbf{x},\mathbf{y})=\chi(\mathbf{x})K(\mathbf{x},\mathbf{y})\chi(\mathbf{y})$, extended by zero to $\mathbb{R}^6$; it equals K on $U_s\times U_s$, and the integral operator with kernel K_χ is denoted by the same symbol. By the Leibniz rule, $|\partial_{\mathbf{x}}^\alpha\partial_{\mathbf{y}}^\beta K_\chi|\le CM$ for $|\alpha|,|\beta|\le 2$, and K_χ is supported in a set of volume at most $(2\,\mathrm{vol}\,B_2(0))^2$.

For the field form, multiplication by the cutoffs and insertion of $h_1^{1/2}$ on both sides give the Hilbert–Schmidt operator $h_1^{1/2}K_\chi h_1^{1/2}$, controlled by the kernel's H^2 norm in each variable. With $\mathcal{F}K_\chi$ the unitary Fourier transform of K_χ on $\mathbb{R}^6$,

$$\begin{aligned}\left\|h_1^{1/2}K_\chi h_1^{1/2}\right\|_2^2 &\overset{\text{Plancherel}}{=} \iint (1+|\mathbf{p}|^2)^{1/2}(1+|\mathbf{q}|^2)^{1/2}|\mathcal{F}K_\chi(\mathbf{p},\mathbf{q})|^2\, d^3p\, d^3q\\ &\le \iint (1+|\mathbf{p}|^2)^2(1+|\mathbf{q}|^2)^2|\mathcal{F}K_\chi(\mathbf{p},\mathbf{q})|^2\, d^3p\, d^3q\\ &\overset{\text{Plancherel}}{\le} C\sum_{|\alpha|,|\beta|\le 2}\left\|\partial_{\mathbf{x}}^\alpha\partial_{\mathbf{y}}^\beta K_\chi\right\|_{L^2(\mathbb{R}^6)}^2 \overset{\text{Leibniz}}{\le} CM^2.\end{aligned}$$

Since $f=\chi f$ and $f'=\chi f'$, and $h_1^{\pm 1/2}$ are self-adjoint, (7.2) completes the estimate:

$$\begin{aligned}\left|\iint fKf'\right| &= |\langle h_1^{-1/2}f, (h_1^{1/2}K_\chi h_1^{1/2})\,h_1^{-1/2}f'\rangle| \le \left\|h_1^{1/2}K_\chi h_1^{1/2}\right\|_2\left\|h_1^{-1/2}f\right\|_2\left\|h_1^{-1/2}f'\right\|_2\\ &\overset{(7.2)}{\le} \max\{1,\mu\}\left\|h_1^{1/2}K_\chi h_1^{1/2}\right\|_2\left\|h_\mu^{-1/2}f\right\|_2\left\|h_\mu^{-1/2}f'\right\|_2 \overset{(2.2)}{\le} C\max\{1,\mu\}M\,\|\mathsf{x}\|\,\|\mathsf{y}\|\,,\end{aligned}$$

where the last step uses $\|h_\mu^{-1/2}f\|_2^2\le 2\|\mathsf{x}\|^2$ and the bound CM of the Hilbert–Schmidt norm.

For the momentum form, the Hilbert–Schmidt kernel norm and (7.3) suffice: the Hilbert–Schmidt norm of K_χ is $\|K_\chi\|_{L^2(\mathbb{R}^6)}\le CM$, and

$$\begin{aligned}\left|\iint gKg'\right| &= |\langle g, K_\chi g'\rangle| \le \|K_\chi\|_{L^2(\mathbb{R}^6)}\,\|g\|_2\,\|g'\|_2\\ &\overset{(7.3)}{\le} 2\,\|K_\chi\|_{L^2(\mathbb{R}^6)}\left\|h_\mu^{1/2}g\right\|_2\left\|h_\mu^{1/2}g'\right\|_2 \overset{(2.2)}{\le} CM\,\|\mathsf{x}\|\,\|\mathsf{y}\|\,.\end{aligned}$$

All cutoff supports and derivative norms have fixed size, so the constants C are independent of $s \geq 5$ and of μ. Density extends the estimates to $\mathcal{X}_{\mu,U_s}$.

7.2. **Polynomial approximation on the disconnected support.** The polynomial selector

$$\mathsf{p}_N(z) = (1-z)^N \sum_{k=0}^{N-1} \binom{N+k-1}{k} z^k \tag{7.4}$$

has degree $2N-1$; $\mathsf{p}_N - 1$ has a zero of order N at zero, and p_N has a zero of order N at one. For the zero of $\mathsf{p}_N - 1$ at zero, the expansion $(1-z)^{-N} = \sum_{k\geq 0} \binom{N+k-1}{k} z^k$, $|z| < 1$, gives

$$\mathsf{p}_N(z) - 1 = -(1-z)^N \sum_{k \geq N} \binom{N+k-1}{k} z^k.$$

For $N \geq 1$ and $0 < r < 1/2$, the identity for $\mathsf{p}_N - 1$ above, the definition (7.4), the binomial bound $\binom{N+k-1}{k} \leq 2^{N+k-1}$ and $\sum_{k=0}^{N-1} \binom{N+k-1}{k} = \binom{2N-1}{N-1} \leq 2^{2N-1}$ give the endpoint bounds

$$|\mathsf{p}_N(z) - 1| \leq (1+r)^N \sum_{k \geq N} 2^{N+k-1} r^k = \frac{(4r(1+r))^N}{2(1-2r)}, \qquad |z| \leq r,$$

$$|\mathsf{p}_N(z)| \leq r^N (1+r)^{N-1} \binom{2N-1}{N-1} \leq (4r(1+r))^N, \qquad |z-1| \leq r.$$

The coefficient sum of a polynomial, the sum of the absolute values of its coefficients, is submultiplicative. Expansion of $(1-z)^N$ gives the coefficient sum 2^N, and the binomial bound gives $\binom{2N-1}{N-1} \leq 2^{2N-1}$ for the sum in (7.4); so the coefficient sum of p_N is at most 2^{3N}. The fixed affine shifts used below are $z \mapsto z + 1/2$ and $z \mapsto 1/2 - z$. Each turns z^k into a polynomial of coefficient sum $(3/2)^k$, so after either shift the coefficient sum of p_N is at most $(3/2)^{2N-1} 2^{3N} \leq C^N$.

Lemma 7.1 (Polynomial increments)**.** *There are universal finite constants $C_1, C_2, c_{\mathrm{pol}}, K_{\mathrm{pol}}$ and a sufficiently large universal S_{pol} with the following properties. For every $\mu > 0$ and $s \geq S_{\mathrm{pol}}$, with $\xi = \mu s$, there are real self-adjoint finite-rank forms F_n with polynomial kernels, $n \geq 1$, with $\mathsf{F}_0 = 0$, such that $\mathsf{F}_n \to \mathsf{D}_{\mu,s}$ in operator norm, where $\mathsf{D}_{\mu,s}$ is the product-minus-joint covariance operator* (7.1)*. Their increments $\mathsf{G}_n = \mathsf{F}_{n+1} - \mathsf{F}_n$, $n \geq 0$, satisfy*

$$\|\mathsf{G}_n\|_1 \leq C_1 (n+1)^3 s^{-2} (1+\xi)^2 e^{-\xi/2} (c_{\mathrm{pol}}/s)^n. \tag{7.5}$$

For $\Lambda_0 = \Lambda_{1/2}$, the constant (5.6) *at $b = 1/2$, and $T = s/2 + 1$, there are finite-rank B_n such that, with the joint reserve $\mathcal{R}_{\mu,s}$,*

$$\mathsf{G}_n = \mathcal{R}_{\mu,s}^{1/2} B_n \mathcal{R}_{\mu,s}^{1/2}, \qquad \|B_n\|_1 \leq C_2 e^{-\xi/2} e^{2\Lambda_0 \mu T} K_{\mathrm{pol}}^n. \tag{7.6}$$

Proof. Step 1: the cross kernels on a polydisc. In dimensionless coordinates $u = \mathbf{x}/s$, $v = \mathbf{y}/s$, the support centres are 0 and $\mathbf{e}_1$. We use a polydisc of radius $r_0 = 1/100$ about the cross centre $(0, \mathbf{e}_1)$, and the slightly larger polydisc of radius $2r_0$. On the latter, $v - u = \mathbf{e}_1 + \boldsymbol{\delta}$ with $\boldsymbol{\delta} \in \mathbb{C}^3$ and $|\boldsymbol{\delta}|_{\mathrm{E}} \leq 4\sqrt{3}\, r_0 < 1/10$, so

$$(v-u)\cdot(v-u) = 1 + \epsilon, \qquad |\epsilon| = |2\delta_1 + \boldsymbol{\delta}\cdot\boldsymbol{\delta}| \leq 2|\boldsymbol{\delta}|_{\mathrm{E}} + |\boldsymbol{\delta}|_{\mathrm{E}}^2 < \tfrac{1}{4}.$$

Hence the square root $\varrho = s\sqrt{1+\epsilon}$ in (6.1) is holomorphic there, and, with $\operatorname{Re}\sqrt{1+\epsilon} \geq \sqrt{1-|\epsilon|}$ as in the proof of Theorem 6.1,

$$\tfrac{s}{2} \leq \operatorname{Re}\varrho \leq |\varrho| \leq 2s.$$

The Bessel bounds (6.2) give the polydisc suprema

$$M_f = C s^{-2} (1+\xi)^{1/2} e^{-\xi/2}, \qquad M_g = C s^{-4} (1+\xi)^{3/2} e^{-\xi/2}$$

of the two cross kernels $\mathfrak{X}_\mu(s(v-u))$ and $\mathfrak{P}_\mu(s(v-u))$: on the polydisc of radius $2r_0$, the pointwise bounds of Step 2 of the proof of Theorem 6.1 hold at $\mathbf{w} = s(v-u)$, and their right-hand sides are at most M_f and M_g because $\frac{s}{2} \le \operatorname{Re}\varrho \le |\varrho| \le 2s$, so that $1 + \mu \operatorname{Re}\varrho \le 2(1+\xi)$ and $e^{-\mu \operatorname{Re}\varrho} \le e^{-\xi/2}$.

Step 2: the polynomial forms. On the polydisc of radius $2r_0$, the two cross kernels are holomorphic, and real at real points; so their Taylor expansions at the cross centre,

$$\mathfrak{X}_\mu(s(v-u)) = \sum_{\alpha,\beta\in\mathbb{N}^3} a^f_{\alpha\beta}\, u^\alpha (v-\mathbf{e}_1)^\beta, \qquad \mathfrak{P}_\mu(s(v-u)) = \sum_{\alpha,\beta\in\mathbb{N}^3} a^g_{\alpha\beta}\, u^\alpha (v-\mathbf{e}_1)^\beta,$$

have real coefficients, and Cauchy's estimate on the polydisc of radius r_0 gives $|a^\bullet_{\alpha\beta}| \le M_\bullet r_0^{-|\alpha|-|\beta|}$.

Let $\mathcal{T}_{n,f}(u, v-\mathbf{e}_1)$ and $\mathcal{T}_{n,g}(u, v-\mathbf{e}_1)$ denote the Taylor polynomials of total degree $n-1$ of these two cross kernels at the cross centre $(0, \mathbf{e}_1)$, in the coordinates $(u, v-\mathbf{e}_1)$; thus $\mathcal{T}_{0,\bullet} = 0$. For $\bullet \in \{f, g\}$, the *oriented polynomial*, whose selector factors equal 1 to high order on the cross component $u \approx 0$, $v \approx \mathbf{e}_1$ and vanish to high order on the other three components of the support, is

$$\mathsf{p}_{2n}(u_1)\, \mathsf{p}_{2n}(1-v_1)\, \mathcal{T}_{n,\bullet}(u, v-\mathbf{e}_1).$$

Adding the transpose of the oriented polynomial and taking the negative sign from (7.1) gives the polynomial kernel

$$f_{n,\bullet}(\mathbf{x},\mathbf{y}) = -\mathsf{p}_{2n}(u_1)\, \mathsf{p}_{2n}(1-v_1)\, \mathcal{T}_{n,\bullet}(u, v-\mathbf{e}_1) - \mathsf{p}_{2n}(v_1)\, \mathsf{p}_{2n}(1-u_1)\, \mathcal{T}_{n,\bullet}(v, u-\mathbf{e}_1),$$

which is real and symmetric in $(\mathbf{x}, \mathbf{y})$, and defines F_n by

$$\langle \mathsf{x}, \mathsf{F}_n \mathsf{y}\rangle_{\mathsf{v}_{\mu,U_s}} = \iint \left[f(\mathbf{x}) f_{n,f}(\mathbf{x},\mathbf{y}) f'(\mathbf{y}) + g(\mathbf{x}) f_{n,g}(\mathbf{x},\mathbf{y}) g'(\mathbf{y}) \right] d^3x\, d^3y$$

for smooth real sources supported in U_s; in particular $\mathsf{F}_0 = 0$.

Since p_{2n} has degree $4n-1$, the degree of $f_{n,\bullet}$ is at most $2(4n-1) + (n-1) = 9n-3 \le 9n$. Expanded in the monomials $\mathbf{x}^\alpha \mathbf{y}^\beta$, the form of F_n is a finite real combination of the products $\ell^f_\alpha(\mathsf{x})\ell^f_\beta(\mathsf{y})$ and $\ell^g_\alpha(\mathsf{x})\ell^g_\beta(\mathsf{y})$, $|\alpha|, |\beta| \le 9n$, of the moments of Lemma 5.4. That lemma, applied at $b = 1/2$ in a ball about 0 containing U_s, bounds each of these moments at x by a finite multiple of $\|\mathcal{R}^{1/2}_{\mu,s}\mathsf{x}\|$, and $\|\mathcal{R}^{1/2}_{\mu,s}\mathsf{x}\| \le \|\mathsf{x}\|$ since $\mathcal{R}_{\mu,s} \le I$. So the moments are bounded on $\mathcal{X}_{\mu,U_s}$, and F_n is a bounded real self-adjoint operator of finite rank. Its form depends on x only through the moments $\ell^f_\alpha(\mathsf{x})$ and $\ell^g_\alpha(\mathsf{x})$, two for each of the $\binom{9n+3}{3}$ multi-indices $\alpha \in \mathbb{N}^3$ with $|\alpha| \le 9n$, so that

$$\operatorname{rank} \mathsf{F}_n \le 2\binom{9n+3}{3} \le C(n+1)^3.$$

Step 3: the differentiated error on the four components. Let $\mathfrak{K}_\bullet$ be the target product-minus-joint kernel, equal to the negative covariance kernel on the two cross components and zero on the two same-region components. The definition (7.1) reads

$$\langle \mathsf{x}, \mathsf{D}_{\mu,s}\mathsf{y}\rangle_{\mathsf{v}_{\mu,U_s}} = \iint \left[f(\mathbf{x}) \mathfrak{K}_f(\mathbf{x},\mathbf{y}) f'(\mathbf{y}) + g(\mathbf{x}) \mathfrak{K}_g(\mathbf{x},\mathbf{y}) g'(\mathbf{y}) \right] d^3x\, d^3y$$

for smooth real sources supported in U_s. Let $\mathcal{N}_s = (B_2(0) \cup B_2(s\mathbf{e}_1))^2$. Then

$$\max_{|\alpha|,|\beta|\le 2} \sup_{\mathcal{N}_s} \left| \partial^\alpha_{\mathbf{x}} \partial^\beta_{\mathbf{y}} (\mathfrak{K}_\bullet - f_{n,\bullet}) \right| \le C M_\bullet (c_{\mathrm{pol}}/s)^n. \tag{7.7}$$

The selector and Taylor estimates for (7.7) hold on slightly larger complex physical neighbourhoods of all four components. Let $\mathcal{N}^{\mathbb{C}}_s \subset \mathbb{C}^6$ be the union of the four open polydiscs of radius 3, in the six coordinates of $(\mathbf{x}, \mathbf{y})$, about the points of $\{0, s\mathbf{e}_1\}^2$. It contains the closed unit polydisc about every point of $\mathcal{N}_s$, and in the coordinates (u, v) its four parts are the polydiscs of radius $3/s$ about the points of $\{0, \mathbf{e}_1\}^2$.

The universal S_{pol} is taken with $3/S_{\mathrm{pol}} \le r_0/2$ and larger than the constant c_{pol} fixed at the end of this step. For $s \ge S_{\mathrm{pol}}$, then, $s > 6$, so the four parts are disjoint, and the cross part about $(0, \mathbf{e}_1)$

lies in the polydisc of radius $r_0/2$ about the cross centre. The exchange $(u,v)\mapsto(v,u)$ carries this cross part onto the other one; $\mathfrak{K}_\bullet$ is symmetric under the exchange, because the kernels in (6.1) are even in $\mathbf{w}$, and so is $f_{n,\bullet}$. Continued as $-\mathfrak{X}_\mu(s(v-u))$ or $-\mathfrak{P}_\mu(s(v-u))$ on the cross parts and as zero on the others, $\mathfrak{K}_\bullet$ is holomorphic on $\mathcal{N}_s^{\mathbb{C}}$ by Step 2.

At points where the six coordinates of $(u,v-\mathbf{e}_1)$ have moduli at most ρ, the bound on $a^\bullet_{\alpha\beta}$ from Step 2 and the number $\binom{k+5}{5}\le(k+1)^5$ of monomials of degree k give

$$\sum_{|\alpha|+|\beta|\ge n}|a^\bullet_{\alpha\beta}u^\alpha(v-\mathbf{e}_1)^\beta|\le M_\bullet\sum_{k\ge n}(k+1)^5\Big(\frac{\rho}{r_0}\Big)^k,\qquad |\mathcal{T}_{n,\bullet}(u,v-\mathbf{e}_1)|\le M_\bullet\sum_{k<n}(k+1)^5\Big(\frac{\rho}{r_0}\Big)^k.$$

On the cross part about $(0,\mathbf{e}_1)$, $\rho=3/s\le r_0/2$; with $(k+1)^5\le(n+1)^5(k-n+1)^5$ for $k\ge n$, the Taylor remainder is at most $C(n+1)^5M_\bullet(C/s)^n$ there, and $|\mathcal{T}_{n,\bullet}(u,v-\mathbf{e}_1)|\le CM_\bullet$. On every part, the arguments $(u,v-\mathbf{e}_1)$ and $(v,u-\mathbf{e}_1)$ of $\mathcal{T}_{n,\bullet}$ have coordinates of moduli at most $1+3/s\le2$, and there $|\mathcal{T}_{n,\bullet}|\le n^6(2/r_0)^nM_\bullet\le(n+1)^6C^nM_\bullet$.

Let $n\ge1$. By the endpoint bounds after (7.4), with $N=2n$ and $r=3/s\le1/200$: at arguments within $3/s$ of zero, $|\mathsf{p}_{2n}-1|\le(C/s)^{2n}\le1$, so $|\mathsf{p}_{2n}|\le2$; at arguments within $3/s$ of one, $|\mathsf{p}_{2n}|\le(C/s)^{2n}$.

On the cross part about $(0,\mathbf{e}_1)$, where $\mathfrak{K}_\bullet=-\sum_{\alpha,\beta}a^\bullet_{\alpha\beta}u^\alpha(v-\mathbf{e}_1)^\beta$, the arguments u_1 and $1-v_1$ of the first two selectors of $f_{n,\bullet}$ lie within $3/s$ of zero, and the arguments v_1 and $1-u_1$ of the other two within $3/s$ of one. There

$$\begin{aligned}\mathfrak{K}_\bullet-f_{n,\bullet}=(\mathfrak{K}_\bullet+\mathcal{T}_{n,\bullet}(u,v-\mathbf{e}_1))+(\mathsf{p}_{2n}(u_1)\mathsf{p}_{2n}(1-v_1)-1)\mathcal{T}_{n,\bullet}(u,v-\mathbf{e}_1)\\+\mathsf{p}_{2n}(v_1)\mathsf{p}_{2n}(1-u_1)\,\mathcal{T}_{n,\bullet}(v,u-\mathbf{e}_1),\end{aligned}$$

where the first term is minus the Taylor remainder, and

$$|\mathsf{p}_{2n}(u_1)\mathsf{p}_{2n}(1-v_1)-1|\le|\mathsf{p}_{2n}(u_1)-1|\,|\mathsf{p}_{2n}(1-v_1)|+|\mathsf{p}_{2n}(1-v_1)-1|\le3(C/s)^{2n}.$$

Hence, on this part and, by the symmetry, on the other cross part,

$$|\mathfrak{K}_\bullet-f_{n,\bullet}|\le C(n+1)^5M_\bullet\Big(\frac{C}{s}\Big)^n+3\Big(\frac{C}{s}\Big)^{2n}CM_\bullet+\Big(\frac{C}{s}\Big)^{4n}(n+1)^6C^nM_\bullet.$$

On the two same-region parts, $\mathfrak{K}_\bullet=0$, and in each of the two terms of $f_{n,\bullet}$ one selector has its argument within $3/s$ of one and is at most $(C/s)^{2n}$, while the other is at most 2; so $|\mathfrak{K}_\bullet-f_{n,\bullet}|\le4(C/s)^{2n}(n+1)^6C^nM_\bullet$ there. Since $(C/s)^{jn}C^n\le(C^{j+1}/s)^n$ for $s\ge1$ and $j\ge1$, the bounds on the four parts combine to

$$\sup_{\mathcal{N}_s^{\mathbb{C}}}|\mathfrak{K}_\bullet-f_{n,\bullet}|\le C(n+1)^6M_\bullet(c/s)^n$$

with a universal c. For $n=0$, where $f_{0,\bullet}=0$, the same bound holds, since $|\mathfrak{K}_\bullet|\le M_\bullet$ on $\mathcal{N}_s^{\mathbb{C}}$ by Step 1.

Cauchy estimates then bound the derivatives in (7.7): for $(\mathbf{x},\mathbf{y})\in\mathcal{N}_s$ and $|\alpha|,|\beta|\le2$, the closed unit polydisc about $(\mathbf{x},\mathbf{y})$ lies in $\mathcal{N}_s^{\mathbb{C}}$, so

$$|\partial_{\mathbf{x}}^\alpha\partial_{\mathbf{y}}^\beta(\mathfrak{K}_\bullet-f_{n,\bullet})(\mathbf{x},\mathbf{y})|\le\alpha!\,\beta!\sup_{\mathcal{N}_s^{\mathbb{C}}}|\mathfrak{K}_\bullet-f_{n,\bullet}|\le4C(n+1)^6M_\bullet(c/s)^n.$$

Polynomial factors in n, here the factor $(n+1)^6$ from the coefficient counts, are absorbed in c_{pol}^n: since $n+1\le2^n$, $(n+1)^6(c/s)^n\le(2^6c/s)^n$, and (7.7) holds with $c_{\mathrm{pol}}=2^6c$. The constants in the Taylor bounds above depend only on the polydisc radius r_0, and those in the selector bounds are absolute; so c and c_{pol} depend only on r_0, and in particular not on μ or s.

Step 4: the midpoint coefficients. If the polynomial kernel $f_{n,\bullet}$ is expressed in midpoint coordinates,

$$f_{n,\bullet}(\mathbf{x},\mathbf{y})=\sum_{\alpha,\beta}c^{(n,\bullet)}_{\alpha\beta}\left(\frac{\mathbf{x}-s\mathbf{e}_1/2}{s}\right)^\alpha\left(\frac{\mathbf{y}-s\mathbf{e}_1/2}{s}\right)^\beta,$$

then, for a universal finite K_{mid},

$$\sum_{\alpha,\beta} |c_{\alpha\beta}^{(n,\bullet)}| \le C M_\bullet K_{\mathrm{mid}}^n. \tag{7.8}$$

Write $u' = u - \mathbf{e}_1/2$ and $v' = v - \mathbf{e}_1/2$ for the midpoint coordinates; then $u = u' + \mathbf{e}_1/2$ and $v - \mathbf{e}_1 = v' - \mathbf{e}_1/2$, so only the first coordinates are shifted, by $\pm 1/2$, and the coefficient sums below are taken in (u', v'). By $|a_{\alpha\beta}^\bullet| \le M_\bullet r_0^{-|\alpha|-|\beta|}$ and the count of monomials, the coefficient sum of $\mathcal{T}_{n,\bullet}$ in $(u, v - \mathbf{e}_1)$ is at most $M_\bullet \sum_{k<n} (k+1)^5 r_0^{-k} \le (n+1)^6 r_0^{-n} M_\bullet$; the shift multiplies it by at most $(3/2)^{n-1}$, since a monomial of degree $k \le n-1$ acquires at most k factors $u_1' + 1/2$ or $v_1' - 1/2$. The two selectors become $\mathsf{p}_{2n}(u_1' + 1/2)$ and $\mathsf{p}_{2n}(1/2 - v_1')$, with coefficient sums at most C^n by the bound after (7.4). The transposed term exchanges u' and v' and has the same bound. The product of these bounds for the Taylor polynomial, the shift and the two selectors gives K_{mid}^n:

$$\sum_{\alpha,\beta} |c_{\alpha\beta}^{(n,\bullet)}| \le 2\, C^n\, C^n \Big(\frac{3}{2}\Big)^{n-1} (n+1)^6 r_0^{-n} M_\bullet \le 2(n+1)^6 \Big(\frac{C}{r_0}\Big)^n M_\bullet \le 2 M_\bullet K_{\mathrm{mid}}^n$$

with $K_{\mathrm{mid}} = 2^6 C / r_0$, by $(n+1)^6 \le 2^{6n}$.

Step 5: the operator-norm error. By Steps 2 and 3, for $n \ge 0$ and smooth real sources supported in U_s, the form of $\mathsf{D}_{\mu,s} - \mathsf{F}_n$ is the sum of the field-source form of the kernel $\mathfrak{K}_f - f_{n,f}$ and the momentum-source form of $\mathfrak{K}_g - f_{n,g}$; at $n = 0$, where $\mathsf{F}_0 = 0$, it is the form (7.1) itself. By (7.7), the smooth-kernel estimate above applies to these kernels with $M = C M_\bullet (c_{\mathrm{pol}}/s)^n$; together with $\max\{1, \mu\} \le 1 + \mu \le 1 + \xi$ for $s \ge 1$, it now gives

$$\begin{aligned} |\langle \mathsf{x}, (\mathsf{D}_{\mu,s} - \mathsf{F}_n)\mathsf{y}\rangle| &\le C\big(\max\{1,\mu\} M_f + M_g\big)(c_{\mathrm{pol}}/s)^n \, \|\mathsf{x}\| \, \|\mathsf{y}\| \\ &\le C\big(s^{-2}(1+\xi)^{3/2} + s^{-4}(1+\xi)^{3/2}\big) e^{-\xi/2} (c_{\mathrm{pol}}/s)^n \, \|\mathsf{x}\| \, \|\mathsf{y}\| \end{aligned}$$

on smooth real sources, and by density on $\mathcal{X}_{\mu,U_s}$. The forms are real and symmetric, so the operators they define on $\mathcal{H}_{\mu,U_s}$ are the complexifications of self-adjoint operators on $\mathcal{X}_{\mu,U_s}$, with the same norms. With $s^{-4} \le s^{-2}$ and $(1+\xi)^{3/2} \le (1+\xi)^2$, the bound of the forms gives

$$\|\mathsf{D}_{\mu,s} - \mathsf{F}_n\| \le C s^{-2} (1+\xi)^2 e^{-\xi/2} (c_{\mathrm{pol}}/s)^n. \tag{7.9}$$

At $n = 0$, (7.9) shows that the form in (7.1) is bounded. It also gives $\mathsf{F}_n \to \mathsf{D}_{\mu,s}$ in operator norm, because $S_{\mathrm{pol}} > c_{\mathrm{pol}}$ (Step 3), so that $(c_{\mathrm{pol}}/s)^n \to 0$.

Step 6: the unweighted increments. The kernel of $\mathsf{G}_n = \mathsf{F}_{n+1} - \mathsf{F}_n$ is $f_{n+1,\bullet} - f_{n,\bullet}$, of degree at most $9(n+1)$; as in Step 2, the form of G_n depends on x only through the moments $\ell_\alpha^f(\mathsf{x})$ and $\ell_\alpha^g(\mathsf{x})$ with $|\alpha| \le 9(n+1)$, so

$$\operatorname{rank} \mathsf{G}_n \le 2 \binom{9n+12}{3} \le C(n+1)^3.$$

The bounds (7.9) at n and $n+1$, multiplied by the rank of $\mathsf{G}_n = (\mathsf{D}_{\mu,s} - \mathsf{F}_n) - (\mathsf{D}_{\mu,s} - \mathsf{F}_{n+1})$, give (7.5) for every $n \ge 0$: the trace norm of a finite-rank operator is at most its rank times its operator norm, and $c_{\mathrm{pol}}/s \le 1$, so

$$\|\mathsf{G}_n\|_1 \le \operatorname{rank}(\mathsf{G}_n)\big(\|\mathsf{D}_{\mu,s} - \mathsf{F}_n\| + \|\mathsf{D}_{\mu,s} - \mathsf{F}_{n+1}\|\big) \le C(n+1)^3 \cdot 2C s^{-2}(1+\xi)^2 e^{-\xi/2} (c_{\mathrm{pol}}/s)^n.$$

Step 7: the weighted increments. In the midpoint expansion of Step 4, the form of G_n is a combination of products of the moments $s^{-|\alpha|}\ell_\alpha^f$ and $s^{-|\alpha|}\ell_\alpha^g$, centred at $\mathbf{x}_0 = s\mathbf{e}_1/2$ and taken over the whole union. The union lies in the ball $B_T(\mathbf{x}_0)$ of radius $T = s/2 + 1$, and $s/2 \le T \le s$ for $s \ge 2$. Lemma 5.4, at $b = 1/2$, applies to each of these moments, for $U = U_s \subset B_T(\mathbf{x}_0)$ and its reserve $\mathcal{R}_{\mu,s}$. Since $T/s \le 1$, every normalized degree costs at most $B_{1/2} = 5\Lambda_{1/2}$:

$$\Big\| s^{-|\alpha|} \ell_\alpha^f \mathcal{R}_{\mu,s}^{-1/2} \Big\| \le M_{1/2} \frac{T}{1+\mu T} e^{\Lambda_0 \mu T} B_{1/2}^{|\alpha|}, \qquad \Big\| s^{-|\alpha|} \ell_\alpha^g \mathcal{R}_{\mu,s}^{-1/2} \Big\| \le M_{1/2} \frac{T^2}{(1+\mu T)^2} e^{\Lambda_0 \mu T} B_{1/2}^{|\alpha|}.$$

The Riesz representation gives vectors $\psi_\alpha^\bullet$, real because $\mathcal{R}_{\mu,s}$ and the moments are real, with

$$s^{-|\alpha|}\ell_\alpha^\bullet(\mathsf{x}) = \langle \psi_\alpha^\bullet, \mathcal{R}_{\mu,s}^{1/2}\mathsf{x}\rangle$$

and norms bounded by the right-hand sides above.

The kernel $f_{n+1,\bullet} - f_{n,\bullet}$ of G_n has the midpoint coefficients $c_{\alpha\beta}^{(n+1,\bullet)} - c_{\alpha\beta}^{(n,\bullet)}$, which vanish unless $|\alpha| + |\beta| \le 9(n+1)$ and, by (7.8), have the coefficient sum at most $CM_\bullet(1+K_{\mathrm{mid}})K_{\mathrm{mid}}^n$. Put

$$B_n = \sum_{\bullet\in\{f,g\}} \sum_{\alpha,\beta} (c_{\alpha\beta}^{(n+1,\bullet)} - c_{\alpha\beta}^{(n,\bullet)})\, |\psi_\alpha^\bullet\rangle\langle\psi_\beta^\bullet|.$$

For smooth real sources, $\langle \mathsf{x}, \mathcal{R}_{\mu,s}^{1/2} B_n \mathcal{R}_{\mu,s}^{1/2}\mathsf{y}\rangle$ is the form of G_n written in the midpoint moments, so $\mathsf{G}_n = \mathcal{R}_{\mu,s}^{1/2} B_n \mathcal{R}_{\mu,s}^{1/2}$ as bounded operators. The two weighted moment norms of each rank-one monomial give its trace norm $\|\psi_\alpha^\bullet\|\,\|\psi_\beta^\bullet\|$, with $B_{1/2}^{|\alpha|+|\beta|} \le B_{1/2}^{9(n+1)}$; multiplied by the coefficient sum, these trace norms give

$$\|B_n\|_1 \le C(1+K_{\mathrm{mid}})K_{\mathrm{mid}}^n B_{1/2}^{9(n+1)} M_{1/2}^2 e^{2\Lambda_0\mu T}\left[M_f\frac{T^2}{(1+\mu T)^2} + M_g\frac{T^4}{(1+\mu T)^4}\right].$$

The dimensional prefactors satisfy, for $s/2 \le T \le s$,

$$s^{-2}\frac{T^2}{(1+\mu T)^2}(1+\xi)^{1/2} \le C,$$
$$s^{-4}\frac{T^4}{(1+\mu T)^4}(1+\xi)^{3/2} \le C,$$

because $T \le s$ and $1+\mu T \ge 1+\xi/2 \ge (1+\xi)/2$. With M_f and M_g from Step 1, the bracket is therefore at most $Ce^{-\xi/2}$, and

$$\|B_n\|_1 \le C(1+K_{\mathrm{mid}})B_{1/2}^9 M_{1/2}^2\, e^{-\xi/2} e^{2\Lambda_0\mu T}\big(K_{\mathrm{mid}}B_{1/2}^9\big)^n.$$

The bound on $\|B_n\|_1$ is thus (7.6) with $K_{\mathrm{pol}} = K_{\mathrm{mid}}B_{1/2}^9$ and $C_2 = C(1+K_{\mathrm{mid}})B_{1/2}^9 M_{1/2}^2$. The operator B_n is real and self-adjoint: the midpoint coefficients are real and symmetric in (α,β), because $f_{n,\bullet}$ is real and symmetric in $(\mathbf{x},\mathbf{y})$, and the vectors $\psi_\alpha^\bullet$ are real. □

7.3. **Interpolation at a small joint reserve power.** The constants in (5.6) fix the parameters

$$\Lambda_0 = 2^{15}, \qquad \theta = \frac{1}{4\Lambda_0} = 2^{-17}, \qquad a = \frac{\theta}{2} = 2^{-18}, \qquad \gamma = 2(1-\theta) = 2(1-2^{-17}). \tag{7.10}$$

In particular, $0 < \theta \le 1/64$, $0 < a \le 1/8$, and $\gamma > 0$. The bound $a \le 1/8$ places a in the range $0 < a \le 1/2$ of Lemmas 4.3 and 4.4, and the order $r_- = 1/(1-a)$ of Section 8 then satisfies $1 < r_- < 2$, as required in Theorem 3.1.

The factor 16 in the moment enlargement (5.6) makes $\Lambda_0 = 16\lambda_{1/4} = 2^{15}$ a power of two, and with it θ and a. The interpolation weight $\theta = 1/(4\Lambda_0)$ is the value at which the growth factor $e^{2\Lambda_0\mu T}$ of (7.6), raised to the power θ, becomes $e^{\mu T/2}$, which the decay $e^{-\xi/2}$ of (7.5) dominates for $T \le 3s/4$; the reserve power $a = \theta/2$ arises because the interpolating family in the proof of Theorem 7.2 is $\mathcal{R}_{\mu,s}^{(1-z)/2} B_n \mathcal{R}_{\mu,s}^{(1-z)/2}$.

Theorem 7.2 (Jointly weighted reverse trace estimate)**.** *There are finite $S_{\mathrm{rev}}, \mathcal{J}_0$, independent of μ, such that for every $\mu > 0$ and $s \ge S_{\mathrm{rev}}$ the product-minus-joint covariance operator $\mathsf{D}_{\mu,s}$ of* (7.1) *factors, with a, γ as in* (7.10)*, as*

$$\mathsf{D}_{\mu,s} = \mathcal{R}_{\mu,s}^a Y_{\mu,s} \mathcal{R}_{\mu,s}^a, \qquad Y_{\mu,s} = Y_{\mu,s}^* \in \mathcal{S}_1, \qquad \|Y_{\mu,s}\|_1 \le \mathcal{J}_0 s^{-\gamma}. \tag{7.11}$$

Here $\mathcal{R}_{\mu,s}$ is the uncertainty reserve of the joint vacuum covariance norm on the union U_s of the two unit balls.

Proof. Step 1: interpolation for each increment. For each increment G_n of Lemma 7.1, let $Z_n = \mathcal{R}_{\mu,s}^{(1-\theta)/2} B_n \mathcal{R}_{\mu,s}^{(1-\theta)/2}$, with B_n from (7.6). Trace-norm interpolation gives

$$\mathsf{G}_n = \mathcal{R}_{\mu,s}^{a} Z_n \mathcal{R}_{\mu,s}^{a}, \qquad \|Z_n\|_1 \le \|\mathsf{G}_n\|_1^{1-\theta} \, \|B_n\|_1^{\theta}. \tag{7.12}$$

Indeed, the analytic family

$$z \longmapsto \mathcal{R}_{\mu,s}^{(1-z)/2} B_n \mathcal{R}_{\mu,s}^{(1-z)/2}, \qquad 0 \le \operatorname{Re} z \le 1,$$

has boundary trace norms $\|\mathsf{G}_n\|_1$ and $\|B_n\|_1$. Since $0 < \mathcal{R}_{\mu,s} \le I$, the powers $\mathcal{R}_{\mu,s}^{w}$ with $\operatorname{Re} w \ge 0$, given by the spectral calculus on $(0,1]$, are contractions, and unitary for $\operatorname{Re} w = 0$. For $\operatorname{Re} z = 0$, $\mathcal{R}_{\mu,s}^{(1-z)/2} = \mathcal{R}_{\mu,s}^{1/2} \mathcal{R}_{\mu,s}^{-z/2}$, and for $\operatorname{Re} z = 1$, $\mathcal{R}_{\mu,s}^{(1-z)/2}$ is unitary; with (7.6),

$$\left\| \mathcal{R}_{\mu,s}^{(1-z)/2} B_n \mathcal{R}_{\mu,s}^{(1-z)/2} \right\|_1 = \begin{cases} \left\| \mathcal{R}_{\mu,s}^{-z/2} \mathsf{G}_n \mathcal{R}_{\mu,s}^{-z/2} \right\|_1 = \|\mathsf{G}_n\|_1 , & \operatorname{Re} z = 0, \\ \|B_n\|_1 , & \operatorname{Re} z = 1. \end{cases}$$

At $z = \theta$, the value of the family is Z_n, and multiplication by $\mathcal{R}_{\mu,s}^{\theta/2} = \mathcal{R}_{\mu,s}^{a}$ on both sides recovers $\mathcal{R}_{\mu,s}^{1/2} B_n \mathcal{R}_{\mu,s}^{1/2} = \mathsf{G}_n$.

Spectral cutoffs and their strong limits prove the continuity and holomorphy of the family, also when zero lies in the spectrum of $\mathcal{R}_{\mu,s}$. Applied to any fixed vector, $w \mapsto \mathcal{R}_{\mu,s}^{w}$ is continuous on $\operatorname{Re} w \ge 0$ and holomorphic on $\operatorname{Re} w > 0$. Since B_n is a finite combination of the rank-one operators $|\psi_\alpha^\bullet\rangle\langle\psi_\beta^\bullet|$ of the proof of Lemma 7.1, the family is the same combination of $|\mathcal{R}_{\mu,s}^{(1-z)/2}\psi_\alpha^\bullet\rangle\langle\mathcal{R}_{\mu,s}^{(1-\bar z)/2}\psi_\beta^\bullet|$. It is therefore continuous in trace norm on the closed strip and holomorphic inside, with trace norm at most $\|B_n\|_1$.

Step 2: the exponent. Inserting (7.5) and (7.6) into the bound of (7.12) gives

$$\begin{aligned} \|Z_n\|_1 &\le \Big(C_1 (n+1)^3 s^{-2} (1+\xi)^2 e^{-\xi/2} (c_{\mathrm{pol}}/s)^n \Big)^{1-\theta} \Big(C_2 e^{-\xi/2} e^{2\Lambda_0 \mu T} K_{\mathrm{pol}}^n \Big)^{\theta} \\ &= C_1^{1-\theta} C_2^{\theta} \, s^{-2(1-\theta)} (1+\xi)^{2(1-\theta)} e^{-\xi/2 + 2\Lambda_0 \theta \mu T} (n+1)^{3(1-\theta)} \left[(c_{\mathrm{pol}}/s)^{1-\theta} K_{\mathrm{pol}}^{\theta} \right]^n , \end{aligned}$$

since $-(1-\theta)\xi/2 - \theta\xi/2 = -\xi/2$. Here $s^{-2(1-\theta)} = s^{-\gamma}$ and $2\Lambda_0\theta = 1/2$ by (7.10). For $s \ge 4$, the enclosing radius satisfies $T \le 3s/4$, so $\mu T/2 \le 3\xi/8$ and

$$-\xi/2 + 2\Lambda_0 \theta \mu T = -\xi/2 + \mu T/2 \le -\xi/8.$$

With this bound on the exponent, the estimate of $\|Z_n\|_1$ above becomes

$$\begin{aligned} \|Z_n\|_1 \le C_1^{1-\theta} C_2^{\theta} s^{-\gamma} (1+\xi)^{2(1-\theta)} e^{-\xi/8} (n+1)^{3(1-\theta)} \\ \times \left[(c_{\mathrm{pol}}/s)^{1-\theta} K_{\mathrm{pol}}^{\theta} \right]^n . \end{aligned} \tag{7.13}$$

Step 3: summation. The threshold S_{rev} is chosen so that every $s \ge S_{\mathrm{rev}}$ satisfies $s \ge S_{\mathrm{pol}}$, $s \ge 5$, and

$$(c_{\mathrm{pol}}/s)^{1-\theta} K_{\mathrm{pol}}^{\theta} \le \tfrac{1}{2}.$$

With $(n+1)^{3(1-\theta)} \le (n+1)^3$ and $(1+\xi)^{2(1-\theta)} \le (1+\xi)^2$, (7.13) then gives

$$\sum_{n \ge 0} \|Z_n\|_1 \le C_1^{1-\theta} C_2^{\theta} s^{-\gamma} \Big(\sup_{\xi \ge 0} (1+\xi)^2 e^{-\xi/8} \Big) \sum_{n \ge 0} (n+1)^3 \, 2^{-n}.$$

The bounds

$$\sum_{n \ge 0} (n+1)^3 \, 2^{-n} = 52, \qquad \sup_{\xi \ge 0} (1+\xi)^2 e^{-\xi/8} \le 256$$

follow from $\sum_{n \ge 0} (n+1)^3 x^n = (1 + 4x + x^2)/(1-x)^4$ at $x = 1/2$, and from the maximum of $(1+\xi)^2 e^{-\xi/8}$ at $\xi = 15$, where it equals $256 e^{-15/8}$. They permit

$$\mathcal{J}_0 = 52 \cdot 256 \, C_1^{1-\theta} C_2^{\theta}. \tag{7.14}$$

Thus $Y_{\mu,s}=\sum_{n\ge 0}Z_n$ converges in trace norm and satisfies the trace bound $\|Y_{\mu,s}\|_1\le \mathcal{J}_0 s^{-\gamma}$.

Step 4: the factorization. Multiplication by the bounded powers $\mathcal{R}^a_{\mu,s}$ on both sides is continuous in trace norm, and it identifies $\mathcal{R}^a_{\mu,s}Y_{\mu,s}\mathcal{R}^a_{\mu,s}$ with $\sum_n \mathsf{G}_n=\mathsf{D}_{\mu,s}$:

$$\mathcal{R}^a_{\mu,s}Y_{\mu,s}\mathcal{R}^a_{\mu,s}=\sum_{n\ge 0}\mathcal{R}^a_{\mu,s}Z_n\mathcal{R}^a_{\mu,s}\overset{(7.12)}{=}\sum_{n\ge 0}\mathsf{G}_n=\lim_{N\to\infty}\mathsf{F}_{N+1}\overset{\text{Lem. 7.1}}{=}\mathsf{D}_{\mu,s},$$

since $\sum_{n=0}^{N}\mathsf{G}_n=\mathsf{F}_{N+1}-\mathsf{F}_0=\mathsf{F}_{N+1}$ and $\mathsf{F}_n\to\mathsf{D}_{\mu,s}$ in operator norm. Each Z_n is self-adjoint, because B_n is, and self-adjointness passes to the trace-norm limit. □

8. Completion of the regional Bell bound

8.1. **Comparisons on every finite source system.** For the parameters (7.10), we fix

$$r_-=\frac{1}{1-a}=\frac{2^{18}}{2^{18}-1},\qquad r_+=\left\lceil\frac{4}{a}\right\rceil=2^{20},\qquad b=1-\frac{1}{r_+}=1-2^{-20}. \tag{8.1}$$

Here r_+ is an integer, as required in (4.11). Then $1<r_-<2$, $r_+\ge 32$, $ar_+=4$, and, by (3.2),

$$\eta=1-\frac{1-b}{ab^2}=1-\frac{1}{4b^2}>0.$$

The reverse exponent a constrains the forward order through the condition $\eta>0$: since $1-b=1/r_+$, positivity of η requires $r_+>1/(ab^2)$, so the small value of a forces a large order; the choice $r_+=4/a$ gives the exponent $b=1-a/4$ close to 1 and, through (8.16), the size of C_*. Theorem 6.1 is used at this fixed b.

For arbitrary finite canonical source systems F_A,F_B in the two unit balls, the symplectic form (2.1) vanishes between a source in F_A and a source in F_B, because the balls are disjoint. On $F=F_A\oplus F_B$, the form is therefore the direct sum of its nondegenerate restrictions to F_A and F_B. Thus F is a finite canonical source system, and, by the Weyl relations, the Weyl operators of F_A commute with those of F_B.

Let ρ_F be the joint Gaussian vacuum state of F, and let

$$\sigma_F=(\rho_F)_{F_A}\otimes(\rho_F)_{F_B}$$

be the product of its marginals. Both states are centred Gaussian states on F. They are faithful by (2.8), applied to F for ρ_F and to F_A and F_B for the two factors of σ_F.

The thermal comparison constants (4.9), the constant F_b of Theorem 6.1 and the constant $\mathcal{J}_0$ of Theorem 7.2 give

$$A_+=c_b^{-1}F_b=(r_+-1)F_b,\qquad A_-=c_a^{-1}\mathcal{J}_0=2^{2a-1}\mathcal{J}_0. \tag{8.2}$$

The comparison envelopes are

$$\varepsilon_+(s)=A_+s^{-2},\qquad \varepsilon_-(s)=A_-s^{-\gamma}. \tag{8.3}$$

For the forward comparison, let $\iota_A\colon F_A\to\mathcal{X}_{\mu,B_1(0)}$ and $\iota_B\colon F_B\to\mathcal{X}_{\mu,B_1(s\mathbf{e}_1)}$ be the inclusions; they are isometries when F_A and F_B carry the marginal covariance inner products $\mathsf{v}_{\mu,B_1(0)}$ and $\mathsf{v}_{\mu,B_1(s\mathbf{e}_1)}$. On F we use the orthogonal sum of these two inner products, which is the product covariance $\mathsf{v}_{\mathrm{prod},\mu,s}$, and coordinates orthonormal for it. In these product-normalized coordinates, $V_{\sigma_F}=I$, because σ_F has the product covariance, and the definition of the cross-covariance operator $C_{\mu,s}$ in Section 6 gives

$$V_{\rho_F}-V_{\sigma_F}=\begin{pmatrix}0 & \iota_A^*C_{\mu,s}\iota_B\\ (\iota_A^*C_{\mu,s}\iota_B)^* & 0\end{pmatrix}.$$

Let $\mathsf{Z}_{+,F}$ denote the operator Z of Theorem 4.1 for $(\rho,\sigma,r)=(\rho_F,\sigma_F,r_+)$ in these coordinates. By Step 1 of the proof of Theorem 4.1, the norms of Z may be computed in these coordinates, and in the joint coordinates below.

Here $\tau = (r_+ - 1)/r_+ = b$. Since σ_F is a product state, its thermal power σ_F^b is the product of the thermal powers of its factors, so the thermal increment of $V_{\sigma_F} = I$ is block diagonal, with blocks $H_{A,F} = \mathsf{H}_b(I_{F_A})$ and $H_{B,F} = \mathsf{H}_b(I_{F_B})$. Hence

$$\mathsf{Z}_{+,F} = \begin{pmatrix} 0 & \mathsf{Z}_{AB} \\ \mathsf{Z}_{AB}^* & 0 \end{pmatrix}, \qquad \mathsf{Z}_{AB} = H_{A,F}^{-1/2} \iota_A^* C_{\mu,s} \iota_B H_{B,F}^{-1/2}.$$

The high-order compression (4.11), applied in each ball with the integer r_+, gives $H_{A,F} \geq c_b \iota_A^* \mathsf{R}_A^{2b} \iota_A$ and $H_{B,F} \geq c_b \iota_B^* \mathsf{R}_B^{2b} \iota_B$, the hypotheses of (4.13). For $s \geq S_b$, the trace estimate (6.3) at this b then gives

$$\begin{aligned} \|\mathsf{Z}_{AB}\|_1 &\overset{(6.3)}{=} \left\| H_{A,F}^{-1/2} \iota_A^* \mathsf{R}_A^b X_{b,\mu,s} \mathsf{R}_B^b \iota_B H_{B,F}^{-1/2} \right\|_1 \\ &\overset{(4.13)}{\leq} c_b^{-1} \|X_{b,\mu,s}\|_1 \overset{(6.3)}{\leq} c_b^{-1} F_b s^{-2} \overset{(8.2),(8.3)}{=} \varepsilon_+(s). \end{aligned}$$

The eigenvalues of $\mathsf{Z}_{+,F}$ are plus and minus the singular values of Z_{AB}, together with zeros. Hence $\|\mathsf{Z}_{+,F}\| = \|\mathsf{Z}_{AB}\| \leq \|\mathsf{Z}_{AB}\|_1$ and $\|\mathsf{Z}_{+,F}\|_1 = 2\|\mathsf{Z}_{AB}\|_1$, which gives

$$\|\mathsf{Z}_{+,F}\| \leq \varepsilon_+(s), \qquad \|\mathsf{Z}_{+,F}\|_1 \leq 2\varepsilon_+(s). \tag{8.4}$$

The number k_r of Theorem 4.1 is here $k_{r_+} = \max\{2, 1/b\} = 2$, since $b \geq 1/2$. If $\varepsilon_+(s) \leq 1/8$, then (8.4) gives the hypothesis $\|\mathsf{Z}_{+,F}\| \leq 1/8 = (4k_{r_+})^{-1}$ of Theorem 4.1, and (4.3) gives $\log \widetilde{Q}_{r_+}(\rho_F \| \sigma_F) \leq 2r_+ k_{r_+} \|\mathsf{Z}_{+,F}\|_1 \leq 2r_+ \cdot 2 \cdot 2\varepsilon_+(s)$, that is,

$$\log \widetilde{Q}_{r_+}(\rho_F \| \sigma_F) \leq 8 r_+ \varepsilon_+(s). \tag{8.5}$$

For the reverse comparison, the combined finite source space F is embedded isometrically in the joint covariance norm by the inclusion $\iota_F \colon F \to \mathcal{X}_{\mu,U_s}$, which preserves ς. The change to coordinates orthonormal for this norm, the joint coordinates, acts on the same finite system and may mix the coordinates of F_A and F_B. In the joint coordinates,

$$\begin{aligned} V_{\rho_F} = I_F, \qquad V_{\sigma_F} - V_{\rho_F} &= \iota_F^* \mathsf{D}_{\mu,s} \iota_F, \\ \mathsf{H}_a(I_F) \geq c_a \mathsf{R}_F^{2a} &\geq c_a \iota_F^* \mathcal{R}_{\mu,s}^{2a} \iota_F. \end{aligned} \tag{8.6}$$

In the first line, $V_{\rho_F} = I_F$ because ι_F is isometric for the joint covariance, and the difference is given by (7.1) because σ_F has the product covariance $\mathsf{v}_{\mathrm{prod},\mu,s}$. In the last line, R_F is the reserve of Lemma 4.4 for $\iota = \iota_F$; the line uses (4.9) on the finite space and (4.10), with $a \leq 1/2$. Theorem 4.1 is now applied with the roles of the two states exchanged: $\mathsf{Z}_{-,F}$ denotes the operator Z of Theorem 4.1 for $(\rho, \sigma, r) = (\sigma_F, \rho_F, r_-)$ in the joint coordinates (8.6).

Here $\tau = (r_- - 1)/r_- = a$. For $s \geq S_{\mathrm{rev}}$, Theorem 7.2 gives the factorization (7.11), and with the first line of (8.6),

$$\begin{aligned} \mathsf{Z}_{-,F} = \mathsf{H}_a(I_F)^{-1/2} \iota_F^* \mathsf{D}_{\mu,s} \iota_F \mathsf{H}_a(I_F)^{-1/2} &\overset{(7.11)}{=} \mathsf{N}^* Y_{\mu,s} \mathsf{N}, \\ \mathsf{N} = \mathcal{R}_{\mu,s}^a \iota_F \mathsf{H}_a(I_F)^{-1/2}&. \end{aligned}$$

The last line of (8.6) is the hypothesis of the one-space normalization bound stated before (4.13), with $H_F = \mathsf{H}_a(I_F)$ and $\iota = \iota_F$. That bound gives $\|\mathsf{N}\|^2 \leq c_a^{-1}$, and therefore

$$\|\mathsf{Z}_{-,F}\|_1 \leq \|\mathsf{N}\|^2 \|Y_{\mu,s}\|_1 \overset{(7.11)}{\leq} c_a^{-1} \mathcal{J}_0 s^{-\gamma} \overset{(8.2),(8.3)}{=} \varepsilon_-(s),$$

that is,

$$\|\mathsf{Z}_{-,F}\|_1 \leq \varepsilon_-(s). \tag{8.7}$$

Here $k_{r_-} = \max\{2, 1/a\} = 1/a$, since $a \le 1/2$. If $\varepsilon_-(s) \le a/4$, then (8.7) gives $\|\mathsf{Z}_{-,F}\| \le \|\mathsf{Z}_{-,F}\|_1 \le a/4 = (4k_{r_-})^{-1}$, the hypothesis of Theorem 4.1, and (4.3) gives $\log \widetilde{Q}_{r_-}(\sigma_F\|\rho_F) \le 2r_-k_{r_-}\|\mathsf{Z}_{-,F}\|_1 \le 2r_-a^{-1}\varepsilon_-(s)$, that is,

$$\log \widetilde{Q}_{r_-}(\sigma_F\|\rho_F) \le \frac{2r_-}{a}\varepsilon_-(s). \tag{8.8}$$

All constants in (8.4)–(8.8) are independent of the finite source systems and their symplectic spectra.

8.2. **Regional events and bounded observables.** Let E and P be effects of the von Neumann algebras generated by the Weyl operators of F_A and of F_B, and write $e_A = \omega(E)$, $e_B = \omega(P)$ and $w = \omega(EP)$. The two algebras commute, so the product EP is an effect. Its expectations in the two states are $\rho_F(EP) = w$, since ρ_F is the vacuum state on F, and $\sigma_F(EP) = e_Ae_B$, since σ_F is the product of the marginals of ρ_F.

When (8.5) and (8.8) hold, the two Rényi quantities there are finite, and the effect bound (4.7) of Lemma 4.2 with $M = EP$, applied to $(\rho, \sigma, r) = (\rho_F, \sigma_F, r_+)$ and to (σ_F, ρ_F, r_-), gives

$$w^{r_+} \le \widetilde{Q}_{r_+}(\rho_F\|\sigma_F)\,(e_Ae_B)^{r_+-1}, \qquad (e_Ae_B)^{r_-} \le \widetilde{Q}_{r_-}(\sigma_F\|\rho_F)\,w^{r_--1}.$$

Bounding the two Rényi quantities by (8.5) and (8.8), we obtain

$$w^{r_+} \le e^{8r_+\varepsilon_+(s)}(e_Ae_B)^{r_+-1}, \tag{8.9}$$

$$(e_Ae_B)^{r_-} \le e^{2r_-\varepsilon_-(s)/a}w^{r_--1}. \tag{8.10}$$

Since (8.9) and (8.10) contain no denominators, they have the form $\Phi(e_A, e_B, w) \le 0$ with Φ continuous on $[0,1]^3$, zero probabilities included, because $r_+ - 1 > 0$ and $r_- - 1 > 0$; their constants depend only on s. Lemma 2.3 therefore passes these continuous inequalities to all effects in the two regional algebras. The suprema in (3.1) range over projections with $e_Ae_B > 0$. For such projections E and P, (8.9) bounds the quotient in $K_{r_+,+}$ by $e^{8r_+\varepsilon_+(s)}$, and (8.10) forces $w > 0$ and bounds the quotient in $K_{r_-,-}$ by $e^{2r_-\varepsilon_-(s)/a}$. Hence the constants $K_{r_+,+}(\mu, s)$ and $K_{r_-,-}(\mu, s)$ of (3.1) for the two regional algebras at mass μ and separation s satisfy

$$\log K_{r_+,+}(\mu, s) \le 8r_+\varepsilon_+(s), \qquad \log K_{r_-,-}(\mu, s) \le \frac{2r_-}{a}\varepsilon_-(s). \tag{8.11}$$

The trace-distance conclusion (4.8), which holds for $r_+ \ge 2$, and (8.5) give

$$\|\rho_F - \sigma_F\|_1 \overset{(4.8)}{\le} \sqrt{\widetilde{Q}_{r_+}(\rho_F\|\sigma_F)^{1/(r_+-1)} - 1} \overset{(8.5)}{\le} \sqrt{\exp\Big(\frac{8r_+\varepsilon_+(s)}{r_+ - 1}\Big) - 1},$$

and, since $r_+/(r_+ - 1) = 1/b$,

$$\|\rho_F - \sigma_F\|_1 \le \sqrt{\exp(8\varepsilon_+(s)/b) - 1}. \tag{8.12}$$

For self-adjoint contractions A and B of the von Neumann algebras generated by the Weyl operators of F_A and of F_B, one has $\omega(AB) = \rho_F(AB)$, $\omega(A)\omega(B) = \sigma_F(AB)$ and $\|AB\| \le 1$, so that

$$|\omega(AB) - \omega(A)\omega(B)| = |\mathrm{Tr}((\rho_F - \sigma_F)AB)| \le \|\rho_F - \sigma_F\|_1 \overset{(8.12)}{\le} \sqrt{\exp(8\varepsilon_+(s)/b) - 1}.$$

The right-hand side depends only on s, and Lemma 2.3 extends the bound to the full regional algebras.

If $\varepsilon_+(s) \le b/8$, then $x = 8\varepsilon_+(s)/b$ lies in $[0,1]$, and the inequality $e^x - 1 \le 2x$ for $0 \le x \le 1$ gives

$$\sqrt{\exp(8\varepsilon_+(s)/b) - 1} \le \sqrt{16\varepsilon_+(s)/b} \overset{(8.3)}{=} \frac{4\sqrt{A_+/b}}{s}.$$

Hence every pair of self-adjoint contractions A and B of the two regional algebras satisfies

$$|\omega(AB) - \omega(A)\omega(B)| \le \frac{Q_0}{s}, \qquad Q_0 = 4\sqrt{A_+/b}. \tag{8.13}$$

8.3. **The sufficient separation ratio.** The geometric thresholds can be chosen to satisfy

$$S_{\mathrm{rev}} \geq \max\left\{S_{\mathrm{pol}},\, 5,\, c_{\mathrm{pol}}\, 2^{1/(1-\theta)} K_{\mathrm{pol}}^{\theta/(1-\theta)}\right\}, \qquad S_{\mathrm{geo}} \geq \max\{5, S_b, S_{\mathrm{rev}}\}. \tag{8.14}$$

The forward geometric condition (6.5), through $S_{\mathrm{geo}} \geq S_b$, requires $S_{\mathrm{geo}} \geq 2\mathfrak{k}_b = (4+80\sqrt{3})\Lambda_b$, because $\mathfrak{k}_b = 2\Lambda_b + 8\sqrt{3}B_b$ and $B_b = 5\Lambda_b$. Since $\Lambda_b = 2^{15}/\sin(\pi b) \geq 2^{15}$ by (5.6), the bound $(4+80\sqrt{3})\Lambda_b$ is at least $(4+80\sqrt{3})2^{15} > 4.6 \times 10^6$ for every $b \in [1/2, 1)$. For the present choice $b = 1 - 2^{-20}$, one has $\sin(\pi b) = \sin(\pi/2^{20}) \leq \pi/2^{20}$, which gives the numerical lower requirement

$$S_{\mathrm{geo}} \geq (4+80\sqrt{3})\Lambda_b = \frac{(4+80\sqrt{3})2^{15}}{\sin(\pi/2^{20})} \geq \frac{(4+80\sqrt{3})2^{35}}{\pi} > 1.5 \times 10^{12}.$$

For the orders (8.1), the probability exponents (3.4) become

$$\nu_+ = \frac{1+b}{4b^2}, \qquad \nu_- = 2^{18} - 1, \qquad \eta = 1 - \frac{1}{4b^2}. \tag{8.15}$$

We denote the probability contribution to the separation ratio by

$$C_{\mathrm{prob}} = Q_0 \exp\left(\frac{2\log 2/(ab) + \nu_+ + \nu_-}{\eta}\right). \tag{8.16}$$

Table 1 gives the dependency chain. Constants defined by finite covers, cutoff norms, and approximation bounds are known only to be finite.

TABLE 1. Constants entering the sufficient ratio C_*.

Constant	Choice or definition	Source
C_c, λ_c	C_c from the finite coarea and submean cover; $\lambda_c = 2048/\sin(2\pi c)$.	Theorem 5.1
Λ_b, B_b	$\Lambda_b = 2^{15}/\sin(\pi b)$, $B_b = 5\Lambda_b$.	(5.6)
M_b	Weighted moment coefficient, depending only on b.	Lemma 5.4
$\mathfrak{d}_b, d_{0,b}, \mathfrak{k}_b$	$2B_b,\ 4\sqrt{3}B_b,\ 2\Lambda_b + 8\sqrt{3}B_b$.	Section 6
S_b, F_b	$S_b \geq \max\{4, 4d_{0,b}, 2\mathfrak{k}_b\}$; F_b from (6.4).	Theorem 6.1
r_0, S_{pol}	$r_0 = 1/100$; S_{pol} covers the fixed analytic neighbourhoods and polynomial error estimates.	Lemma 7.1
$C_1, C_2, c_{\mathrm{pol}}, K_{\mathrm{pol}}$	Constants of (7.5) and (7.6); c_{pol} depends only on r_0, and the others include the fixed cutoff norms, coefficient bound K_{mid}, rank bound, and $M_{1/2}$.	Lemma 7.1
$\Lambda_0, \theta, a, \gamma$	$2^{15},\ 2^{-17},\ 2^{-18},\ 2(1-2^{-17})$.	(7.10)
$S_{\mathrm{rev}}, \mathcal{J}_0$	S_{rev} as in (8.14); $\mathcal{J}_0 = 52 \cdot 256\, C_1^{1-\theta} C_2^{\theta}$.	(8.14), (7.14)
r_-, r_+, b	$2^{18}/(2^{18}-1),\ 2^{20},\ 1-2^{-20}$.	(8.1)
c_a, c_b	$2^{1-2a},\ (1-b)/b = 1/(2^{20}-1)$.	(4.9)
η, ν_+, ν_-	$1 - 1/(4b^2),\ (1+b)/(4b^2),\ 2^{18}-1$.	(8.15)
A_+, A_-	$(r_+ - 1)F_b,\ 2^{2a-1}\mathcal{J}_0$.	(8.2)
Q_0	$4\sqrt{A_+/b}$.	(8.13)
S_{geo}	Any constant at least $\max\{5, S_b, S_{\mathrm{rev}}\}$.	(8.14)
C_{prob}	$Q_0 \exp[(2\log 2/(ab) + \nu_+ + \nu_-)/\eta]$.	(8.16)
C_*	Any finite number satisfying the four requirements below.	(8.17)

We choose a finite C_* satisfying

$$C_* \geq \max\left\{S_{\mathrm{geo}},\ \sqrt{8r_+A_+},\ \left(\frac{4A_-}{a}\right)^{1/\gamma},\ C_{\mathrm{prob}}\right\}. \tag{8.17}$$

8.4. **Proof of the main theorem and the fixed-regions corollary.**

Theorem 1.1 (Separation threshold)**.** *There is a finite constant $C_* > 2$, independent of m and R, such that*

$$\forall m, R > 0, \qquad L \geq C_* R \quad \Longrightarrow \quad \beta_m(R, L) = 1.$$

Proof of Theorem 1.1. We first use unit-radius balls and arbitrary mass $\mu > 0$, and we fix $s \geq C_*$.

Step 1: the range of s. By the first entry of (8.17) and by (8.14),

$$s \geq C_* \geq S_{\mathrm{geo}} \geq \max\{5, S_b, S_{\mathrm{rev}}\}.$$

Hence $s \geq S_b$ and $s \geq S_{\mathrm{rev}}$, the ranges of Theorems 6.1 and 7.2, so (8.4) and (8.7) hold for all finite canonical source systems in the two balls. The same chain gives $C_* \geq 5 > 2$.

Step 2: the envelopes. The second and third entries of (8.17) give $s^2 \geq 8r_+A_+$ and $s^\gamma \geq 4A_-/a$. By (8.3),

$$\varepsilon_+(s) = \frac{A_+}{s^2} \leq \frac{1}{8r_+} \leq \frac{b}{8} < \frac{1}{8}, \qquad \varepsilon_-(s) = \frac{A_-}{s^\gamma} \leq \frac{a}{4},$$

where $1/r_+ \leq b$ because $r_+ b = r_+ - 1 \geq 1$. The bounds $\varepsilon_+(s) \leq 1/8$ and $\varepsilon_-(s) \leq a/4$ are the conditions of (8.5) and (8.8), which therefore hold, and with them (8.9)–(8.12). The bound $\varepsilon_+(s) \leq b/8$ is the condition of (8.13), which therefore holds for all self-adjoint contractions $A \in \mathfrak{A}_\mu(D_1)$ and $B \in \mathfrak{A}_\mu(D_1 + (0, s\mathbf{e}_1))$, the double-cone algebras of the two balls.

Step 3: the event constants. By (8.11) and Step 2,

$$\log K_{r_+,+} \leq 8r_+\varepsilon_+(s) \leq 8r_+ \cdot \frac{1}{8r_+} = 1, \qquad \log K_{r_-,-} \leq \frac{2r_-}{a}\varepsilon_-(s) \leq \frac{2r_-}{a} \cdot \frac{a}{4} = \frac{r_-}{2} < 1,$$

the last inequality because $r_- < 2$. In particular, $K_{r_+,+}$ and $K_{r_-,-}$ are finite.

Step 4: the cutoff (3.5). By Step 2, the correlation bound (3.3) holds with $q_{\mathrm{cl}} = Q_0/s$. The exponents η, ν_+, ν_- of (8.15) are positive, and the last entry of (8.17) is the number C_{prob} of (8.16), so $s \geq C_{\mathrm{prob}}$. Hence

$$\left(\frac{Q_0}{s}\right)^\eta K_{r_+,+}^{\nu_+} K_{r_-,-}^{\nu_-} \overset{\text{Step 3}}{\leq} \left(\frac{Q_0}{s}\right)^\eta e^{\nu_+ + \nu_-} \overset{s \geq C_{\mathrm{prob}}}{\leq} \left(\frac{Q_0}{C_{\mathrm{prob}}}\right)^\eta e^{\nu_+ + \nu_-} \overset{(8.16)}{=} 2^{-2/(ab)},$$

which is (3.5) with $q_{\mathrm{cl}} = Q_0/s$.

Step 5: the unit-radius conclusion. By (2.3), the algebras $\mathfrak{A}_\mu(D_1)$ and $\mathfrak{A}_\mu(D_1 + (0, s\mathbf{e}_1))$ are generated by the Weyl operators of sources in the disjoint balls $B_1(0)$ and $B_1(s\mathbf{e}_1)$. These Weyl operators commute, because the symplectic form vanishes between the two balls, so the two algebras commute. The orders (8.1) satisfy $r_+ > 2$ and $1 < r_- < 2$, and they give $b = 1 - 1/r_+$ and $a = 1 - 1/r_-$, as in (3.2), with $\eta > 0$. With Steps 3 and 4, every hypothesis of Theorem 3.1 holds, and that theorem gives

$$\forall \mu > 0, \qquad s \geq C_* \quad \Longrightarrow \quad \beta_\mu(1, s) = 1.$$

Step 6: scaling. Let $m, R > 0$ and $L \geq C_* R$. Then $mR > 0$ and $L/R \geq C_*$, so Lemma 2.1 and Step 5 give

$$\beta_m(R, L) = \beta_{mR}(1, L/R) = 1.$$

This proves (1.1). Every entry of (8.17) is independent of μ, so the finite constant C_* does not depend on m or R. □

Proof of Corollary 1.2. Let $m > 0$, let O_1, O_2 be bounded regions, and let e be the spacelike vector in the given direction with $e \cdot e = -1$, so that the translates of O_2 in that direction are $O_2 + Le$, $L > 0$.

Step 1: a spatial direction. Since $e \cdot e = -1$, a boost followed by a rotation gives a proper orthochronous Lorentz transformation Λ with $\Lambda e = (0, \mathbf{e}_1)$. By Poincaré covariance of the free field,

a unitary operator that fixes the vacuum vector carries $\mathfrak{A}_m(O)$ onto $\mathfrak{A}_m(\Lambda O)$ for every region O, so, as in Step 4 of the proof of Lemma 2.1,

$$\beta(\omega;\mathfrak{A}_m(O_1),\mathfrak{A}_m(O_2+Le)) = \beta(\omega;\mathfrak{A}_m(\Lambda O_1),\mathfrak{A}_m(\Lambda O_2+(0,L\mathbf{e}_1))).$$

Step 2: double cones centred at time zero. The bounded set $\Lambda O_1 \cup \Lambda O_2$ lies in a double cone D_R centred at the origin, so that

$$\Lambda O_1 \subset D_R, \qquad \Lambda O_2 + (0,L\mathbf{e}_1) \subset D_R + (0,L\mathbf{e}_1).$$

The radius R does not depend on L, and the centres 0 and $(0,L\mathbf{e}_1)$ lie on the time-zero slice at distance L. For $L > 2R$, the bases $B_R(0)$ and $B_R(L\mathbf{e}_1)$ are disjoint, and the algebras $\mathfrak{A}_m(D_R)$ and $\mathfrak{A}_m(D_R+(0,L\mathbf{e}_1))$ commute, as in Step 5 of the proof of Theorem 1.1.

Step 3: isotony. If one region is contained in another, its local algebra is contained in that of the other. The supremum (2.4) can only increase when both algebras are enlarged, so the inclusions of Step 2 give

$$\beta(\omega;\mathfrak{A}_m(\Lambda O_1),\mathfrak{A}_m(\Lambda O_2+(0,L\mathbf{e}_1))) \le \beta(\omega;\mathfrak{A}_m(D_R),\mathfrak{A}_m(D_R+(0,L\mathbf{e}_1))) = \beta_m(R,L).$$

Step 4: conclusion. Let $L \ge C_* R$, which exceeds $2R$. By Steps 1 to 3, one unitary carries $\mathfrak{A}_m(O_1)$ and $\mathfrak{A}_m(O_2+Le)$ into the commuting algebras $\mathfrak{A}_m(D_R)$ and $\mathfrak{A}_m(D_R+(0,L\mathbf{e}_1))$, so $\mathfrak{A}_m(O_1)$ and $\mathfrak{A}_m(O_2+Le)$ commute, and their Bell correlation is at most $\beta_m(R,L)$, which equals 1 by Theorem 1.1. The identities $A_0 = A_1 = B_0 = B_1 = I$ give the value 1 in (2.4). Hence the Bell correlation of $\mathfrak{A}_m(O_1)$ and $\mathfrak{A}_m(O_2+Le)$ equals 1 for all $L \ge C_* R$. □

9. Open problems

(1) **Wedges at a positive gap and growing regions.** For complementary wedges, the Summers–Werner results give maximal Bell correlation [39, 40]. Wedges separated by a positive gap are unbounded localization regions, and their Bell correlation at large gap is open. Determining this Bell correlation requires control of source families extending arbitrarily far along the wedge. The Bell correlation at large separation is also open for bounded regions whose diameters grow in proportion to their separation, at ratios that Theorem 1.1 does not cover.

(2) **Other Bell inequalities, measurement settings, and local filtering.** The probability criterion of Theorem 3.1 uses the four correlators of CHSH, or equivalently two binary settings at each site. For larger numbers of settings or outcomes, the problem is to find analogous event criteria. The relation between the eventual value $\beta = 1$ of the CHSH correlation (2.4), other Bell inequalities, and filtered output states remains to be determined.

(3) **Other dimensions and fields.** The null-disc construction of Lemma 5.3 needs at least three complex spatial coordinates, so one or two spatial dimensions require a replacement for that construction. Every dimension other than three requires the appropriate source-space and covariance estimates. Extensions to other free fields and to interacting models pose corresponding questions about local modular structure and covariance comparison. The independence of C_* from the mass also motivates a continuity analysis of regional Bell correlations at $m = 0$, including the source-space identifications at that endpoint.

(4) **Numerical enclosure of the sufficient ratio.** The finite Lorentz cover determines C_c existentially; enclosing the Jacobians and submean constants of the cover is the first quantitative task. These constants then enter $M_b, F_b, \mathcal{J}_0$, and ultimately C_*. Even before those enclosures, the probability contribution (8.16) obeys, along $r_+ = 4/a$ (so that $b = 1 - a/4$, $2\log 2/(ab) = 2\log 2/a + O(1)$, and, by (3.2) and (3.4), $\nu_+ = (1+b)/(4b^2) = O(1)$, $\nu_- = 1/a - 1$ and

$\eta = 1 - 1/(4b^2) = 3/4 + O(a)$),

$$\log_{10}(C_{\mathrm{prob}}/Q_0) \overset{(8.16)}{=} \frac{2\log 2/(ab) + \nu_+ + \nu_-}{\eta \log 10} = \frac{(2\log 2 + 1)/a + O(1)}{(3/4 + O(a))\log 10}$$
$$= \frac{4(2\log 2 + 1)}{3\log 10}\,\frac{1}{a} + O(1) = \frac{1.38180596\ldots}{a} + O(1).$$

For the parameters in Table 1,

$$C_{\mathrm{prob}} = Q_0 \cdot 10^{362\,232.284166\ldots}, \qquad C_* \geq C_{\mathrm{prob}}.$$

A numerical enclosure of the full C_* is an open problem, and so is the true threshold, the infimum of the L_0 such that $\beta_m(R, L) = 1$ for all $L \geq L_0$.

(5) **The regime $L < C_* R$.** The dependence of $\beta_m(R, L)$ on mR and L/R, including the region of strict violation and the approach to the classical value, remains to be determined for $2R < L < C_* R$. Tangent-region maximality provides the short-distance reference point [40]. Results on selected smeared modes, such as the two-mode separability scale in [45], give mode-dependent scales. Relating these scales to the supremum over the full local algebras is a further quantitative problem.

Appendix A. Bargmann-kernel derivation of the Gaussian sandwich formulas

The formulas (4.4) follow from the Bargmann kernel [2]; the resulting covariance identities agree with the Gaussian product formalism by which Seshadreesan, Lami and Wilde compute $\widetilde{Q}_r(\rho\|\sigma)$ [32, Theorem 22]. The sandwich $\Theta = \sigma^{-\tau/2}\rho\sigma^{-\tau/2}$ of (4.4), which contains inverse powers of σ, is defined as a positive quadratic form, and its coherent-state diagonal supplies the trace-class criterion.

For the Bargmann kernel and the matrices built from it, quadratures are ordered as $(q_1, \ldots, q_n, p_1, \ldots, p_n)$. This ordering differs from the ordering $(q_1, p_1, \ldots, q_n, p_n)$, in which the commutation matrix is J_n, by an orthogonal permutation matrix Π. The change of coordinates conjugates every covariance matrix by Π (Step 1 of the proof of Theorem 4.1), and conjugation by Π commutes with sums, products, inverses and positive square roots and preserves determinants; so (A.5) and (A.6), whose matrices are formed from covariance matrices in this way, hold in either ordering.

The annihilation and number operators are $a_j = (q_j + ip_j)/\sqrt{2}$ and $N_j = a_j^* a_j$. The Fock vacuum $|0\rangle$ is annihilated by every a_j and has covariance $I/2$, and $|k\rangle$, $k \in \mathbb{N}^n$, is the number basis, $N_j|k\rangle = k_j|k\rangle$.

For a centred Gaussian state ρ, define

$$\mathsf{C}_\rho = V_\rho + \tfrac{1}{2}I, \qquad D_{\mathrm{B}} = \frac{1}{\sqrt{2}}\begin{pmatrix} I & I \\ iI & -iI \end{pmatrix}, \qquad S_{\mathrm{B}} = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}.$$

For $z \in \mathbb{C}^n$, the unnormalized exponential vectors and the normalized coherent states are

$$|e(z)\rangle = \exp\left(\sum_j z_j a_j^*\right)|0\rangle = \sum_{k\in\mathbb{N}^n} \frac{z^k}{\sqrt{k!}}\,|k\rangle, \qquad |z\rangle = e^{-|z|^2/2}|e(z)\rangle,$$

with $z^k = \prod_j z_j^{k_j}$ and $k! = \prod_j k_j!$. The kernel of ρ between exponential vectors is

$$\langle e(z)|\rho|e(w)\rangle = \frac{1}{\sqrt{\det \mathsf{C}_\rho}} \exp\left[\frac{1}{2}\begin{pmatrix}\bar{z}\\ w\end{pmatrix}^T M_\rho \begin{pmatrix}\bar{z}\\ w\end{pmatrix}\right], \tag{A.1}$$
$$M_\rho = S_{\mathrm{B}} - D_{\mathrm{B}}^T \mathsf{C}_\rho^{-1} D_{\mathrm{B}}.$$

Step 1: the kernel of a centred Gaussian state. To obtain this formula, the overlap with a normalized coherent state is computed as the Fourier transform of the product of the Gaussian

characteristic function and the vacuum characteristic function. With the characteristic function $\chi_A(u) = \mathrm{Tr}(Ae^{iu^T(q,p)})$, $u \in \mathbb{R}^{2n}$, of a trace-class operator A, where (q,p) is the vector of quadratures in this order, any two trace-class operators satisfy the trace formula

$$\mathrm{Tr}(AB) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^{2n}} \chi_A(u)\, \chi_B(-u)\, d^{2n}u.$$

The state ρ has $\chi_\rho(u) = e^{-\frac{1}{2}u^T V_\rho u}$, and the coherent projection has $\chi_{|z\rangle\langle z|}(u) = e^{-\frac{1}{4}u^T u} e^{iu^T r(z)}$, the characteristic function of $|0\rangle$ times a phase, where $r(z) = \sqrt{2}(\mathrm{Re}\, z, \mathrm{Im}\, z)$ is the vector of quadrature means of $|z\rangle$. With $e^{-\frac{1}{2}u^T V_\rho u} e^{-\frac{1}{4}u^T u} = e^{-\frac{1}{2}u^T \mathsf{C}_\rho u}$, the trace formula and the Fourier Gaussian integral give

$$\begin{aligned}\langle z|\rho|z\rangle = \mathrm{Tr}(\rho\, |z\rangle\langle z|) &= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^{2n}} e^{-\frac{1}{2}u^T V_\rho u}\, e^{-\frac{1}{4}u^T u}\, e^{-iu^T r(z)}\, d^{2n}u \\ &= \frac{1}{\sqrt{\det \mathsf{C}_\rho}} \exp[-\tfrac{1}{2} r(z)^T \mathsf{C}_\rho^{-1} r(z)].\end{aligned}$$

Since $r(z) = D_{\mathrm{B}}(\bar{z}, z)$ and $D_{\mathrm{B}}^T D_{\mathrm{B}} = S_{\mathrm{B}}$, the exponent of (A.1) at $w = z$ is

$$\frac{1}{2}\begin{pmatrix}\bar{z}\\ z\end{pmatrix}^T M_\rho \begin{pmatrix}\bar{z}\\ z\end{pmatrix} = \frac{1}{2} r(z)^T r(z) - \frac{1}{2} r(z)^T \mathsf{C}_\rho^{-1} r(z) = |z|^2 - \frac{1}{2} r(z)^T \mathsf{C}_\rho^{-1} r(z).$$

Multiplication of the overlap by $e^{|z|^2}$, which turns $\langle z|\rho|z\rangle$ into $\langle e(z)|\rho|e(z)\rangle$, therefore gives (A.1) on the diagonal. Both sides of (A.1) are entire functions of $(\bar{z}, w) \in \mathbb{C}^{2n}$: the left side is the series $\sum_{k,l} \bar{z}^k w^l \langle k|\rho|l\rangle / \sqrt{k!\, l!}$, which converges everywhere because $|\langle k|\rho|l\rangle| \le 1$, and the right side is the exponential of a quadratic polynomial. Polarization of the entire sesquianalytic kernel gives (A.1) off the diagonal.

Step 2: the kernel and the coherent diagonal of the sandwich. In Williamson coordinates, the faithful reference state is

$$\sigma = Z_\sigma \prod_j \zeta_j^{N_j}, \qquad \zeta_j = \frac{\upsilon_j - 1/2}{\upsilon_j + 1/2} \in (0,1), \qquad Z_\sigma = \prod_j (1 - \zeta_j).$$

For $0 < \tau < 1$, let

$$\mathbb{T}_\sigma = \mathrm{diag}(\zeta_j^{-\tau/2}) \oplus \mathrm{diag}(\zeta_j^{-\tau/2}).$$

Since $\sigma|k\rangle = Z_\sigma \zeta^k |k\rangle$, with $\zeta^k = \prod_j \zeta_j^{k_j}$ as for z^k, the series of $|e(z)\rangle$ gives $\sigma^{-\tau/2}|e(z)\rangle = Z_\sigma^{-\tau/2}|e(z')\rangle$ with $z'_j = \zeta_j^{-\tau/2} z_j$; in particular, $|e(z)\rangle$ lies in the domain of $\sigma^{-\tau/2}$. With $w'_j = \zeta_j^{-\tau/2} w_j$ likewise, $(\bar{z}', w') = \mathbb{T}_\sigma(\bar{z}, w)$, and the quadratic form $\Theta = \sigma^{-\tau/2}\rho\sigma^{-\tau/2}$ has the kernel

$$\langle e(z)|\Theta|e(w)\rangle = \langle \sigma^{-\tau/2} e(z)|\rho|\sigma^{-\tau/2} e(w)\rangle = Z_\sigma^{-\tau} \langle e(z')|\rho|e(w')\rangle,$$

which is (A.1) with z, w replaced by z', w' and its prefactor multiplied by $Z_\sigma^{-\tau}$.

On the diagonal $w = z$, the commutation $D_{\mathrm{B}} \mathbb{T}_\sigma = \mathbb{T}_\sigma D_{\mathrm{B}}$ gives $D_{\mathrm{B}}(\bar{z}', z') = \mathbb{T}_\sigma r(z)$, while $(\bar{z}', z')^T S_{\mathrm{B}}(\bar{z}', z') = 2\sum_j \zeta_j^{-\tau} |z_j|^2 = r(z)^T \mathbb{T}_\sigma^2 r(z)$ and $2|z|^2 = r(z)^T r(z)$. The coherent diagonal of Θ is therefore

$$\begin{aligned}\langle z|\Theta|z\rangle &= e^{-|z|^2} \langle e(z)|\Theta|e(z)\rangle \\ &\overset{(A.1)}{=} \frac{Z_\sigma^{-\tau}}{\sqrt{\det \mathsf{C}_\rho}} \exp\Big[-\tfrac{1}{2} r(z)^T r(z) + \tfrac{1}{2} r(z)^T \mathbb{T}_\sigma^2 r(z) - \tfrac{1}{2} r(z)^T \mathbb{T}_\sigma \mathsf{C}_\rho^{-1} \mathbb{T}_\sigma r(z)\Big] \\ &= \frac{Z_\sigma^{-\tau}}{\sqrt{\det \mathsf{C}_\rho}} \exp[-\tfrac{1}{2} r(z)^T \mathsf{C}_\tau^{-1} r(z)],\end{aligned}$$

where, in real quadrature coordinates, C_τ^{-1} denotes the real symmetric matrix

$$\mathsf{C}_\tau^{-1} = I - \mathbb{T}_\sigma^2 + \mathbb{T}_\sigma \mathsf{C}_\rho^{-1} \mathbb{T}_\sigma. \tag{A.2}$$

Step 3: positivity of C_τ^{-1}. The matrix $\mathbb{T}_\sigma$ is diagonal with entries $\zeta_j^{-\tau/2} > 1$, so $I - \mathbb{T}_\sigma^{-2}$ is diagonal with entries $1 - \zeta_j^\tau \in (0,1)$, and (A.2) reads

$$\mathsf{C}_\tau^{-1} = \mathbb{T}_\sigma \big(\mathsf{C}_\rho^{-1} - (I - \mathbb{T}_\sigma^{-2})\big) \mathbb{T}_\sigma.$$

Congruence by $\mathbb{T}_\sigma^{-1}$, which gives $\mathsf{C}_\tau^{-1} > 0$ exactly when $\mathsf{C}_\rho^{-1} > I - \mathbb{T}_\sigma^{-2}$, and inversion of positive matrices, which reverses their order, show that

$$\mathsf{C}_\tau^{-1} > 0 \quad \Longleftrightarrow \quad \mathsf{C}_\rho < (I - \mathbb{T}_\sigma^{-2})^{-1} \quad \Longleftrightarrow \quad V_\rho < \mathsf{T}_\tau, \qquad \mathsf{T}_\tau = V_\sigma^{[\tau]}, \tag{A.3}$$

where $(I - \mathbb{T}_\sigma^{-2})^{-1} = \mathsf{T}_\tau + I/2$. This identity holds because $\sigma^\tau / \operatorname{Tr} \sigma^\tau = \prod_j (1 - \zeta_j^\tau) \zeta_j^{\tau N_j}$ is σ with every ζ_j replaced by ζ_j^τ, while the definition of ζ_j is equivalent to $\upsilon_j + \frac{1}{2} = (1 - \zeta_j)^{-1}$. Hence $\mathsf{T}_\tau = V_\sigma^{[\tau]}$ is diagonal in Williamson coordinates, with entries $\upsilon_j^{[\tau]}$ satisfying $\upsilon_j^{[\tau]} + \frac{1}{2} = (1 - \zeta_j^\tau)^{-1}$, the diagonal entries of $(I - \mathbb{T}_\sigma^{-2})^{-1}$.

Step 4: the trace and the operator Θ. On finite linear combinations ψ of number states, $\langle \psi | \Theta | \psi \rangle = \langle \sigma^{-\tau/2} \psi | \rho | \sigma^{-\tau/2} \psi \rangle \geq 0$, so the sandwich defines a positive matrix form, with entries

$$\langle k | \Theta | l \rangle = Z_\sigma^{-\tau} (\zeta^k \zeta^l)^{-\tau/2} \langle k | \rho | l \rangle, \qquad k, l \in \mathbb{N}^n.$$

By the series of $|e(z)\rangle$, the coherent diagonal of Θ is the absolutely convergent series $\langle z | \Theta | z \rangle = e^{-|z|^2} \sum_{k,l} \bar{z}^k z^l \langle k | \Theta | l \rangle / \sqrt{k!\, l!}$. Angular integration over the phases of $z_1, \dots, z_n$ removes the terms with $k \neq l$, and $\pi^{-n} \int_{\mathbb{C}^n} e^{-|z|^2} |z^k|^2 \, d^{2n} z = k!$, where $d^{2n} z = \prod_j d \operatorname{Re} z_j \, d \operatorname{Im} z_j$. Monotone convergence of the nonnegative diagonal coefficients then identifies the integral with the sum of the number-state diagonal entries:

$$\sum_{k \in \mathbb{N}^n} \langle k | \Theta | k \rangle = \frac{1}{\pi^n} \int_{\mathbb{C}^n} \langle z | \Theta | z \rangle \, d^{2n} z = \frac{Z_\sigma^{-\tau}}{(2\pi)^n \sqrt{\det \mathsf{C}_\rho}} \int_{\mathbb{R}^{2n}} e^{-\frac{1}{2} r^T \mathsf{C}_\tau^{-1} r} \, d^{2n} r,$$

the last step by Step 2 and $d^{2n} z = 2^{-n} d^{2n} r$ for $r = r(z)$.

Under (A.3), the matrix (A.2) is positive definite, with inverse C_τ, and the last integral equals $(2\pi)^n (\det \mathsf{C}_\tau)^{1/2}$, so the diagonal sum is finite. A positive matrix form with finite diagonal sum extends uniquely to a positive trace-class operator. Hence Θ is a positive trace-class operator whose trace is the diagonal sum, and

$$\operatorname{Tr} \Theta = Z_\sigma^{-\tau} \sqrt{\frac{\det \mathsf{C}_\tau}{\det \mathsf{C}_\rho}} < \infty. \tag{A.4}$$

Since $\sigma^{\tau/2} |k\rangle = Z_\sigma^{\tau/2} (\zeta^k)^{\tau/2} |k\rangle$, the entries of Θ give

$$\langle k | \sigma^{\tau/2} \Theta \sigma^{\tau/2} | l \rangle = Z_\sigma^\tau (\zeta^k \zeta^l)^{\tau/2} \langle k | \Theta | l \rangle = \langle k | \rho | l \rangle,$$

so $\rho = \sigma^{\tau/2} \Theta \sigma^{\tau/2}$, both sides being bounded operators with the same matrix in the number basis.

Step 5: the covariance of the normalized sandwich. The matrices $\mathbb{T}_\sigma$, T_τ and L are diagonal, and $\mathbb{T}_\sigma^2 > I$. By Step 3, $\mathsf{T}_\tau + \frac{1}{2} I = (I - \mathbb{T}_\sigma^{-2})^{-1} = \mathbb{T}_\sigma^2 (\mathbb{T}_\sigma^2 - I)^{-1} = I + (\mathbb{T}_\sigma^2 - I)^{-1}$, and $\mathsf{L}^2 = (\mathsf{T}_\tau - \frac{1}{2} I)(\mathsf{T}_\tau + \frac{1}{2} I)$; hence

$$\mathsf{T}_\tau - \tfrac{1}{2} I = (\mathbb{T}_\sigma^2 - I)^{-1}, \qquad \mathsf{L} = (\mathsf{T}_\tau^2 - I/4)^{1/2} = \mathbb{T}_\sigma (\mathbb{T}_\sigma^2 - I)^{-1}.$$

These identities and the Woodbury matrix identity, in the form of [32, Lemma 5], applied to (A.2) with the invertible summands $I-\mathbb{T}_\sigma^2$ and $\mathbb{T}_\sigma \mathsf{C}_\rho^{-1}\mathbb{T}_\sigma$, give

$$\begin{aligned}\mathsf{C}_\tau &= (I-\mathbb{T}_\sigma^2)^{-1} - (I-\mathbb{T}_\sigma^2)^{-1}\big((I-\mathbb{T}_\sigma^2)^{-1}+\mathbb{T}_\sigma^{-1}\mathsf{C}_\rho\mathbb{T}_\sigma^{-1}\big)^{-1}(I-\mathbb{T}_\sigma^2)^{-1}\\ &= -(\mathsf{T}_\tau-\tfrac12 I) - \mathsf{L}(V_\rho-\mathsf{T}_\tau)^{-1}\mathsf{L} = -(\mathsf{T}_\tau-\tfrac12 I) + \mathsf{L}(\mathsf{T}_\tau-V_\rho)^{-1}\mathsf{L}.\end{aligned}$$

The second line uses $(I-\mathbb{T}_\sigma^2)^{-1} = -(\mathsf{T}_\tau-\frac12 I)$, $(I-\mathbb{T}_\sigma^2)^{-1}\mathbb{T}_\sigma = -\mathsf{L}$ and

$$(I-\mathbb{T}_\sigma^2)^{-1}+\mathbb{T}_\sigma^{-1}\mathsf{C}_\rho\mathbb{T}_\sigma^{-1} = \mathbb{T}_\sigma^{-1}\big(\mathsf{C}_\rho-\mathsf{T}_\tau-\tfrac12 I\big)\mathbb{T}_\sigma^{-1} = \mathbb{T}_\sigma^{-1}(V_\rho-\mathsf{T}_\tau)\mathbb{T}_\sigma^{-1},$$

which is invertible under (A.3).

By Step 2 and (A.4), the density operator $\Theta/\operatorname{Tr}\Theta$ has the coherent diagonal

$$\frac{\langle z|\Theta|z\rangle}{\operatorname{Tr}\Theta} = \frac{1}{\sqrt{\det \mathsf{C}_\tau}}\exp\big[-\tfrac12 r(z)^T\mathsf{C}_\tau^{-1}r(z)\big] = \frac{1}{(2\pi)^n}\int_{\mathbb{R}^{2n}} e^{-\frac12 u^T\mathsf{C}_\tau u}\, e^{-iu^T r(z)}\, d^{2n}u.$$

For every density operator, as in Step 1, the coherent diagonal is $(2\pi)^{-n}$ times the Fourier transform at $r(z)$ of the product of its characteristic function and the vacuum characteristic function $e^{-\frac14 u^T u}$, which has no zeros. Uniqueness of the Fourier transform therefore gives $\Theta/\operatorname{Tr}\Theta$ the characteristic function $e^{-\frac12 u^T(\mathsf{C}_\tau-\frac12 I)u}$, so $\Theta/\operatorname{Tr}\Theta$ is the centred Gaussian state with covariance $V_\tau = \mathsf{C}_\tau - \frac12 I$. With the formula for C_τ above, the covariance of the normalized sandwich is

$$V_\tau = -\mathsf{T}_\tau + \mathsf{L}(\mathsf{T}_\tau - V_\rho)^{-1}\mathsf{L}. \tag{A.5}$$

Step 6: the trace formula. The form of (A.2) in Step 3 and $I-\mathbb{T}_\sigma^{-2} = (\mathsf{T}_\tau+\frac12 I)^{-1}$ give

$$\mathsf{C}_\tau^{-1} = \mathbb{T}_\sigma\big(\mathsf{C}_\rho^{-1}-(\mathsf{T}_\tau+\tfrac12 I)^{-1}\big)\mathbb{T}_\sigma = \mathbb{T}_\sigma\mathsf{C}_\rho^{-1}(\mathsf{T}_\tau-V_\rho)(\mathsf{T}_\tau+\tfrac12 I)^{-1}\mathbb{T}_\sigma,$$

since $\mathsf{T}_\tau+\frac12 I-\mathsf{C}_\rho = \mathsf{T}_\tau - V_\rho$. Taking determinants in the product form of (A.2), and using $\mathsf{T}_\tau+\frac12 I = \mathbb{T}_\sigma^2(\mathbb{T}_\sigma^2-I)^{-1}$ from Step 5, gives

$$\frac{\det \mathsf{C}_\tau}{\det \mathsf{C}_\rho} = \frac{\det\big((\mathsf{T}_\tau+\frac12 I)\mathbb{T}_\sigma^{-2}\big)}{\det(\mathsf{T}_\tau-V_\rho)} = \frac{1}{\det(\mathbb{T}_\sigma^2-I)\det(\mathsf{T}_\tau-V_\rho)}.$$

All factors of $\operatorname{Tr}\Theta$ except $\det(\mathsf{T}_\tau-V_\rho)$ depend only on σ,τ:

$$\operatorname{Tr}\Theta \overset{(A.4)}{=} \frac{Z_\sigma^{-\tau}}{\sqrt{\det(\mathbb{T}_\sigma^2-I)}}\,\frac{1}{\sqrt{\det(\mathsf{T}_\tau-V_\rho)}}.$$

At $\rho=\sigma$, $\Theta=\sigma^{1-\tau}$, and (A.3) holds because $\mathsf{T}_\tau - V_\sigma = \mathsf{H}_\tau(V_\sigma) > 0$. Evaluation of the formula for $\operatorname{Tr}\Theta$ at $\rho=\sigma$ therefore gives $Z_\sigma^{-\tau}\det(\mathbb{T}_\sigma^2-I)^{-1/2} = \operatorname{Tr}\sigma^{1-\tau}(\det\mathsf{H})^{1/2}$, with

$$\mathsf{H} = \mathsf{T}_\tau - V_\sigma, \qquad \mathsf{Z} = \mathsf{H}^{-1/2}(V_\rho - V_\sigma)\mathsf{H}^{-1/2}.$$

The factorization $\mathsf{T}_\tau - V_\rho = \mathsf{H} - (V_\rho - V_\sigma) = \mathsf{H}^{1/2}(I-\mathsf{Z})\mathsf{H}^{1/2}$ gives $\det(\mathsf{T}_\tau - V_\rho) = \det\mathsf{H}\,\det(I-\mathsf{Z})$. With this value of $Z_\sigma^{-\tau}\det(\mathbb{T}_\sigma^2-I)^{-1/2}$, the formula for $\operatorname{Tr}\Theta$ becomes the trace formula

$$\operatorname{Tr}\Theta = \operatorname{Tr}\sigma^{1-\tau}\sqrt{\frac{\det\mathsf{H}}{\det(\mathsf{T}_\tau-V_\rho)}} = \frac{\operatorname{Tr}\sigma^{1-\tau}}{\sqrt{\det(I-\mathsf{Z})}}. \tag{A.6}$$

Equations (A.5) and (A.6) are the two identities in (4.4).

Data availability

This article has no associated data.

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